PEARSON EDEXCEL INTERNATIONAL A LEVEL CHEMISTRY · WCH15

Unit 5: Transition Metals and Organic Nitrogen Chemistry

Revision notes, spec point by spec point (Topics 16 to 20)

How to use these notes. Every section starts with a spec-point badge and a short summary of what that spec point asks, so you always know which point you are revising. The header at the top of each page names the spec point(s) on that page.

BEYOND THE SPEC, BUT EXAMINED

Content the specification does not spell out, but that past-paper mark schemes have rewarded.

EXAM TIP

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MARK-SCHEME LANGUAGE

Words and phrases mark schemes credit.

COMMON MISTAKE

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Topic 16 – Redox Equilibria

SPEC 16.1 Oxidation and reduction: electrons and oxidation numbers

WHAT THIS SPEC POINT ASKS

Define oxidation and reduction as electron transfer and as changes in oxidation number, and apply this to s-, p- and d-block elements.

16.1 The two definitions you use at A level

	In terms of electrons	In terms of oxidation number
Oxidation	**loss** of electrons	oxidation number **increases**
Reduction	**gain** of electrons	oxidation number **decreases**

- Memory aid: **OIL RIG** (Oxidation Is Loss, Reduction Is Gain).
- An **oxidising agent** takes electrons, so it is itself **reduced**. A **reducing agent** gives electrons, so it is itself **oxidised**.
- Oxygen and hydrogen definitions (gain of O, loss of H) still work for simple cases, but only the electron and oxidation-number definitions work for everything.

16.1 Across the blocks

Block	Typical behaviour	Example
s-block metals	oxidised to 1+ or 2+ ions	$Na \rightarrow Na^+ + e^-$; $Ca \rightarrow Ca^{2+} + 2e^-$
p-block metals	oxidised to positive ions	$Al \rightarrow Al^{3+} + 3e^-$; Sn shows +2 and +4
p-block non-metals	usually reduced to negative ions (charge = group number − 8)	$Cl_2 + 2e^- \rightarrow 2Cl^-$; $O_2 + 4e^- \rightarrow 2O^{2-}$
d-block metals	oxidised to ions in **several** oxidation states	Fe^{2+} and Fe^{3+}; Mn from +2 to +7

High d-block oxidation states rarely exist as bare ions. They are found in **oxo-ions**: V(+5) is VO_2^+, V(+4) is VO^{2+}, Mn(+7) is MnO_4^-, Cr(+6) is $Cr_2O_7^{2-}$.

WORKED EXAMPLE: OXIDATION NUMBERS IN UNIT 5 IONS

MnO_4^-: Mn + 4(−2) = −1 → **Mn = +7**
MnO_4^{2-}: Mn + 4(−2) = −2 → **Mn = +6**
$Cr_2O_7^{2-}$: 2Cr + 7(−2) = −2 → **Cr = +6** (the same in CrO_4^{2-})
VO_2^+: V + 2(−2) = +1 → **V = +5**; **VO^{2+}**: V + (−2) = +2 → **V = +4**
$S_2O_3^{2-}$ (thiosulfate): 2S + 3(−2) = −2 → **S = +2**
$S_4O_6^{2-}$ (tetrathionate): 4S + 6(−2) = −2 → **S = +2.5** (an average)
$C_2O_4^{2-}$ (ethanedioate): 2C + 4(−2) = −2 → **C = +3**
$[Rh(CO)_2I_2]^-$: CO is neutral, each I is −1 → Rh − 2 = −1 → **Rh = +1**

16.1 Balancing a half-equation in acid

Use this every time; it works for any oxo-ion.

1. Balance the element that changes oxidation number.

2. Balance **O** by adding **H_2O**.
3. Balance **H** by adding **H^+**.
4. Balance **charge** by adding **electrons** (e^-) to the more positive side.

WORKED EXAMPLE: MnO_4^- TO Mn^{2+}

Mn already balanced: $MnO_4^- \rightarrow Mn^{2+}$
Add $4H_2O$ for the oxygens: $MnO_4^- \rightarrow Mn^{2+} + 4H_2O$
Add $8H^+$ for the hydrogens: $MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O$
Charges: left +7, right +2. Add $5e^-$ to the left:
$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)$
Check: Mn goes +7 → +2, a gain of 5 electrons. ✓

To build the **overall** equation, multiply each half-equation so the electrons are equal, add them, then cancel electrons and any H^+ or H_2O that appear on both sides.

MARK-SCHEME LANGUAGE

- Name the **element** and give both oxidation numbers: "Mn reduced from (+)4 to (+)2"; "Cl oxidised from −1 to 0 (in Cl_2)" (WCH15 Jan 2023).
- DO NOT ACCEPT "HNO_3 is reduced": say "nitrogen is reduced from +5 to +2" (WCH15 Jun 2023).
- Not redox: "the oxidation number of vanadium is +5 in both species". IGNORE "there are no electrons in the equation" (WCH15 Oct 2020).

Often asked as: which element is oxidised/reduced and how you know; is this reaction redox?; oxidation number of a metal in an unfamiliar complex.

COMMON MISTAKE

Naming the compound or ion ("nitric acid is reduced") instead of the element.
Saying H or O is oxidised or reduced when it is really the metal that changes.
Calling a reaction redox just because the formulae change: $Cr_2O_7^{2-} \rightarrow CrO_4^{2-}$ is **not** redox (Cr is +6 in both).

BEYOND THE SPEC, BUT EXAMINED

Using the **change** in oxidation number from titration data to find an unknown final oxidation state (WCH15 Oct 2021).
Oxidation numbers in organometallic catalysts, e.g. Rh is +1 in $[Rh(CO)_2I_2]^-$ (WCH15 Jan 2026).

SPEC 16.2 Standard electrode potential, E°

WHAT THIS SPEC POINT ASKS

Know what a standard electrode potential is.

16.2 What it measures

- A **half-cell** is an element in contact with its ions (or two ions of one element), set up so an equilibrium like $Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$ is established at the electrode surface.
- You can't measure the potential of one half-cell on its own. You can only measure the **potential difference** between two half-cells.

- So every half-cell is compared with the same reference: the **standard hydrogen electrode** (SHE), defined as **0.00 V**.

KEY IDEA

The **standard electrode potential**, E°, of a half-cell is the **potential difference (emf)** of a cell made from that half-cell and a **standard hydrogen electrode**, measured under **standard conditions** (298 K, 1.00 mol dm^{-3} ions, 100 kPa gases).

16.2 What the sign tells you

- Half-equations are always written as **reductions** (electrons on the left).
- **More positive E°**: the left-hand species is **more easily reduced** than H^+. It is a better **oxidising agent**. Example: $Cl_2 + 2e^- \rightleftharpoons 2Cl^-$, E° = +1.36 V.
- **More negative E°**: the right-hand species is **more easily oxidised**. It is a better **reducing agent**. Example: $Zn^{2+} + 2e^- \rightleftharpoons Zn$, E° = −0.76 V, so zinc readily gives electrons.

EXAM TIP

E° is never asked as a bare definition in WCH15 papers so far. It is tested through the conditions and the SHE inside a "draw the cell" or "describe how to measure" question. Learn the definition anyway: it is the backbone of those answers.

COMMON MISTAKE

Writing a half-equation as an oxidation when quoting E°. Data-book half-equations are always reductions.

SPEC 16.3 Standard conditions for E°

WHAT THIS SPEC POINT ASKS

Know the standard conditions for measuring E°: 298 K, 100 kPa for gases, 1.00 mol dm^{-3} for ions.

Condition	Value	Why it matters
Temperature	**298 K** (25 °C)	E depends on temperature
Pressure of any gas	**100 kPa**	changes the position of an equilibrium involving a gas
Concentration of each ion	**1.00 mol dm^{-3}**	changes the position of the half-cell equilibrium

- In a half-cell with two ions (e.g. Fe^{3+}/Fe^{2+}), **each** ion is at 1.00 mol dm^{-3}.
- If H^+ appears in the half-equation (MnO_4^-, $Cr_2O_7^{2-}$), $[H^+]$ must also be 1.00 mol dm^{-3}.
- A **high-resistance voltmeter** is used so almost no current flows. If current flowed, the concentrations would change and the reading would drift.

MARK-SCHEME LANGUAGE

- "1 mol dm^{-3}" for each ion (stated once if all the same); "hydrogen gas at 100 kPa"; "298 K / 25 °C". The temperature "may be shown anywhere" on a diagram, but it must be there (WCH15 Jun 2023).
- ACCEPT 101 kPa, 1 atm, 1 bar or 1×10^5 Pa for the pressure.

Often asked as: the conditions labelled on a cell diagram; MCQ on the pressure in Pa (100 000 Pa).

COMMON MISTAKE

Leaving out the temperature because it is not "attached" to a piece of apparatus (WCH15 Jun 2023 examiner report). Forgetting the H_2 pressure or the H^+ concentration (WCH15 Oct 2023 examiner report).

BEYOND THE SPEC, BUT EXAMINED

Dilution on mixing. If you mix equal volumes of two solutions to make the half-cell electrolyte, each concentration halves. $Fe_2(SO_4)_3$ supplies **two** Fe^{3+} per formula unit, so 0.5 mol dm^{-3} $Fe_2(SO_4)_3$ gives 1.0 mol dm^{-3} Fe^{3+} (WCH15 Jun 2021).

SPEC 16.4 The standard hydrogen electrode and why a reference is needed

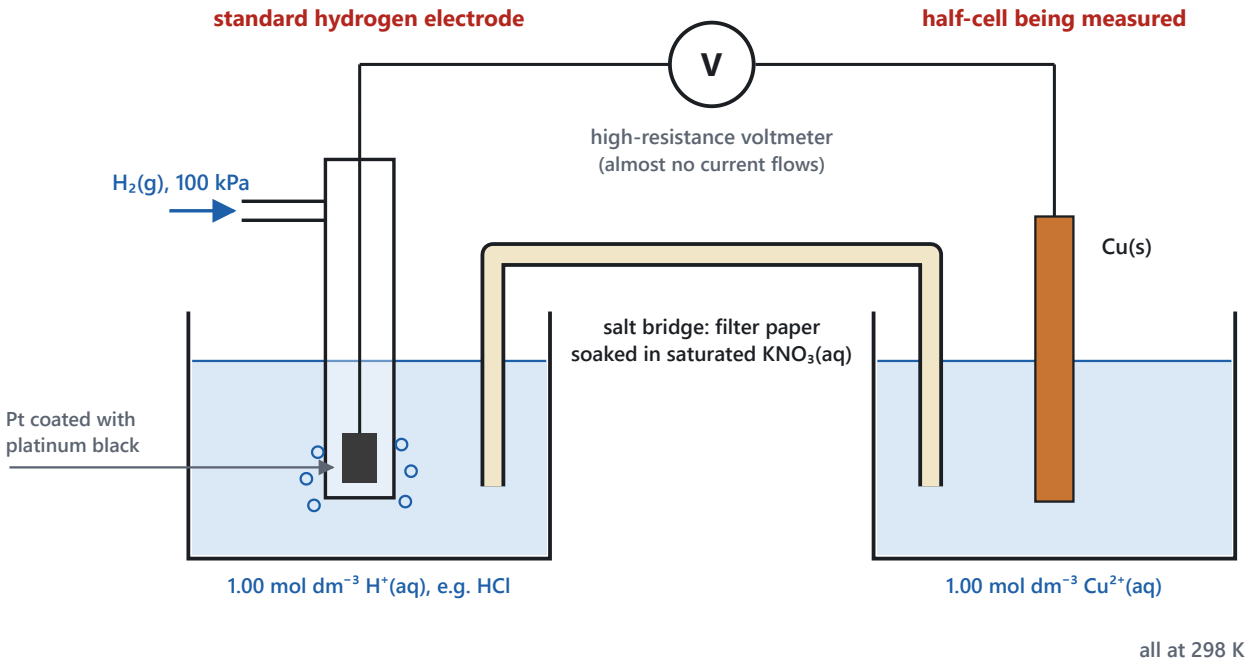
WHAT THIS SPEC POINT ASKS

Know the features of the standard hydrogen electrode (SHE) and why a reference electrode is needed.

16.4 Why you need a reference

- A voltmeter needs **two** connections. Putting a second wire into the same solution would create a second electrode with its own potential, so a single electrode potential can never be measured.
- The fix: agree on one half-cell as the zero and measure everything against it. That reference is the SHE, given a potential of **exactly 0.00 V**.

16.4 Features of the SHE



Measuring E° for Cu^{2+}/Cu. The SHE (left) is the reference; the voltmeter reads +0.34 V, with the copper electrode positive.

- **Hydrogen gas at 100 kPa** bubbled over the electrode.
- A solution with **$[H^+]$ = 1.00 mol dm^{-3}** (e.g. 1.00 mol dm^{-3} HCl).
- An inert **platinum** electrode coated with **platinum black** (finely divided Pt). The large surface area lets the equilibrium $2H^+(aq) + 2e^- \rightleftharpoons H_2(g)$ be reached **quickly**, and platinum conducts electrons without reacting.
- **298 K**.

16.4 Why it's awkward in practice

The SHE needs a steady supply of hydrogen at exactly 100 kPa, the platinum is easily "poisoned" by impurities, and equilibrium is slow to set up. Labs often use a **secondary standard** (such as a calomel electrode) whose potential against the SHE is known.

MARK-SCHEME LANGUAGE

- "1 mol dm^{-3} H^+(aq) and platinum (black) electrode"; "hydrogen gas … 100 kPa (at 298 K)"; "hydrogen gas must be able to enter the tube" (WCH15 Oct 2023).
- Platinum black is there "to increase the rate of the equilibrium between hydrogen gas and hydrogen ions" (WCH15 Jan 2021).
- Advantage of a calomel electrode (any one): "does not require a continuous supply of hydrogen"; "platinum is easily poisoned"; "difficult to ensure the hydrogen electrode is at equilibrium". IGNORE cheaper, safer, quicker, "hydrogen is explosive" (WCH15 Jun 2023).

Often asked as: draw and label the SHE as part of a cell; purpose of platinum black; advantage of another reference electrode.

COMMON MISTAKE

Hydrogen drawn with no way into the tube, or flowing away from the platinum; one beaker with no liquid; a salt bridge that doesn't dip into both solutions (WCH15 Oct 2023 examiner report).
Using 1 mol dm^{-3} H_2SO_4: that gives 2 mol dm^{-3} H^+. Use 1 mol dm^{-3} HCl. (0.5 mol dm^{-3} H_2SO_4 gives 1 mol dm^{-3} H^+; one mark scheme refused it, two later ones accepted it, so HCl is the safe answer.)

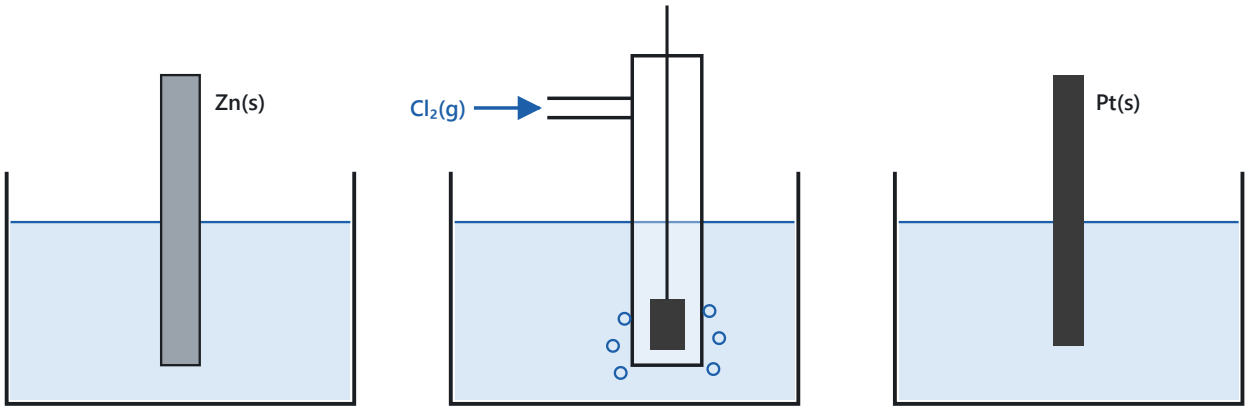
BEYOND THE SPEC, BUT EXAMINED

Calomel electrode (Hg_2Cl_2 | Hg in saturated KCl), E = +0.24 V against the SHE. You are given its value and work out the other electrode from the measured emf. The KCl is saturated to keep **[Cl^-] constant** (not "to act as a salt bridge") (WCH15 Jun 2023).

SPEC 16.5 Measuring E° for different kinds of half-cell

WHAT THIS SPEC POINT ASKS

Understand the different set-ups for (i) a metal or non-metal in contact with its ions and (ii) two ions of the same element in different oxidation states.



Each half-cell is joined to a SHE by a salt bridge and a high-resistance voltmeter.

Type	Electrode	Solution	Example
Metal / its ion	the **metal itself**	1.00 mol dm^{-3} of its ion	Zn in $Zn^{2+}(aq)$: $Zn^{2+} + 2e^- \rightleftharpoons Zn$, −0.76 V
Non-metal / its ion	**platinum** (the non-metal can't conduct, or is a gas or liquid)	1.00 mol dm^{-3} of the ion; gas at 100 kPa	Cl_2 bubbled over Pt in $Cl^-(aq)$: +1.36 V; Br_2/Br^- (both in solution): +1.09 V
Two ions of one element	**platinum**	**both** ions at 1.00 mol dm^{-3} (plus 1.00 mol dm^{-3} H^+ if H^+ is in the half-equation)	Fe^{3+}/Fe^{2+}: +0.77 V; MnO_4^-/Mn^{2+} with H^+: +1.51 V

- Platinum is **inert**: it doesn't react, it just provides a surface and carries electrons.
- The **salt bridge** completes the circuit by letting **ions** move between the half-cells. A wire can't do this, because it only carries electrons and would set up extra electrode potentials.
- The salt in the bridge must not react with either solution. KNO_3 is the usual choice. KCl is avoided with Ag^+ ($AgCl$ precipitates) and never use something like $KMnO_4$ (it reacts).

MARK-SCHEME LANGUAGE

- Electrode for Fe^{3+}/Fe^{2+}: "platinum / Pt" (ACCEPT platinised platinum, platinum black). DO NOT ACCEPT "iron" (WCH15 Oct 2020).
- Solution: "Fe^{2+} and Fe^{3+} ions, 1 mol dm^{-3} of each", e.g. "1 mol dm^{-3} $FeSO_4$ and 0.5 mol dm^{-3} $Fe_2(SO_4)_3$" (WCH15 Jan 2025).
- Salt bridge: "(saturated) potassium nitrate"; function: "ions can flow through a salt bridge (but not through a wire)". DO NOT ACCEPT "electrons flow through the salt bridge" (WCH15 Jan 2025).

Often asked as: label the parts of a Fe^{3+}/Fe^{2+} half-cell; choose the electrode for an unfamiliar couple; why a salt bridge and not a wire.

COMMON MISTAKE

Giving "Fe^{2+} and Fe^{3+}" with no concentrations or no named compound. Explaining the salt bridge as "balancing charge" without saying **ions move** (WCH15 Jan 2025 examiner report).
Using a metal electrode for a couple of two ions: the Fe^{3+}/Fe^{2+} half-cell needs Pt, not iron.

SPEC 16.6 Core Practical 12: investigating electrochemical cells

WHAT THIS SPEC POINT ASKS

Set up electrochemical cells, measure their emf, and compare the results with values calculated from standard electrode potentials.

CORE PRACTICAL 12: METHOD IN BRIEF

1. Clean strips of metal (e.g. Cu, Zn, Fe, Ag) with **emery paper** to remove the oxide layer, then rinse.
2. Put each strip in a small beaker of a **1.0 mol dm^{-3}** solution of its own ions (e.g. $CuSO_4$, $ZnSO_4$, $FeSO_4$; $AgNO_3$ for silver).
3. Make a salt bridge from a strip of **filter paper soaked in saturated KNO_3(aq)**, dipping into both beakers.
4. Connect the two metals to a **high-resistance voltmeter** with leads and crocodile clips. Record the reading once it is steady.
5. Repeat for other pairs, using a fresh salt bridge each time.
6. For a two-ion half-cell, use a **platinum** (or carbon) electrode in a solution containing both ions, e.g. Fe^{2+} and Fe^{3+} at 1.0 mol dm^{-3} each.
7. Compare each measured emf with E°cell calculated from the Data Booklet.

16.6 Making sense of the results

- The metal connected to the **negative** terminal (when the reading is positive) is the half-cell with the more negative E°: it is where oxidation happens.
- Measured values are usually a little **lower** than the Data Booklet values, because:
  - conditions are not standard (room temperature isn't exactly 298 K; concentrations aren't exactly 1.00 mol dm^{-3});
  - oxide layers or impurities remain on the metal surface;
  - the voltmeter draws a small current.
- School labs often use Cu^{2+}/Cu (+0.34 V) as the reference instead of a SHE, then add 0.34 V to convert.

16.6 Safety

$AgNO_3$ stains skin and is an irritant; $CuSO_4$ is harmful and toxic to aquatic life; wear eye protection and gloves where needed. Dispose of metal-ion solutions in the residues container, not the sink.

MARK-SCHEME LANGUAGE

For "describe how you would measure E° of $Co^{2+}|Co$" (6-mark, WCH15 Oct 2024), the indicative points are:
- SHE: H_2 gas, platinum electrode, "1 atm, 1.0 mol dm^{-3} HCl and 298 K";
- salt bridge "filter paper soaked in (saturated) KNO_3";
- "1.0 mol dm^{-3} of a named cobalt salt, e.g. $Co(NO_3)_2$", with a cobalt electrode;
- "(high-resistance) voltmeter … to complete the circuit";
- "voltmeter reading = standard electrode potential".

Often asked as: draw and label the apparatus (3–5 marks); 6-mark description of measuring E° for a metal or for Fe^{3+}/Fe^{2+}.

COMMON MISTAKE

Writing "Co^{2+}(aq), 1.0 mol dm^{-3}" without naming a cobalt **compound** (WCH15 Oct 2024 examiner report). Using platinum when the question gives you copper and zinc electrodes (WCH15 Jun 2022 examiner report). Drawing the salt bridge as an unlabelled line: label it and say what is in it.

BEYOND THE SPEC, BUT EXAMINED

Reading the sign of the voltmeter. The right-hand electrode is connected to the positive terminal, so a positive reading means the left electrode is the negative one (WCH15 Jun 2023).

SPEC 16.7

Calculating E°cell

WHAT THIS SPEC POINT ASKS

Calculate the standard emf of a cell, E°cell, from two standard electrode potentials.

KEY IDEA

E°cell = E°(right) − E°(left) = E°(reduction half-cell) − E°(oxidation half-cell)
For a working (feasible) cell, this is **more positive minus less positive**, so E°cell is positive.

- The half-cell with the **more positive** E° is where **reduction** happens: it is the **positive** terminal.
- The half-cell with the **more negative** E° is where **oxidation** happens: it is the **negative** terminal.
- Never multiply an E° value when you multiply a half-equation to balance electrons. E° is a property of the half-cell, not of the amount.

WORKED EXAMPLE: THE DANIELL CELL

$Zn^{2+} + 2e^- \rightleftharpoons Zn$, E° = −0.76 V; $Cu^{2+} + 2e^- \rightleftharpoons Cu$, E° = +0.34 V
Cu^{2+}/Cu is more positive, so Cu^{2+} is reduced (right); Zn is oxidised (left).
E°cell = (+0.34) − (−0.76) = **+1.10 V**. Copper is the positive electrode.

WORKED EXAMPLE: WORKING BACK TO AN UNKNOWN E°

A cell is made from $Zn^{2+}|Zn$ (left, −0.76 V) and an unknown half-cell (right). E°cell = +1.56 V.
E°cell = E°right − E°left → 1.56 = E°right − (−0.76) → **E°right = +0.80 V**.
The Data Booklet shows this is $Ag^+ + e^- \rightleftharpoons Ag$.

MARK-SCHEME LANGUAGE

- "E°cell = E°R − E°L (or most positive − least positive)". The **sign** must be shown: "+1.17 V"; "−1.17 (V) scores 1 mark" (WCH15 Jun 2025).
- Choose half-equations only from the Data Booklet when told to: "No other metals will score" (WCH15 Jan 2025).

Often asked as: one-mark MCQ calculations; find an unknown E° from a measured E°cell; choose a metal that gives the largest E°cell.

COMMON MISTAKE

Doubling E° when you double the half-equation. Dropping the + sign. Comparing two E° values in words when the question asks for E°cell (WCH15 Jun 2024 examiner report).
Using the wrong half-equation for a species that appears in two of them, e.g. thiosulfate/sulfur instead of thiosulfate/tetrathionate (WCH15 Oct 2022 examiner report).

BEYOND THE SPEC, BUT EXAMINED

Using a **numbered table** of unfamiliar half-equations, including alkaline ones, and picking the right pair (WCH15 Jun 2021; Jun 2024). Alkaline cells: Zn/MnO_2 (WCH15 Jan 2023).

SPEC 16.8

Cell diagrams

WHAT THIS SPEC POINT ASKS

Write cell diagrams using the conventional representation of half-cells.

16.8 The rules

- A **single vertical line** (|) marks a phase boundary, e.g. between a solid metal and a solution.
- A **double vertical line** (||), or a dashed one, is the salt bridge.
- A **comma** separates two species in the **same** phase.
- **Oxidation** half-cell on the **left**, **reduction** half-cell on the **right** (the two r's go together: Reduction on the Right).
- Electrodes go on the **outside** edges. Pt is written in wherever neither species is a metal.
- Write each side in the **direction it reacts**: on the left, reactant (the reduced form) then product; on the right, reactant (the oxidised form) then product.
- Include **state symbols** and, where the half-equations have them, the **molar ratios**, H^+ and H_2O.

WORKED EXAMPLES: CELL DIAGRAMS

Zn/Cu: **$Zn(s) \mid Zn^{2+}(aq) \,||\, Cu^{2+}(aq) \mid Cu(s)$** E°cell = +1.10 V
SHE with copper: **$Pt(s) \mid H_2(g) \mid 2H^+(aq) \,||\, Cu^{2+}(aq) \mid Cu(s)$** E°cell = +0.34 V
Fe^{2+}/Fe^{3+} with chlorine: **$Pt(s) \mid Fe^{2+}(aq), Fe^{3+}(aq) \,||\, Cl_2(g) \mid 2Cl^-(aq) \mid Pt(s)$** E°cell = 1.36 − 0.77 = +0.59 V
Chloride with dichromate: **$Pt(s) \mid 2Cl^-(aq) \mid Cl_2(g) \,||\, [Cr_2O_7^{2-}(aq) + 14H^+(aq)], [2Cr^{3+}(aq) + 7H_2O(l)] \mid Pt(s)$**

MARK-SCHEME LANGUAGE

- Allowed: a comma or a vertical line between ions in one phase; a dashed line for the salt bridge.
- Penalised (once each): missing molar ratios; a single solid line between the half-cells; electrons written in the diagram (WCH15 Oct 2023).
- Part marks: "1 for the whole diagram reversed"; "1 if Pt missing on both sides but otherwise correct".

Often asked as: pick the correct diagram (MCQ); write the diagram for two given half-equations (2 marks).

COMMON MISTAKE

Leaving out the molar ratios, the H^+ and H_2O, or one of the platinum electrodes. Examiners say it is "very rare for both marks to be awarded" (WCH15 Oct 2023 examiner report).
Writing an overall equation or two half-equations when a cell diagram was asked for.

SPEC 16.9 Why the conditions matter: non-standard electrode potentials

WHAT THIS SPEC POINT ASKS

Understand why the conditions matter when measuring an electrode potential, E.

16.9 Concentration: think Le Chatelier

Each half-cell is an equilibrium written as a reduction: oxidised form + ne^- ⇌ reduced form.

- **Increase the oxidised form** (e.g. more Cu^{2+}), or decrease the reduced form → equilibrium shifts **right**, electrons are removed → E becomes **more positive**.
- **Decrease the oxidised form** (e.g. less Zn^{2+}) → equilibrium shifts **left**, electrons are released → E becomes **more negative**.
- **H^+ in the half-equation** (MnO_4^-, $Cr_2O_7^{2-}$): more acid shifts the equilibrium right, so E is **more positive** and the ion is a stronger oxidising agent. That is why these oxidants are used in **acid**.

Change	Effect on E of that half-cell	Effect on E°cell (Zn/Cu cell)
$[Cu^{2+}]$ increased (right-hand cell)	more positive	increases
$[Zn^{2+}]$ decreased (left-hand cell)	more negative	increases
$[Zn^{2+}]$ increased	less negative	decreases

16.9 Temperature and pressure

- Temperature changes the position of every equilibrium, so E changes with temperature too.
- For a gas electrode, a higher gas pressure has the same effect as a higher concentration.

MARK-SCHEME LANGUAGE

- Lower $[Zn^{2+}]$ in a Zn/Cu cell: "Ecell would increase / become more positive"; "$Zn^{2+} + 2e^-$ ⇌ Zn would shift to the left, making the Zn half-cell more negative". Standalone marks (WCH15 Jan 2025).
- Concentrated HCl with dichromate (E°cell = 1.33 − 1.36 = −0.03 V under standard conditions): "increases $[H^+]$ in the first equilibrium (shifting it right) so increases E"; "increases $[Cl^-]$ in the second equilibrium (shifting it left) so decreases E"; "the values are very close, so Ecell becomes positive". Every mark needs **both** the shift and its effect on E (WCH15 Oct 2023).
- The practical reason concentrated HCl is not used to acidify dichromate: it would produce **toxic chlorine** (WCH15 Oct 2024).

Often asked as: effect of changing one concentration on E or Ecell; why a reaction with a slightly negative E°cell can still happen.

COMMON MISTAKE

A bare "the conditions are non-standard" earns nothing. Say **which** equilibrium shifts, **which way**, and what that does to E (WCH15 Jan 2025 examiner report).
Saying a concentration above 1 mol dm^{-3} makes a reduction potential less positive. More of the oxidised form makes it **more** positive (WCH15 Jun 2022 examiner report).

BEYOND THE SPEC, BUT EXAMINED: THE NERNST EQUATION

The formula is always given. For oxidised form + ne^- ⇌ reduced form:

$E = E° + (RT / nF) \ln([\text{oxidised}] / [\text{reduced}])$

R = 8.31 J K^{-1} mol^{-1}, T = 298 K, F = 96 500 C mol^{-1}, n = electrons transferred. For a metal/metal-ion half-cell, [reduced] (the solid metal) is taken as 1, so $E = E° + (RT/nF) \ln[M^{n+}]$.

Asked in WCH15 Oct 2020, Jun 2022 and Oct 2024.

WORKED EXAMPLE: NERNST EQUATION BOTH WAYS

Find E for $Zn^{2+}|Zn$ with $[Zn^{2+}]$ = 0.010 mol dm^{-3} ($E°$ = −0.76 V, n = 2).

$RT/nF = (8.31 \times 298) \div (2 \times 96\,500) = 0.01283$ V

$E = -0.76 + 0.01283 \times \ln 0.010 = -0.76 + 0.01283 \times (-4.605) = -0.76 - 0.059 =$ **−0.82 V** (more negative, as Le Chatelier predicts).

Find the concentration when E = −0.79 V.

$\ln[Zn^{2+}] = (-0.79 - (-0.76)) \div 0.01283 = -2.338$

$[Zn^{2+}] = e^{-2.338} =$ **0.097 mol dm^{-3}**

Keep unrounded values in your calculator: one paper refused an answer of 9×10^{-4} when the working gave 8.99×10^{-4} (WCH15 Oct 2024).

SPEC **16.10**

Using E° to predict whether a reaction is feasible

WHAT THIS SPEC POINT ASKS

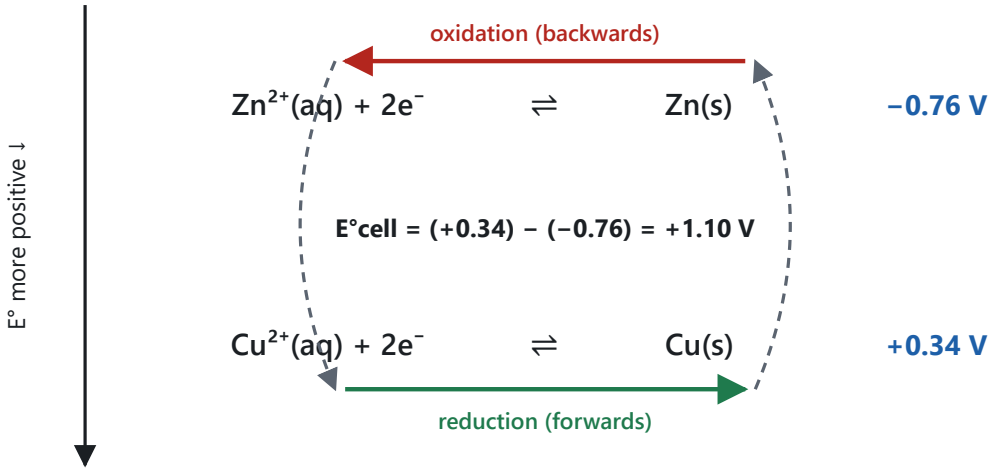
Use standard electrode potentials to predict the thermodynamic feasibility of a reaction.

KEY IDEA

A reaction is **thermodynamically feasible** if **E°cell is positive**, where E°cell = E°(half-cell being reduced) − E°(half-cell being oxidised).

16.10 The anticlockwise rule

Write the two half-equations as reductions, the **more negative** one on top. The reaction that goes is **anticlockwise**: the lower (more positive) one goes **forwards** (reduction), the upper one goes **backwards** (oxidation).



Overall: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

The more positive couple is reduced; the more negative couple runs in reverse. Zn reduces Cu^{2+}, not the other way round.

16.10 A method that always works

1. Find the two half-equations in the Data Booklet.
2. Decide which species must be **reduced** for the reaction you're asked about.
3. E°cell = E°(that reduction) − E°(the other half-equation).
4. Positive → feasible. Negative → not feasible (the **reverse** reaction is feasible).
5. Write the overall equation: reverse the oxidation half-equation, balance electrons, add, cancel.

WORKED EXAMPLE: WILL IRON(III) OXIDISE IODIDE? BROMIDE?

$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$, +0.77 V; $I_2 + 2e^- \rightleftharpoons 2I^-$, +0.54 V; $Br_2 + 2e^- \rightleftharpoons 2Br^-$, +1.09 V

Fe^{3+} with I^-: Fe^{3+} reduced, I^- oxidised. E°cell = 0.77 − 0.54 = **+0.23 V** → **feasible**.

$2Fe^{3+}(aq) + 2I^-(aq) \rightarrow 2Fe^{2+}(aq) + I_2(aq)$

Fe^{3+} with Br^-: E°cell = 0.77 − 1.09 = **−0.32 V** → **not feasible**. (Instead, Br_2 oxidises Fe^{2+}: +0.32 V.)

WORKED EXAMPLE: CHOOSING AN OXIDISING AGENT

Which halogen could oxidise Fe^{2+} to Fe^{3+}? It needs E° more positive than +0.77 V.

Cl_2 (+1.36 V) and Br_2 (+1.09 V) can; I_2 (+0.54 V) can't.

$Cl_2(aq) + 2Fe^{2+}(aq) \rightarrow 2Cl^-(aq) + 2Fe^{3+}(aq)$, E°cell = 1.36 − 0.77 = **+0.59 V**

MARK-SCHEME LANGUAGE

- E°cell must carry a **+ sign or the word "positive"** (WCH15 Oct 2025).
- Equations and E°cell values score **separately**: four correct half-equations score even if not combined (WCH15 Oct 2022).
- "VO^{2+} is not oxidised any further as E°cell is −0.2 V / negative" (WCH15 Oct 2020). Uncancelled electrons, H^+ or H_2O are penalised once.

Often asked as: write the equation, calculate E°cell and say whether it is feasible (3–5 marks); 6-mark on which reducing agents can take vanadium to which oxidation state.

COMMON MISTAKE

Calculating every E°cell but writing no equations, or the reverse. Give **both** (WCH15 Jan 2023 examiner report).
Not saying where a reduction **stops**, e.g. "so V^{3+} is not reduced to V^{2+} because E°cell is negative" (WCH15 Oct 2022 examiner report).

BEYOND THE SPEC, BUT EXAMINED

A reaction with a small negative E°cell, e.g. −0.01 V, may still happen to some extent if conditions change (WCH15 Oct 2022). The anticlockwise rule with an unfamiliar middle couple, e.g. Fe^{3+}/Fe^{2+} (+0.77 V) between I_2/I^- and $S_2O_8^{2-}/SO_4^{2-}$ (WCH15 Jan 2026).

SPEC 16.11

E°cell, total entropy change and the equilibrium constant

WHAT THIS SPEC POINT ASKS

Understand that E°cell is directly proportional to ΔS°total and to ln K for a reaction.

16.11 Three ways of saying the same thing

Measure	Feasible when...	Link
E°cell	positive	—
ΔS°total	positive	ΔS°total ∝ E°cell
ln K	positive (K > 1)	ln K ∝ E°cell

- The **bigger** E°cell, the **bigger** ΔS°total and the **bigger** K. A large positive E°cell means the equilibrium lies far to the right: the reaction goes almost to completion.
- A small positive E°cell (a few tenths of a volt) still gives K > 1, but both reactants and products may be present at equilibrium.
- A negative E°cell means ΔS°total is negative and K < 1: the reverse reaction is the feasible one.

EXAM TIP

Think of three gauges of one quantity. If one is positive, the others are too. MCQs often show a straight line through the origin for E°cell against ΔS°total or ln K: that is the "directly proportional" idea.

MARK-SCHEME LANGUAGE

- "E°cell is proportional to ΔS°total"; "so the total entropy change is positive as E°cell is positive"; "a positive ΔS°total means the reaction is (thermodynamically) feasible" (WCH15 Jan 2026).
- ACCEPT ΔS°total = nFE°cell ÷ T; "(R) ln K is proportional to E°cell".

Often asked as: MCQ on the signs of ΔS°total and ln K for a positive E°cell; a 3-mark explanation linking E°cell to feasibility.

BEYOND THE SPEC, BUT EXAMINED: THE EQUATIONS BEHIND THE PROPORTIONALITY

$\Delta S°total = nFE°cell \div T$ and $\ln K = nFE°cell \div RT$ (accepted in mark schemes, but numerical use has not been required).
For the Daniell cell (n = 2, E°cell = +1.10 V): $\Delta S°total = 2 \times 96\,500 \times 1.10 \div 298 =$ **+712 J K^{-1} mol^{-1}**; ln K = 85.7, so $K \approx 10^{37}$: the reaction goes to completion.

SPEC **16.12**

Limitations of predictions from E°

WHAT THIS SPEC POINT ASKS

Understand why predictions from standard electrode potentials can be wrong: kinetic stability and non-standard conditions.

16.12 E° tells you "can it?", never "how fast?"

- A positive E°cell means the reaction is **thermodynamically feasible**. It says **nothing about the rate**.
- If the **activation energy is high**, a feasible reaction may be too slow to see. The reactants are **kinetically stable** (thermodynamically unstable, but kinetically inert).

Feasible but slow	E°cell	Why it's slow
$S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2$	2.01 − 0.54 = +1.47 V	two **negative ions repel**, so high Ea (see 17.31)
$2H_2O_2 \rightarrow 2H_2O + O_2$	1.77 − 0.68 = +1.09 V	high Ea; fast only with a catalyst such as MnO_2
MnO_4^- + $C_2O_4^{2-}$ at room temperature	positive	anions repel; needs heating and Mn^{2+} (see 17.32)

16.12 Conditions are rarely standard

- Real concentrations are not 1.00 mol dm^{-3}, temperatures are not 298 K, and many reactions don't happen in aqueous solution at all. Under different conditions E changes (16.9).
- If two E° values are **close** (within about 0.1 V), a modest change in concentration can reverse the prediction.
- E° values describe aqueous ions. They can't predict reactions between solids or gases, or in concentrated solutions.

MARK-SCHEME LANGUAGE

- Feasible but doesn't happen: "the activation energy for the reaction is high" and "the reactants repel (because they are both negatively charged)" (WCH15 Jan 2026); "reaction rate must be slow / activation energy high" (WCH15 Oct 2022).
- Prediction wrong the other way: "the conditions must be non-standard and the E° values are so close" (WCH15 Jun 2022). ACCEPT "high kinetic energy barrier".

Often asked as: explain why a reaction with a positive E°cell doesn't happen; explain why an unexpected reaction does happen.

COMMON MISTAKE

"The reaction is feasible despite a negative E°cell because conditions are non-standard" with no further reasoning (WCH15 Oct 2023 examiner report). Name the equilibrium and the direction it shifts.
Writing only "no reaction" when asked why, with no reference to rate or activation energy (WCH15 Oct 2022 examiner report).

SPEC 16.13

Standard reduction potentials and the electrochemical series

WHAT THIS SPEC POINT ASKS

Know that standard electrode potentials are also called standard reduction potentials and can be listed as an electrochemical series.

- Because every half-equation is written as a **reduction**, E° values are also called **standard reduction potentials**.
- Listing them in order of E° gives the **electrochemical series**.

Half-equation	E° / V	
$Li^+ + e^- \rightleftharpoons Li$	−3.03	← strongest **reducing agent** (Li)
$Mg^{2+} + 2e^- \rightleftharpoons Mg$	−2.37	
$Zn^{2+} + 2e^- \rightleftharpoons Zn$	−0.76	
$Fe^{2+} + 2e^- \rightleftharpoons Fe$	−0.44	
$2H^+ + 2e^- \rightleftharpoons H_2$	0.00	reference
$Cu^{2+} + 2e^- \rightleftharpoons Cu$	+0.34	
$I_2 + 2e^- \rightleftharpoons 2I^-$	+0.54	
$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$	+0.77	
$Ag^+ + e^- \rightleftharpoons Ag$	+0.80	
$Br_2 + 2e^- \rightleftharpoons 2Br^-$	+1.09	
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightleftharpoons 2Cr^{3+} + 7H_2O$	+1.33	
$Cl_2 + 2e^- \rightleftharpoons 2Cl^-$	+1.36	
$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O$	+1.51	
$F_2 + 2e^- \rightleftharpoons 2F^-$	+2.87	← strongest **oxidising agent** (F_2)

- **Oxidising agents** are on the **left**; they get stronger **down** the table.
- **Reducing agents** are on the **right**; they get stronger **up** the table.

MARK-SCHEME LANGUAGE

- "reduction" for "what type of potential is this?". IGNORE "redox", "standard electrode potential" (WCH15 Jun 2024).
- "electrochemical (series)" (ACCEPT reduction potential series). "Reactivity series" does not score (WCH15 Jun 2024).

Often asked as: two one-mark recall items.

COMMON MISTAKE

Writing "reactivity series". Under a quarter of candidates got "electrochemical series" (WCH15 Jun 2024 examiner report): it is straight recall of the spec wording.

SPEC 16.14 Disproportionation and E°

WHAT THIS SPEC POINT ASKS

Use standard electrode potentials to predict whether a disproportionation reaction is thermodynamically feasible.

16.14 What disproportionation is

Disproportionation: one species is **both oxidised and reduced** in the same reaction, so one element ends up in a higher **and** a lower oxidation state.

- Example: $2Cu^{+}(aq) \rightarrow Cu^{2+}(aq) + Cu(s)$. Copper goes +1 → +2 **and** +1 → 0.
- The reverse, where two oxidation states combine to make one in between, is **comproportionation**.

16.14 How to test it with E°

The species appears in **two** half-equations: once on the **left** (being reduced) and once on the **right** (being oxidised).

1. Find the half-equation where the species is **reduced** (it's on the left).
2. Find the half-equation where the species is **formed** (it's on the right); this one runs **backwards** as the oxidation.
3. E°cell = E°(reduction of the species) − E°(the one that forms it).
4. Positive → the species **disproportionates**. Negative → it is **stable** to disproportionation.

WORKED EXAMPLE: COPPER(I) DISPROPORTIONATES

$Cu^{+} + e^{-} \rightleftharpoons Cu$, E° = +0.52 V ($Cu^{+}$ reduced)
$Cu^{2+} + e^{-} \rightleftharpoons Cu^{+}$, E° = +0.15 V (run backwards: Cu^{+} oxidised)
E°cell = 0.52 − 0.15 = **+0.37 V → feasible**
$2Cu^{+}(aq) \rightarrow Cu(s) + Cu^{2+}(aq)$
This is why simple Cu^{+} salts dissolved in water give a brown copper precipitate and a blue solution. Cu(I) is only stable in water when it is in a precipitate (CuI) or a complex.

WORKED EXAMPLE: IRON(II) DOES NOT DISPROPORTIONATE

$Fe^{2+} + 2e^{-} \rightleftharpoons Fe$, E° = −0.44 V ($Fe^{2+}$ reduced)
$Fe^{3+} + e^{-} \rightleftharpoons Fe^{2+}$, E° = +0.77 V (run backwards: Fe^{2+} oxidised)
E°cell = −0.44 − 0.77 = **−1.21 V → not feasible**. $3Fe^{2+} \rightarrow Fe + 2Fe^{3+}$ doesn't happen. The reverse ($Fe + 2Fe^{3+} \rightarrow 3Fe^{2+}$, comproportionation) does.

WORKED EXAMPLE: MANGANATE(VI) IN ACID

$MnO_4^{2-} + 4H^{+} + 2e^{-} \rightleftharpoons MnO_2 + 2H_2O$, E° = +2.26 V ($MnO_4^{2-}$ reduced, Mn +6 → +4)
$MnO_4^{-} + e^{-} \rightleftharpoons MnO_4^{2-}$, E° = +0.56 V (run backwards: MnO_4^{2-} oxidised, Mn +6 → +7)
E°cell = 2.26 − 0.56 = **+1.70 V → feasible**
Electrons: the reduction takes $2e^{-}$, the oxidation gives $1e^{-}$, so double the oxidation:
$2MnO_4^{2-} \rightarrow 2MnO_4^{-} + 2e^{-}$
$MnO_4^{2-} + 4H^{+} + 2e^{-} \rightarrow MnO_2 + 2H_2O$
$3MnO_4^{2-}(aq) + 4H^{+}(aq) \rightarrow 2MnO_4^{-}(aq) + MnO_2(s) + 2H_2O(l)$
Green manganate(VI) turns purple, with a brown-black solid.

EXAM TIP

Hydrogen peroxide disproportionates too: O goes −1 → −2 (in H_2O) and −1 → 0 (in O_2). E°cell = 1.77 − 0.68 = +1.09 V, so $2H_2O_2 \rightarrow 2H_2O + O_2$ is feasible. It is slow without a catalyst: a link to 16.12.

16.14 Latimer-style diagrams

Some questions give the E° values on a line of species from highest to lowest oxidation state, with the E° for each step written over the arrow:

Cu^{2+} —(+0.15 V)→ Cu^{+} —(+0.52 V)→ Cu

A species in the middle **disproportionates if the E° on its right is more positive than the E° on its left** (here 0.52 > 0.15, so Cu^{+} disproportionates). This is the same calculation as above in a compact form.

MARK-SCHEME LANGUAGE

- "$Cu^{2+} + e^- \rightleftharpoons Cu^{+}$ E° = +0.15 V and $Cu^{+} + e^- \rightleftharpoons$ Cu E° = +0.52 V"; "E°cell = 0.52 − 0.15 = (+)0.37 (V) and therefore the reaction is (thermodynamically) feasible". "No TE on incorrect half-equations" (WCH15 Jan 2021).
- "Unbalanced equation with all species correct scores (1)" (WCH15 Jun 2021).
- Comproportionation with E°cell = −0.03 V: "the E° values are close"; "the reaction may be shifted in the required direction using concentrated alkali" (WCH15 Jun 2021).

Often asked as: write the ionic equation and use E° values to show the disproportionation is feasible; derive the equation for an unfamiliar Mn species from given clues.

COMMON MISTAKE

Choosing the wrong pair of half-equations: you need the two in which the **same** species appears, once on each side (WCH15 Jan 2021 examiner report).
Getting the three Mn species right but failing to balance. Write the two half-equations first, then combine (WCH15 Jan 2024 examiner report).
Not saying the E°cell is **positive** after calculating it.

BEYOND THE SPEC, BUT EXAMINED

Disproportionation in **alkali** using the alkaline half-equation: $3MnO_4^{2-} + 2H_2O \rightarrow 2MnO_4^{-} + MnO_2 + 4OH^{-}$ (WCH15 Jun 2021).
Unfamiliar species from a Latimer-style table, e.g. H_3MnO_4 (Mn +5) → $MnO_4^{-} + MnO_2$, E°cell = +1.62 V (WCH15 Jan 2024).
Spotting a disproportionation in an MCQ: $2Ag^{2+} \rightarrow Ag^{+} + Ag^{3+}$ (WCH15 Oct 2021).

SPEC 16.15 Redox titration calculations

WHAT THIS SPEC POINT ASKS

Carry out structured and unstructured redox titration calculations, including iron(II) with manganate(VII) and iodine with thiosulfate.

16.15 The two key reactions

Titration	Ionic equation	Mole ratio
Manganate(VII) and iron(II)	$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$	**$MnO_4^- : Fe^{2+} = 1 : 5$**
Manganate(VII) and ethanedioate	$2MnO_4^- + 16H^+ + 5C_2O_4^{2-} \rightarrow 2Mn^{2+} + 8H_2O + 10CO_2$	**$MnO_4^- : C_2O_4^{2-} = 2 : 5$**
Dichromate(VI) and iron(II)	$Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 7H_2O + 6Fe^{3+}$	**$Cr_2O_7^{2-} : Fe^{2+} = 1 : 6$**
Iodine and thiosulfate	$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$	**$I_2 : S_2O_3^{2-} = 1 : 2$**

16.15 The route through any titration calculation

1. Write (or find) the balanced equation and the mole ratio.
2. **Moles of the titrant** = concentration × titre (in dm^3).
3. Use the ratio to get **moles of the other reactant** in the portion titrated.
4. **Scale up** to the whole sample (e.g. × 10 for 25.0 cm^3 taken from 250.0 cm^3).
5. Convert to what is asked: mass (× M), concentration (÷ volume), percentage.

Iodine–thiosulfate titrations are usually **indirect**: an oxidising agent reacts with **excess** iodide to release I_2, and you titrate the I_2. Chain the ratios: oxidant → I_2 → $S_2O_3^{2-}$.

WORKED EXAMPLE: PURITY OF IRON(II) SULFATE (MNO_4^-, STRUCTURED)

5.00 g of impure $FeSO_4 \cdot 7H_2O$ (M = 277.9 g mol^{-1}) is dissolved in dilute sulfuric acid and made up to 250.0 cm^3. 25.0 cm^3 portions need a mean titre of 17.60 cm^3 of 0.0200 mol dm^{-3} $KMnO_4$.
$n(MnO_4^-) = 0.0200 \times 17.60/1000 = 3.52 \times 10^{-4}$ mol
$n(Fe^{2+})$ in 25.0 $cm^3 = 5 \times 3.52 \times 10^{-4} = 1.76 \times 10^{-3}$ mol
$n(Fe^{2+})$ in 250.0 $cm^3 = 1.76 \times 10^{-2}$ mol
mass $FeSO_4 \cdot 7H_2O = 1.76 \times 10^{-2} \times 277.9 = 4.891$ g
purity = 4.891 ÷ 5.00 × 100 = **97.8%**

WORKED EXAMPLE: COPPER IN BRASS (IODINE–THIOSULFATE)

0.950 g of brass is dissolved in nitric acid, neutralised and made up to 250.0 cm^3. A 25.0 cm^3 portion is added to excess KI: $2Cu^{2+} + 4I^- \rightarrow 2CuI + I_2$. The iodine needs 19.00 cm^3 of 0.0500 mol dm^{-3} $Na_2S_2O_3$.
$n(S_2O_3^{2-}) = 0.0500 \times 0.01900 = 9.50 \times 10^{-4}$ mol
$n(I_2) = 9.50 \times 10^{-4} \div 2 = 4.75 \times 10^{-4}$ mol
$n(Cu^{2+}) = 2 \times 4.75 \times 10^{-4} = 9.50 \times 10^{-4}$ mol in 25.0 cm^3 (overall $Cu^{2+} : S_2O_3^{2-} = 1 : 1$)
in 250.0 cm^3: 9.50×10^{-3} mol; mass = $9.50 \times 10^{-3} \times 63.5 = 0.603$ g
% Cu = 0.603 ÷ 0.950 × 100 = **63.5%**

WORKED EXAMPLE: AN UNKNOWN OXIDATION STATE (UNSTRUCTURED)

25.0 cm^3 of 0.100 mol dm^{-3} V^{2+} needs 30.0 cm^3 of 0.0500 mol dm^{-3} $KMnO_4$. What is the final oxidation state of vanadium?
$n(MnO_4^-) = 0.0500 \times 0.0300 = 1.50 \times 10^{-3}$ mol; each takes $5e^-$, so electrons accepted = 7.50×10^{-3} mol
$n(V^{2+}) = 0.100 \times 0.0250 = 2.50 \times 10^{-3}$ mol
electrons lost per V = $7.50 \times 10^{-3} \div 2.50 \times 10^{-3} = 3$
V goes from +2 to **+5** (as VO_2^+).

WORKED EXAMPLE: A BACK TITRATION

0.200 g of an ore containing MnO_2 is warmed with 50.0 cm^3 of 0.100 mol dm^{-3} Fe^{2+} in acid (excess): $MnO_2 + 4H^+ + 2Fe^{2+} \rightarrow Mn^{2+} + 2Fe^{3+} + 2H_2O$. The leftover Fe^{2+} needs 20.00 cm^3 of 0.0200 mol dm^{-3} $KMnO_4$.
$n(MnO_4^-) = 0.0200 \times 0.02000 = 4.00 \times 10^{-4}$ mol → leftover $Fe^{2+} = 5 \times 4.00 \times 10^{-4} = 2.00 \times 10^{-3}$ mol
Fe^{2+} added $= 0.100 \times 0.0500 = 5.00 \times 10^{-3}$ mol
Fe^{2+} that reacted with $MnO_2 = 5.00 \times 10^{-3} - 2.00 \times 10^{-3} = 3.00 \times 10^{-3}$ mol
$n(MnO_2) = 3.00 \times 10^{-3} \div 2 = 1.50 \times 10^{-3}$ mol; mass $= 1.50 \times 10^{-3} \times 86.9 = 0.130$ g
% $MnO_2 = 0.130 \div 0.200 \times 100 =$ **65.2%**

MARK-SCHEME LANGUAGE

- Marks are by **process**: "moles of MnO_4^- in the titre"; "use of 1:5 ratio"; "scaling to 250.0 cm^3"; "conversion of moles to g"; "percentage … to 3 SF". "TE at each stage"; "ignore premature rounding if the final answer is correct".
- "Correct answer with no working scores zero" on some papers: always show the route.
- Use Ar from the periodic table in the Data Booklet (55.8 or 56 for Fe both accepted). Atomic numbers used as Ar score nothing (WCH15 Jun 2021).

Often asked as: % of iron in tablets, wire or moss killer (5–6 marks); Cu in a complex salt by iodine–thiosulfate; dissolved oxygen (Winkler); find an unknown oxidation state.

COMMON MISTAKE

Using 25.0 cm^3 where the **titre** belongs, or forgetting the × 10 scale-up (WCH15 Jan 2025 examiner report).
The wrong ratio, or none at all (WCH15 Jan 2024 examiner report). If you get a percentage over 100, **recheck the ratio**: don't fudge it (WCH15 Oct 2020 examiner report).
Rounding intermediate values, and quoting 4 or more SF (WCH15 Oct 2024 examiner report). Using the formula mass of VO_3^- when the question asks for mass of vanadium (WCH15 Oct 2022 examiner report).

BEYOND THE SPEC, BUT EXAMINED

Back titrations: excess reagent added, the leftover titrated, the amount used found by difference (WCH15 Jun 2023; Oct 2022). Many candidates lose marks by missing the back-titration step.
Winkler method for dissolved oxygen: moles of $S_2O_3^{2-} \div 2$ = moles of I_2; $I_2 : MnO_2 = 1 : 1$; $MnO_2 : O_2 = 2 : 1$ (WCH15 Oct 2025).
ppm and ppb. For dilute aqueous solutions 1 $dm^3 \approx 1$ kg, so 1 mg $dm^{-3} \approx 1$ ppm and 1 µg $dm^{-3} \approx 1$ ppb. Example: 2.15×10^{-5} mol dm^{-3} Fe × 55.8 $= 1.20 \times 10^{-3}$ g $dm^{-3} = 1.20$ mg dm^{-3} = **1.20 ppm** (WCH15 Jan 2025; Jun 2021).
A formula of a hydrate from titration data, e.g. $KCr(SO_4)_2 \cdot 12H_2O$ (WCH15 Jan 2025).

SPEC 16.16

Uncertainty in titration results

WHAT THIS SPEC POINT ASKS

Discuss the uncertainty of measurements and what it means for how valid the final result is.

16.16 Percentage uncertainty

% uncertainty = (uncertainty ÷ measured value) × 100

Apparatus	Typical uncertainty	Readings	Notes
Burette	±0.05 cm^3 per reading	**2** (start and end)	titre uncertainty = ±0.10 cm^3
Pipette (25.0 cm^3)	±0.06 cm^3	1	
Volumetric flask (250.0 cm^3)	±0.3 cm^3	1	
Balance (2 d.p.)	±0.005 g per reading	**2** if weighing by difference	

- A **bigger titre** gives a **smaller** % uncertainty: ±0.10 on 5.00 cm^3 is 2%, but on 25.00 cm^3 it is only 0.4%. Choose concentrations so the titre is large (ideally 20–30 cm^3).
- Add the % uncertainties of each measurement to get the total % uncertainty of the result.
- **% uncertainty is not % error.** Error compares your result with an accepted value.

WORKED EXAMPLE: IS 97.8% JUSTIFIED?

From the $FeSO_4$ titration in 16.15:
burette: 0.10 ÷ 17.60 × 100 = 0.57%
pipette: 0.06 ÷ 25.0 × 100 = 0.24%
flask: 0.3 ÷ 250.0 × 100 = 0.12%
balance: 2 × 0.005 ÷ 5.00 × 100 = 0.20%
total ≈ **1.1%** → 97.8% ± 1.1%, so the result lies between about 96.7% and 98.9%.
The third significant figure is uncertain: quote **98%** (2 SF) or 97.8% (3 SF), never 97.82%.

MARK-SCHEME LANGUAGE

- "100 × 0.05 × 2 ÷ 27.35 = 0.37(%)" for the burette; "100 × 0.06 ÷ 25 = 0.24(%)" for the pipette (WCH15 Jun 2021).
- "the total percentage uncertainty is much bigger than the rounding, so the answer should be to no more than 2 / 3 SF". IGNORE "data given to 3 SF".

Often asked as: calculate % uncertainties for several pieces of apparatus; decide how many SF are sensible; effect of an impurity or a damp sample on the result.

COMMON MISTAKE

Forgetting that a burette (and weighing by difference) involves **two** readings.
Too many significant figures: the examiners' rule is one fewer than the least precise datum, or the same number (WCH15 Jan 2021 examiner report).

SPEC 16.17

Core Practicals 13a and 13b: redox titrations

WHAT THIS SPEC POINT ASKS

Carry out redox titrations with (i) iron(II) and potassium manganate(VII) and (ii) sodium thiosulfate and iodine.

CORE PRACTICAL 13A: IRON(II) WITH POTASSIUM MANGANATE(VII)

1. Weigh the iron(II) sample accurately (e.g. ground iron tablets or ammonium iron(II) sulfate) by difference.
2. Dissolve in about 100 cm^3 of **dilute sulfuric acid**, transfer with washings to a 250.0 cm^3 volumetric flask, make up to the mark with distilled water and invert to mix.

3. Pipette 25.0 cm^3 into a conical flask and add about 10 cm^3 more dilute H_2SO_4 (excess acid).
4. Fill the burette with $KMnO_4(aq)$ (≈0.02 mol dm^{-3}). Read the **top** of the meniscus, because the purple solution is too dark to see the bottom.
5. Titrate until the first **permanent pale pink** colour. No indicator is needed: MnO_4^- is **self-indicating**.
6. Do a rough titre, then accurate titres until two are **concordant** (within 0.10 cm^3). Average the concordant titres.

CORE PRACTICAL 13B: IODINE WITH SODIUM THIOSULFATE

1. Pipette 25.0 cm^3 of the oxidising agent (e.g. KIO_3 standard, bleach, or a Cu^{2+} solution) into a conical flask.
2. Add **excess** potassium iodide (and acid where needed). Iodine is released: the solution turns brown.
3. Titrate with $Na_2S_2O_3(aq)$ from the burette until the brown fades to **pale straw yellow**.
4. Add about 1 cm^3 of **starch** solution: it turns **blue-black**.
5. Continue dropwise until the blue-black colour **just disappears** (colourless; with copper, an off-white CuI precipitate remains).
6. Repeat to concordant titres.

16.17 Why each step matters

Step	Reason
Acidify MnO_4^- with **dilute H_2SO_4**	MnO_4^- needs H^+ to be reduced fully to Mn^{2+}. Too little acid gives a brown MnO_2 precipitate and the ratio changes.
Not HCl	MnO_4^- (+1.51 V) oxidises Cl^- to Cl_2 (+1.36 V), so extra MnO_4^- is used and Cl_2 is toxic.
Not nitric acid	It is itself an oxidising agent and would oxidise Fe^{2+}.
Excess KI	ensures all the oxidant reacts, so moles of I_2 truly reflect the oxidant; it also keeps I_2 dissolved (as I_3^-).
Starch added **near the end**	added too early, a starch–iodine complex forms that breaks up only slowly, so the end point is late and unclear.

MARK-SCHEME LANGUAGE

- MnO_4^- in the burette: "colourless (or pale green) to (pale) pink". MnO_4^- in the flask: "(pale) pink to colourless". DO NOT ACCEPT purple, magenta or "permanent" alone (WCH15 Jan 2024; Jun 2021).
- Iodine–thiosulfate: "blue-black to colourless" (WCH15 Jan 2022).
- Self-indicating: advantage "no indicator needed"; disadvantage "the intense colour of manganate(VII) makes reading the burette difficult" (WCH15 Jan 2026).
- Dichromate needs an indicator because "the colour change (orange to green) is not distinctive / sharp enough". $Cr_2O_7^{2-}$ orange, Cr^{3+} green; DO NOT ACCEPT blue (WCH15 Jun 2022).

Often asked as: the end-point colour change; why KI must be in excess; advantage/disadvantage of a self-indicating titrant.

COMMON MISTAKE

The colour change the wrong way round (it depends on which solution is in the burette), or giving the colours of acid–base indicators (WCH15 Jan 2024 examiner report).
Leaving out the word "colour" when asked why an indicator is needed (WCH15 Jun 2022 examiner report).

BEYOND THE SPEC, BUT EXAMINED

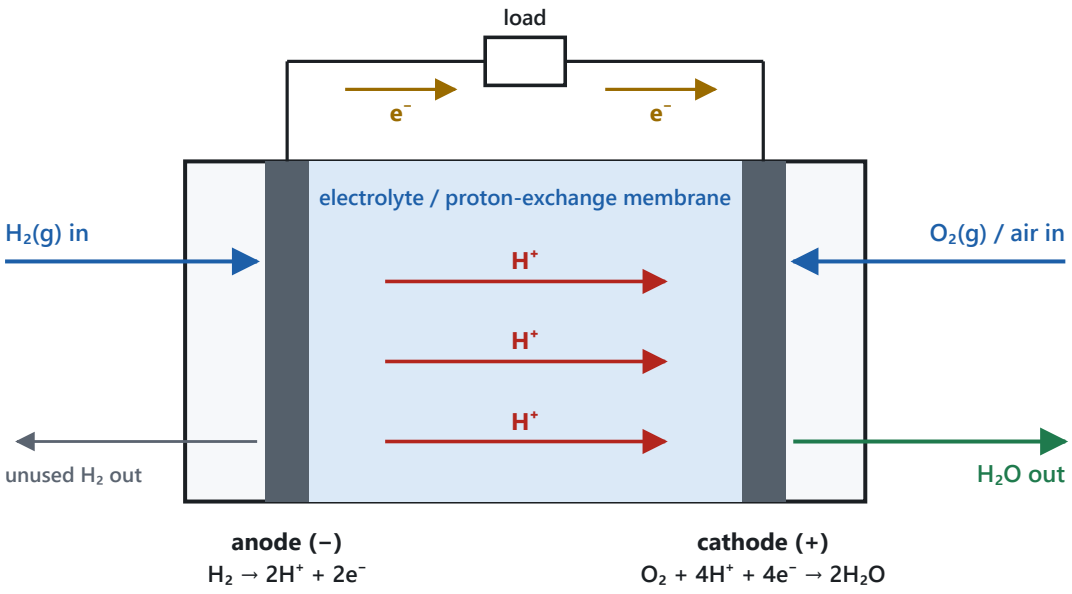
Titrations with **Ce^{4+}** or **Ti^{3+}** as titrant, and heating the flask "to speed up the reaction" (WCH15 Oct 2021; Jan 2026).

SPEC 16.18 Fuel cells

WHAT THIS SPEC POINT ASKS

Understand that fuel cells use the energy released when a fuel reacts with oxygen to generate a voltage; methanol and other hydrogen-rich fuels can be used.

16.18 How a fuel cell works



electrodes: porous, coated with a platinum catalyst

Acid (proton-exchange membrane) fuel cell. H^+ ions move through the electrolyte; electrons move through the external circuit.

- The **fuel** (H_2, methanol, ethanol…) is **oxidised** at the **anode** (negative electrode). Electrons flow round the external circuit.
- **Oxygen** is **reduced** at the **cathode** (positive electrode).
- The overall reaction is the same as burning the fuel, but the energy comes out as **electricity** instead of heat, so no flame is needed.
- Fuel and oxygen are supplied **continuously**, so the cell keeps working as long as they are fed in. It doesn't need recharging, and it doesn't store energy.
- Electrodes are porous and coated with a **platinum catalyst**.

16.18 Methanol and other hydrogen-rich fuels

Hydrogen is hard to store (a gas with low energy per volume) and flammable. A liquid **hydrogen-rich fuel** such as methanol is easier to handle:

- Anode (acid): $CH_3OH + H_2O \rightarrow CO_2 + 6H^+ + 6e^-$
- Cathode: $1½O_2 + 6H^+ + 6e^- \rightarrow 3H_2O$
- Overall: $\mathbf{CH_3OH + 1½O_2 \rightarrow CO_2 + 2H_2O}$

16.18 Advantages and disadvantages

Advantages	Disadvantages
H_2 cell: only **water** made; no CO_2 at the point of use (combustion engines give CO_2, CO, NO_x)	making H_2 needs energy, often from fossil fuels (electrolysis or from methane)
more **efficient** than burning the fuel in an engine	H_2 is **flammable**; storing and transporting it is difficult (high pressure or low temperature)
quiet, few moving parts	expensive platinum catalyst; fuel cells have a limited lifetime

MARK-SCHEME LANGUAGE

- How it works: fuel cells "use the energy released on the reaction of a fuel / hydrogen with oxygen to generate a voltage" (WCH15 Jun 2024).
- Advantage: "only water is produced, whereas CO_2 is produced in an internal combustion engine". Disadvantage: "production / transport / storage of hydrogen is difficult". Another fuel: "e.g. methanol (as it is hydrogen-rich)".
- Methanol oxidation: "$CH_3OH + H_2O \rightarrow 6H^+ + CO_2 + 6e^-$". DO NOT ACCEPT the formula CH_2OH (WCH15 Oct 2024).

Often asked as: 6-mark "explain how a hydrogen fuel cell works, with an advantage, a disadvantage and another fuel"; MCQs on electrode half-equations.

COMMON MISTAKE

Forgetting to explain **how** the cell works (the first indicative point was the least seen, WCH15 Jun 2024 examiner report). Not answering the last part of the question ("name another fuel").
Treating "flammable" and "explosive" as the same thing. Writing methanol as CH_2OH.

SPEC 16.19 Electrode reactions in a hydrogen–oxygen fuel cell

WHAT THIS SPEC POINT ASKS

Know the electrode reactions in a hydrogen–oxygen fuel cell with an acidic and with an alkaline electrolyte.

	Acidic electrolyte	Alkaline electrolyte (e.g. KOH)
Ion carrying charge	H^+ (anode → cathode)	OH^- (cathode → anode)
Anode (−), oxidation	$H_2 \rightarrow 2H^+ + 2e^-$	$H_2 + 2OH^- \rightarrow 2H_2O + 2e^-$
Cathode (+), reduction	$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$	$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$
E° values (as reductions)	$2H^+/H_2$ 0.00 V; O_2/H_2O +1.23 V	H_2O/H_2, OH^- −0.83 V; O_2/OH^- +0.40 V
E°cell	1.23 − 0.00 = **+1.23 V**	0.40 − (−0.83) = **+1.23 V**
Overall	**$2H_2 + O_2 \rightarrow 2H_2O$**	**$2H_2 + O_2 \rightarrow 2H_2O$**

- The overall reaction and E°cell are the **same** in acid and in alkali: only the ions in the half-equations differ.
- Check which medium the question gives: use **H^+** in acid and **OH^-** in alkali, never a mixture.

MARK-SCHEME LANGUAGE

- Acid: anode "$H_2(g) \rightleftharpoons 2H^+(aq) + 2e^-$"; cathode "$O_2(g) + 4H^+(aq) + 4e^- \rightleftharpoons 2H_2O(l)$" (WCH15 Jun 2024).
- Alkaline: anode "$H_2 + 2OH^- \rightarrow 2H_2O + 2e^-$"; cathode "$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$" (WCH15 Jan 2021; Jun 2025).
- Overall "$H_2 + ½O_2 \rightarrow H_2O$", E° = +1.23 V (WCH15 Jan 2024).

Often asked as: MCQs picking the anode or cathode half-equation in acid or alkali; writing the reduction half-equation for an unfamiliar cell.

COMMON MISTAKE

Mixing acid and alkaline half-equations, or balancing with OH^- when H^+ and H_2O were specified (WCH15 Oct 2024 examiner report).
Mixing up anode and cathode. In any cell, the **an**ode is where **oxidation** happens (both start with a vowel); in a fuel cell it is the **negative** terminal.

BEYOND THE SPEC, BUT EXAMINED

Other cells built from the same ideas: zinc–air ($Zn + 2OH^- \rightarrow Zn(OH)_2 + 2e^-$, WCH15 Jan 2022) and the alkaline Zn/MnO_2 cell ($2MnO_2 + H_2O + 2e^- \rightarrow Mn_2O_3 + 2OH^-$, WCH15 Jan 2023). A Nernst-type expression applied to a fuel cell (WCH15 Oct 2024).

Topic 17 – Transition Metals and their Chemistry

SPEC 17.1 What makes a transition metal

WHAT THIS SPEC POINT ASKS

Know that transition metals are d-block elements forming one or more stable ions with incompletely filled d-orbitals.

KEY IDEA

A **transition metal** is a **d-block element** that forms **at least one stable ion with an incomplete d subshell** (between 1 and 9 d electrons).

- **d-block element**: its highest-energy electron goes into a **d subshell**. For Period 4, that is scandium to zinc.
- Not every d-block element is a transition metal. The test is about the **ions**, not the atom.

Element	Only stable ion	d electrons	Transition metal?
Scandium	Sc^{3+} = [Ar]	$3d^0$ (empty)	**No**
Titanium to copper	e.g. Ti^{3+} $3d^1$, Fe^{2+} $3d^6$, Cu^{2+} $3d^9$	partly filled	**Yes**
Zinc	Zn^{2+} = [Ar]$3d^{10}$	$3d^{10}$ (full)	**No**

- Copper counts even though Cu^+ is $3d^{10}$, because **Cu^{2+}** ($3d^9$) has an incomplete d subshell. One such ion is enough.
- Typical transition-metal properties all come from that partly filled d subshell: **variable oxidation states**, **coloured** ions, **complex** formation and **catalytic** activity.

MARK-SCHEME LANGUAGE

- Mercury: "Hg^+ is [Xe]$4f^{14}5d^{10}6s^1$ and Hg^{2+} is [Xe]$4f^{14}5d^{10}$"; "d-block element as the last electron goes into a (5)d orbital"; "not a transition element as Hg^+ and Hg^{2+} do not have incompletely filled d orbitals" (WCH15 Jun 2023).
- DO NOT ACCEPT an answer based only on the **atom's** configuration. Use "d orbitals" (plural) or "d subshell".
- Nickel "forms stable ions with partially filled d orbitals" (WCH15 Jan 2021).

Often asked as: is Zn / Sc / Hg a transition metal? (use the configurations of the ions); MCQs on the definition.

COMMON MISTAKE

Defining a transition metal by the atom ("it has d electrons") instead of by its **ions**.
Not explaining what "d-block" means when asked (WCH15 Jun 2023 examiner report).

BEYOND THE SPEC, BUT EXAMINED

Applying the definition to **4d and 5d** metals (Ag^+ is [Kr]$4d^{10}$; Hg^{2+} is [Xe]$4f^{14}5d^{10}$), so neither forms a coloured simple ion (WCH15 Jan 2022; Jun 2023).

SPEC 17.2 Electronic configurations of Sc–Zn and their ions

WHAT THIS SPEC POINT ASKS

Deduce the electronic configurations of the atoms and ions of Sc to Zn from atomic number and charge.

17.2 Rules

1. **4s fills before 3d** (4s is lower in energy in the empty atom).
2. **Chromium and copper** are exceptions: Cr is **[Ar]3d^{5}4s^1** and Cu is **[Ar]3d^{10}4s^1**. A half-full or full d subshell is especially stable.
3. When the atom forms an ion, **4s electrons are lost first**, before any 3d electrons.

	Atom	Common ions
Sc (21)	[Ar]3d^{1}4s^2	Sc^{3+} [Ar]
Ti (22)	[Ar]3d^{2}4s^2	Ti^{3+} [Ar]3d^1; Ti^{4+} [Ar]
V (23)	[Ar]3d^{3}4s^2	V^{3+} [Ar]3d^2; V^{2+} [Ar]3d^3
Cr (24)	**[Ar]3d^{5}4s^1**	Cr^{3+} [Ar]3d^3; Cr^{2+} [Ar]3d^4
Mn (25)	[Ar]3d^{5}4s^2	Mn^{2+} [Ar]3d^5
Fe (26)	[Ar]3d^{6}4s^2	Fe^{2+} [Ar]3d^6; Fe^{3+} [Ar]3d^5
Co (27)	[Ar]3d^{7}4s^2	Co^{2+} [Ar]3d^7
Ni (28)	[Ar]3d^{8}4s^2	Ni^{2+} [Ar]3d^8
Cu (29)	**[Ar]3d^{10}4s^1**	Cu$^+$ [Ar]3d^{10}; Cu^{2+} [Ar]3d^9
Zn (30)	[Ar]3d^{10}4s^2	Zn^{2+} [Ar]3d^{10}

WORKED EXAMPLE: Mn^{3+} AND Fe^{3+}

Mn is 25: [Ar]3d^{5}4s^2. Remove 3 electrons: the two 4s first, then one 3d → **Mn^{3+} = [Ar]3d^4**.
Fe is 26: [Ar]3d^{6}4s^2. Remove 3: 4s^2, then one 3d → **Fe^{3+} = [Ar]3d^5**.

17.2 Electrons-in-boxes

- Each box is one orbital. Fill each 3d box singly first with **parallel spins**, then pair up with **opposite** spins.
- Fe^{2+} (3d^6): five boxes, the first holding a pair (↑↓) and the other four single (↑). The 4s box is **empty**.

MARK-SCHEME LANGUAGE

- "[Ar]3d^5" for Fe^{3+} and Mn^{2+}; "[Ar]3d^6" for Fe^{2+}; "[Ar]3d^{10}" for Zn^{2+} and Cu$^+$.
- Boxes: half-arrows accepted; "allow all arrows pointing downwards" in the 3d subshell, but two arrows in one orbital must point **opposite** ways (WCH15 Jan 2025).

Often asked as: MCQ on a configuration; complete the boxes for an atom and its ion; evidence from successive ionisation energies.

COMMON MISTAKE

Removing 3d electrons before 4s, or leaving 4s occupied in Fe^{2+} (WCH15 Jan 2025 examiner report). Writing 3d for silver instead of 4d (WCH15 Jan 2022 examiner report).

SPEC 17.3 Why transition metals have variable oxidation numbers

WHAT THIS SPEC POINT ASKS

Understand why transition metals show variable oxidation numbers.

17.3 The reason

- The **4s and 3d** subshells have **similar energies**.
- So after the 4s electrons are lost, the 3d electrons can be lost too without a huge jump in ionisation energy. Successive ionisation energies rise **gradually**, with no big jump until all 4s and 3d electrons are gone.
- Losing a different number of electrons gives a different ion. The energy needed is paid back by **lattice energy**, **hydration energy** or **covalent bonding**, so several oxidation states can be stable.
- The **maximum** oxidation state up to Mn equals the **total of 4s + 3d electrons**: V +5 ($3d^3 4s^2$), Cr +6, Mn +7.

Metal	Common oxidation states
Ti	+2, +3, **+4**
V	+2, +3, +4, **+5**
Cr	+2, **+3**, **+6**
Mn	**+2**, +4, +6, **+7**
Fe	**+2**, **+3**
Co	**+2**, +3
Ni	**+2**
Cu	+1, **+2**

17.3 Which state is more stable?

- **Fe^{3+} ($3d^5$)** is more stable than Fe^{2+} ($3d^6$). In $3d^6$, one orbital holds a pair, and the extra repulsion makes that sixth electron easy to remove. So Fe^{2+} is readily oxidised by air.
- **Mn^{2+} ($3d^5$)** is more stable than Mn^{3+} ($3d^4$). Removing an electron would break the stable half-filled subshell. So Mn^{2+} is hard to oxidise, and the Mn^{3+}/Mn^{2+} E° is very positive.
- High oxidation states exist only in **oxo-ions or covalent compounds**, never as bare ions. A V^{5+} ion would be tiny and highly charged: it would polarise the surrounding water molecules so strongly that they lose H^+, giving VO_2^+.

MARK-SCHEME LANGUAGE

- Highest state of V: "+5"; "vanadium has 3 electrons in the 3d and 2 in the 4s"; "the total of 3d and 4s electrons gives the highest oxidation state" (WCH15 Oct 2022).
- Mn^{2+} v Fe^{2+}: "there is stability associated with a half-full set of (3)d orbitals"; "Mn^{2+} has five d electrons so is more stable than Mn^{3+}"; "Fe^{2+} has six d electrons so is less stable than Fe^{3+}" (WCH15 Jan 2023); "$3d^6$ has two paired electrons which results in repulsion" (WCH15 Jun 2021).
- V^{5+}: "small / highly charged"; "so it would polarise water molecules / O–H bonds"; "causing them to lose hydrogen ions" (WCH15 Jan 2023).

Often asked as: why Fe^{2+} solutions go brown in air but Mn^{2+} solutions don't; the highest oxidation state from a configuration; why V^{5+} doesn't exist in water.

COMMON MISTAKE

Writing [Ar]$3d^3$ for vanadium (forgetting 4s) and deducing +3 as the maximum (WCH15 Oct 2022 examiner report).
Comparing two **elements** when the question is about two **ions** of the same element. Mixing up orbital, subshell and shell (WCH15 Jan 2023; Jan 2025 examiner reports).

BEYOND THE SPEC, BUT EXAMINED

The $Fe^{2+} \rightarrow Fe^{3+}$ argument with energies: the third ionisation energy ($Fe^{2+}(g) \rightarrow Fe^{3+}(g) + e^-$) is large, but it is **recovered by the more exothermic hydration** of Fe^{3+}, which has the greater charge (charge density) (WCH15 Jun 2021).

SPEC 17.4

Ligands

WHAT THIS SPEC POINT ASKS

Know what a ligand is.

KEY IDEA

A **ligand** is a molecule or ion with **at least one lone pair of electrons** that it donates to a central metal ion to form a **dative (coordinate) bond**.

- Ligands are **Lewis bases** (electron-pair donors).
- Common ones: H_2O and NH_3 (lone pair on O and N), Cl^-, OH^-, CN^-.
- A species **without a lone pair** can't be a ligand. NH_4^+ has none left (all are used in N–H bonds).

MARK-SCHEME LANGUAGE

- "(a species with a) lone pair (of electrons) that can form a dative (covalent) bond to a (central transition) metal (ion)". ACCEPT "coordinate" for dative (WCH15 Oct 2021).
- MCQ answer: a ligand is an "electron-pair donor" (WCH15 Jun 2021).

Often asked as: define a ligand / a monodentate ligand (2 marks); why a species can't act as a ligand.

COMMON MISTAKE

Leaving out the words "lone pair". Saying a monodentate ligand "contains only one lone pair": H_2O has two lone pairs but uses **one** (WCH15 Oct 2021 examiner report).

SPEC 17.5 Dative covalent bonding in complex ions

WHAT THIS SPEC POINT ASKS

Understand that dative (coordinate) bonds form when complex ions are made.

- In a **dative covalent bond**, **both** electrons of the shared pair come from the **same** atom: the ligand's donor atom.
- In $[Cu(H_2O)_6]^{2+}$, each water uses **one lone pair on its oxygen atom** to bond to Cu^{2+}. The metal ion has empty orbitals that accept the pairs.
- Once formed, a dative bond is the same as any other covalent bond. In diagrams, draw it as an arrow **from the donor atom to the metal** (→) or as a plain line.
- A complex ion therefore contains **covalent** bonds (inside the ligands) and **dative covalent** bonds (ligand to metal), but no ionic bonds inside the brackets.

MARK-SCHEME LANGUAGE

- "forms one dative (covalent) / coordinate bond (to Fe^{2+}) using a lone pair (of electrons) on **oxygen**" (WCH15 Oct 2023).
- Bonding in a complex: covalent and dative covalent only (WCH15 Jan 2022).

COMMON MISTAKE

Saying the lone pair comes from "the water molecule" instead of **the oxygen atom** (WCH15 Oct 2023 examiner report).
Drawing a dative arrow from H in NH_3: the bond comes from **N** (WCH15 Jun 2025).

SPEC 17.6 Complex ions

WHAT THIS SPEC POINT ASKS

Know that a complex ion is a central metal ion surrounded by ligands.

- A **complex** is a central metal atom or ion with ligands bonded to it by dative bonds. If it carries a charge, it is a **complex ion**.
- The whole complex is written in **square brackets**, with the overall charge outside.
- **Overall charge = charge on the metal ion + total charge of the ligands.**

Complex	Metal ion	Ligands	Overall charge
$[Cu(H_2O)_6]^{2+}$	Cu^{2+}	6 H_2O (neutral)	2+
$[CuCl_4]^{2-}$	Cu^{2+}	4 Cl^-	+2 − 4 = 2−
$[Fe(CN)_6]^{3-}$	Fe^{3+}	6 CN^-	+3 − 6 = 3−
$[Cr(OH)_6]^{3-}$	Cr^{3+}	6 OH^-	3−
$[Pt(NH_3)_2Cl_2]$	Pt^{2+}	2 NH_3, 2 Cl^-	0 (neutral)

WORKED EXAMPLE: WORKING BACK TO THE METAL'S CHARGE

$[Ni(C_2O_4)_2Cl_2]^x$: Ni^{2+}, two $C_2O_4^{2-}$ (−4) and two Cl^- (−2) → x = +2 − 4 − 2 = **4−**.
$[Co(NH_3)_5Cl]^{2+}$: NH_3 neutral, Cl^- is −1, so Co − 1 = +2 → **Co is +3**.

MARK-SCHEME LANGUAGE

- Formula of the complex in $FeCl_2 \cdot 4H_2O$ dissolved in water: $[Fe(H_2O)_6]^{2+}$ (ACCEPT $[FeCl(H_2O)_5]^+$) (WCH15 Oct 2021).
- Overall charge and coordination number deduced from the ligands (WCH15 Oct 2020; Jun 2024).

Often asked as: deduce a charge; write the formula of the complex ion in a solution.

COMMON MISTAKE

Giving just "Fe^{2+}", a neutral complex, or $Fe(OH)_2$ when asked for the complex ion in solution (WCH15 Oct 2021 examiner report).
Forgetting the charge on an anionic ligand such as SCN^- when writing the formula (WCH15 Oct 2024 examiner report).

BEYOND THE SPEC, BUT EXAMINED

Mass of chlorine **in the complex ion only**, not in the whole compound (e.g. in $[Co(NH_3)_5Cl]Cl_2$ only one Cl is a ligand) (WCH15 Oct 2024).

SPEC 17.7

Aqueous transition-metal ions are usually coloured

WHAT THIS SPEC POINT ASKS

Know that aqueous solutions of transition-metal ions are usually coloured.

Ion (as hexaaqua ion unless shown)	d electrons	Colour
Sc^{3+}	$3d^0$	colourless
Ti^{3+}	$3d^1$	violet
VO_2^+ (+5) / VO^{2+} (+4) / V^{3+} / V^{2+}	$3d^0$ / $3d^1$ / $3d^2$ / $3d^3$	yellow / blue / green / violet
Cr^{3+}	$3d^3$	green (violet when pure hexaaqua)
$Cr_2O_7^{2-}$ / CrO_4^{2-} (+6)	$3d^0$	orange / yellow
Mn^{2+}	$3d^5$	very pale pink (looks colourless)
MnO_4^- (+7)	$3d^0$	purple
Fe^{2+}	$3d^6$	pale green
Fe^{3+}	$3d^5$	yellow / orange-brown
Co^{2+}	$3d^7$	pink
Ni^{2+}	$3d^8$	green
Cu^{2+}	$3d^9$	blue
Cu^+ (in complexes)	$3d^{10}$	colourless
Zn^{2+}	$3d^{10}$	colourless

The colours of d^0 oxo-ions (MnO_4^-, $Cr_2O_7^{2-}$) come from a different process, charge transfer, and need no explanation at this level.

EXAM TIP

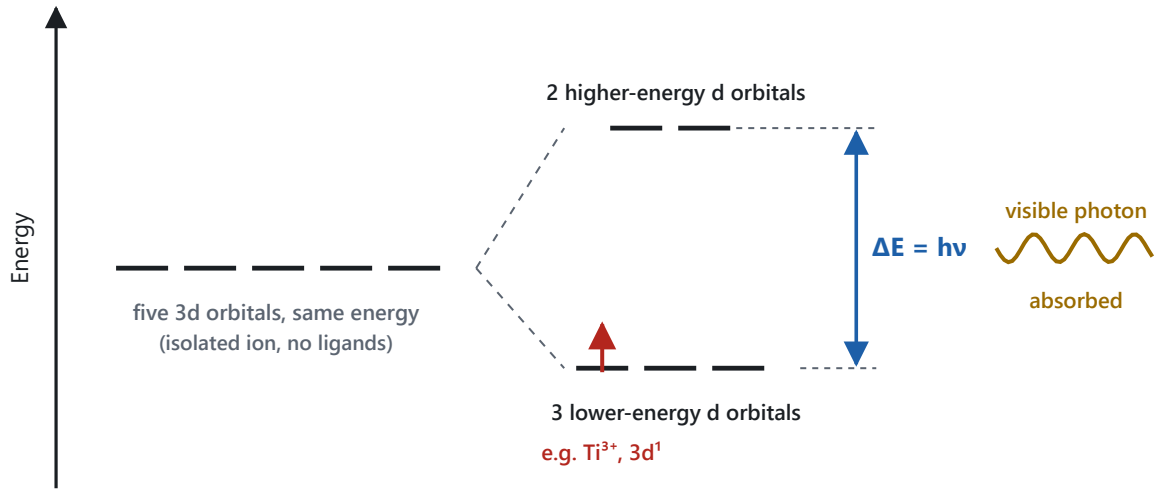
Learn these as a block: they feed straight into the NaOH/NH_3 table (17.22) and the vanadium and chromium sections. Every "describe what you see" question needs the **starting colour**.

SPEC 17.8

Why complex ions are coloured: d-orbital splitting

WHAT THIS SPEC POINT ASKS

Understand that colour arises because ligands split the energy levels of the d orbitals.



Ligands split the d orbitals. An electron in the lower set absorbs a photon of energy ΔE and is promoted.

In an octahedral complex the five 3d orbitals split into a lower set of three and a higher set of two.

17.8 The explanation, step by step

1. In an isolated ion, the five 3d orbitals all have the **same energy**.
2. When **ligands** bond to the ion, their lone pairs repel the d electrons unequally, so the d orbitals **split** into two groups with an **energy gap, ΔE**.
3. An electron in the lower group can absorb a photon of **visible light** whose energy exactly matches ΔE, and is **promoted** to the higher group (a **d–d transition**).
4. The light of that frequency is **absorbed**. The rest is **transmitted** (or reflected), and that is the colour you see: the **complementary colour** of the light absorbed.

$\Delta E = h\nu = hc/\lambda$, where $h = 6.63 \times 10^{-34}$ J s, ν = frequency (Hz), $c = 3.00 \times 10^{8}$ m s^{-1}, λ = wavelength (m).

- A **bigger** ΔE means **higher frequency** (shorter wavelength) light is absorbed.

Light absorbed	Colour seen
red	green / blue-green
orange	blue
yellow	violet
green	purple / magenta
blue	orange
violet	yellow

WORKED EXAMPLE: ΔE = HN

$[Ti(H_2O)_6]^{3+}$ absorbs most strongly at $\lambda = 500$ nm (green-yellow light).
$\nu = c/\lambda = 3.00 \times 10^{8} \div 500 \times 10^{-9} =$ **6.00×10^{14} Hz**
$\Delta E = h\nu = 6.63 \times 10^{-34} \times 6.00 \times 10^{14} =$ **3.98×10^{-19} J** per ion
Per mole: $3.98 \times 10^{-19} \times 6.02 \times 10^{23} = 2.39 \times 10^{5}$ J mol^{-1} = **239 kJ mol^{-1}**
Green-yellow light is absorbed, so the solution looks **violet**.
For comparison, $[Cu(H_2O)_6]^{2+}$ absorbs around 800 nm (red/orange): $\Delta E = 2.49 \times 10^{-19}$ J, a **smaller** gap, and the solution looks blue.

MARK-SCHEME LANGUAGE

The chain the marks follow (WCH15 Jun 2021; Oct 2024; Jan 2026):
- "ligands **split** the energy of the **d subshell / d orbitals**";
- "(visible) light is **absorbed** and promotes electrons from lower to higher energy (d orbitals)";
- "colour seen is the light **transmitted / reflected** / the wavelengths not absorbed / the complementary colour".
- Different colours: "different ligands / oxidation states → different **energy gap** → different wavelength / frequency absorbed".
- Always "d orbitals" (plural) or "d subshell". DO NOT ACCEPT "d orbital" (singular); penalise "shell" for subshell once.

Often asked as: explain why $[M(H_2O)_6]^{n+}$ is coloured (3 marks); 6-mark on why two complexes of one metal differ in colour; why an indicator-free titration has a visible end point.

COMMON MISTAKE

"Light is **emitted**" (that's a flame test): DO NOT ACCEPT. "**Adsorbed**" for absorbed. "A d orbital splits" (WCH15 Oct 2021; Jan 2024; Oct 2024 examiner reports).
Vague "different energy levels" instead of "a different **energy gap**" (WCH15 Oct 2024 examiner report).

BEYOND THE SPEC, BUT EXAMINED

Colorimetry: the absorbance of a coloured complex is proportional to its concentration, so a **calibration graph** of absorbance against known concentration gives an unknown concentration. Use a filter of the **complementary** colour (WCH15 Jun 2021; Oct 2024).
f-block ions: lanthanide colours come from f–f transitions; La^{3+} (no f electrons) is colourless (WCH15 Oct 2020; Oct 2025).

SPEC 17.9

Why some ions are colourless

WHAT THIS SPEC POINT ASKS

Understand why some aqueous ions and complex ions have no colour.

A d–d transition needs an electron in the lower set **and** a space in the upper set. So there's no colour when:

Case	Example	Reason
d^0: empty d subshell	Sc^{3+}, Ti^{4+}	no d electrons to promote
d^{10}: full d subshell	Zn^{2+}, Cu^+, Ag^+	upper set already full, so no transition possible
ΔE outside the visible range	some complexes	absorption is in the UV, so no visible light is absorbed

MARK-SCHEME LANGUAGE

- Cu(I): "the d orbitals are **full**, so no (electron) transitions are possible / no d–d transitions" (WCH15 Jan 2021).
- Cr(VI): "chromium(VI) has an **empty** d subshell" (WCH15 Jun 2022).
- DO NOT ACCEPT "the subshell is not split", "the light is outside the visible region" or "no electrons in d orbitals" **for Cu(I)**.

Often asked as: explain why a Cu(I) or Ag(I) complex is colourless; why Zn^{2+} solutions are colourless.

COMMON MISTAKE

Saying a d^{10} ion has "no d electrons". It has ten: the point is that there's **nowhere to promote** them.

SPEC 17.10 Coordination number

WHAT THIS SPEC POINT ASKS

Understand what coordination number means.

KEY IDEA

The **coordination number** is the **number of dative bonds** from ligands to the central metal ion.

- For monodentate ligands, it equals the number of ligands. For multidentate ligands, it **doesn't**: $[Cu(en)_3]^{2+}$ has 3 ligands but coordination number **6** (each en forms 2 bonds).

Coordination number	Shape	Typical ligands	Example
6	octahedral	small ligands: H_2O, NH_3, OH^-	$[Fe(H_2O)_6]^{2+}$
4	tetrahedral	large ligands, e.g. Cl^-	$[CuCl_4]^{2-}$
4	square planar	Pt^{2+} and Ni^{2+} complexes	$[Pt(NH_3)_2Cl_2]$
2	linear	Ag^+	$[Ag(NH_3)_2]^+$

MARK-SCHEME LANGUAGE

- "6 and octahedral" for three bidentate ligands; coordination number is **not** the number of ligands (WCH15 Jun 2021; Jun 2024).
- Deducing a formula from data: "a 1 : 1 ratio indicates one thiocyanate ion in an octahedral complex, so 5 water molecules": $[Fe(H_2O)_5SCN]^{2+}$ (WCH15 Oct 2024).

COMMON MISTAKE

Giving 3 for $[M(en)_3]^{n+}$. Count **bonds**, not ligands (WCH15 Jun 2024 examiner report).

SPEC 17.11 Why colours change: oxidation number, ligand and coordination number

WHAT THIS SPEC POINT ASKS

Understand that the colour of a transition-metal ion can change because of a change in (i) oxidation number, (ii) ligand, (iii) coordination number.

Each change alters the **size of ΔE**, so a different frequency of light is absorbed and a different colour is seen.

Change	Example	Colour change
Oxidation number	$[Fe(H_2O)_6]^{2+} \rightarrow [Fe(H_2O)_6]^{3+}$	pale green → yellow-brown
	$VO_2^+ \rightarrow VO^{2+} \rightarrow V^{3+} \rightarrow V^{2+}$	yellow → blue → green → violet
Ligand (same coordination number)	$[Cu(H_2O)_6]^{2+} \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+}$	blue → deep blue
	$[Co(H_2O)_6]^{2+} \rightarrow [Co(NH_3)_6]^{2+}$	pink → straw / yellow-brown
Coordination number (and ligand)	$[Cu(H_2O)_6]^{2+}$ (6) → $[CuCl_4]^{2-}$ (4)	blue → yellow (green in between)
	$[Co(H_2O)_6]^{2+}$ (6) → $[CoCl_4]^{2-}$ (4)	pink → blue

MARK-SCHEME LANGUAGE

- "the ammonia ligands split the two groups of orbitals by a different amount than water"; "the cobalt is in a different oxidation state, which splits … by a different amount"; "so different wavelengths / frequencies of light are absorbed" (WCH15 Jan 2026).
- ALLOW "a different number of electrons in the d orbitals" for the oxidation-state effect.

Often asked as: MCQ colour sequences; 6-mark on why two cobalt complexes have different colours.

COMMON MISTAKE

Saying the colour changes "because the ligand is different" and stopping. Link it to a **different ΔE** and **different frequency absorbed**.

BEYOND THE SPEC, BUT EXAMINED

$[Co(NH_3)_6]^{2+}$ is slowly **oxidised by air** to $[Co(NH_3)_6]^{3+}$: the straw solution darkens to brown. A change of oxidation state with the same ligands (WCH15 Jan 2026).

SPEC 17.12

H_2O, OH^- and NH_3 as monodentate ligands

WHAT THIS SPEC POINT ASKS

Understand that H_2O, OH^- and NH_3 act as monodentate ligands.

- A **monodentate** ligand forms **one** dative bond to the metal ion, through **one** donor atom.
- **H_2O**: lone pair on **O** (it has two lone pairs but can only use one at a time).
- **OH^-**: lone pair on **O**; it carries a negative charge, so it lowers the complex's overall charge.
- **NH_3**: the lone pair on **N**.
- H_2O and NH_3 are similar in size and both neutral, so swapping one for the other doesn't change the coordination number. Cl^- is also monodentate, but bigger.

MARK-SCHEME LANGUAGE

- "monodentate: forms a single / one dative (covalent) bond"; ACCEPT "occupies one coordination site" (WCH15 Oct 2021).

COMMON MISTAKE

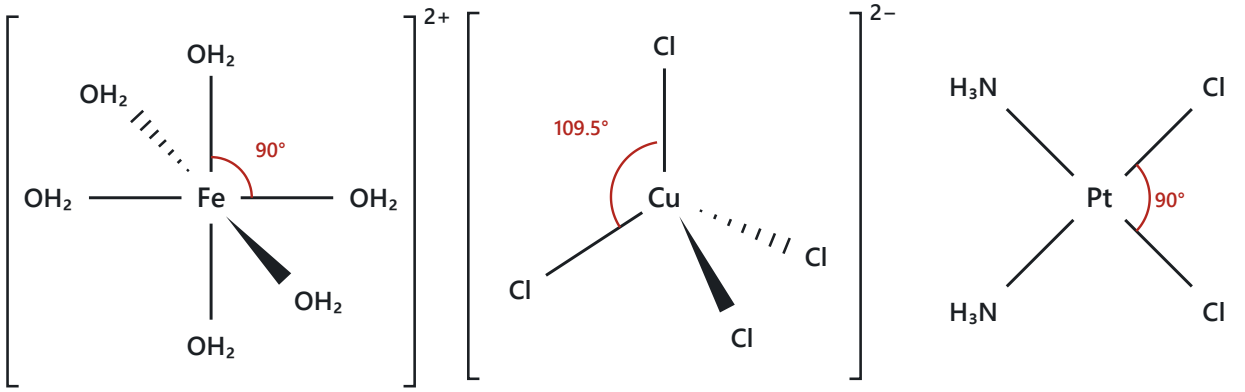
"Monodentate means it has one lone pair." It means it forms **one** bond.

SPEC 17.13 Octahedral complexes

WHAT THIS SPEC POINT ASKS

Understand why six-fold coordination gives an octahedral shape, as in complexes with H_2O, OH^- and NH_3.

wedge = towards you; dashes = away from you; plain lines = in the plane of the paper



octahedral: 6 dative bonds, 90°
$[Fe(H_2O)_6]^{2+}$

tetrahedral: 4 large Cl^-, 109.5°
$[CuCl_4]^{2-}$, $[CoCl_4]^{2-}$

square planar: 4 bonds, 90°
cisplatin, $[Pt(NH_3)_2Cl_2]$

The three shapes you need, with bond angles. Draw wedges and dashes for 3D shapes.

- Six ligands mean **six bonding pairs** around the metal ion.
- Electron pairs **repel**, so they get **as far apart as possible**: an **octahedron**, all bond angles **90°** (180° between opposite ligands).
- Small ligands (H_2O, NH_3, OH^-) fit six around a Period 4 ion: $[Fe(H_2O)_6]^{2+}$, $[Ni(NH_3)_6]^{2+}$, $[Cr(OH)_6]^{3-}$.
- Three bidentate ligands or one hexadentate ligand also give an octahedral complex.

17.13 Drawing it

Four bonds in the plane of the paper (two vertical, two horizontal, plain lines), one **wedge** (towards you) and one **dashed** bond (away). Show the metal symbol, all six donor atoms bonded through the right atom (O of H_2O, N of NH_3), square brackets and the overall charge.

MARK-SCHEME LANGUAGE

- "**six pairs of (bonding) electrons**, which are as far apart as possible to **minimise repulsion**" (WCH15 Oct 2023).
- 3D diagrams: wedge and dash bonds, 90°, correct charge. "Penalise lack of / incorrect wedges once only"; dative arrows accepted (WCH15 Oct 2025).

Often asked as: draw a 3D diagram of an octahedral complex with bond angles; explain the shape.

COMMON MISTAKE

Explaining the shape by "six ligands" alone, instead of "six **pairs of electrons** repelling" (WCH15 Oct 2023 examiner report).

BEYOND THE SPEC, BUT EXAMINED

VSEPR for other complexes: $[Ag(NH_3)_2]^+$ is **linear, 180°**, because it has two bonding pairs that minimise repulsion (WCH15 Jan 2022).
Isomerism in octahedral complexes: cis/trans isomers (e.g. two Cl^- next to or opposite each other in $[Co(NH_3)_4Cl_2]^+$) and **optical isomers** with three bidentate ligands ($[Ni(en)_3]^{2+}$): two non-superimposable mirror images (WCH15 Jan 2024).

SPEC 17.14 Tetrahedral complexes with large ligands such as Cl^-

WHAT THIS SPEC POINT ASKS

Know that transition-metal ions may form tetrahedral complexes with relatively large ions such as Cl^-.

- Chloride ions are **much larger** than H_2O or NH_3. Only **four** fit around a Period 4 metal ion.
- Four bonding pairs repel to be as far apart as possible: **tetrahedral**, **109.5°**.
- Examples: **$[CuCl_4]^{2-}$** (yellow), **$[CoCl_4]^{2-}$** (blue). Both are made by adding **concentrated HCl** to the aqua ion (17.24).

MARK-SCHEME LANGUAGE

- "**chloride ions are large**, so steric hindrance is too great for six ligands to fit around the central ion". ACCEPT "chloride ions are large". DO NOT ACCEPT "chlorine atoms". IGNORE "repulsion between negative chloride ions" (WCH15 Jun 2021).
- "$[CoCl_4]^{2-}$ is tetrahedral because Cl^- are too large for six to coordinate around Co^{2+}" (WCH15 Oct 2022).

Often asked as: why the chloro complex is tetrahedral (1–2 marks); draw $[CuCl_4]^{2-}$ in 3D with 109.5°.

COMMON MISTAKE

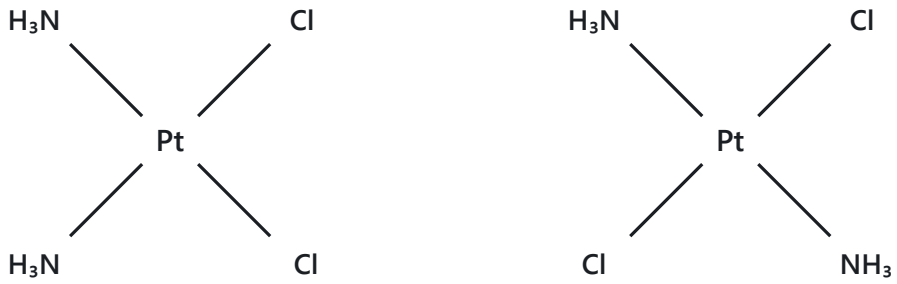
Writing "chlorine" or "chlorine atoms" for the chloride **ion** (WCH15 Oct 2022 examiner report).

SPEC 17.15 Square planar complexes and cisplatin

WHAT THIS SPEC POINT ASKS

Know that square planar complexes exist and that cisplatin is one, used in cancer treatment as a single isomer (not mixed with the trans form).

- Some four-coordinate complexes are **square planar**: all four bonds in one plane, **90°** apart. Pt^{2+} and Ni^{2+} form them, e.g. $[Pt(NH_3)_2Cl_2]$, $[Ni(CN)_4]^{2-}$.
- **Cisplatin**, $[Pt(NH_3)_2Cl_2]$, is a neutral platinum(II) complex. The two Cl^- ligands are **next to each other** (cis, 90° apart).
- In **transplatin** the two Cl^- are **opposite** each other (180°). Same formula, different arrangement: **cis/trans (geometric) isomers**.



cis-platin (the drug)
both Cl^- on the same side

trans-platin (not used)
Cl^- opposite each other

Only the cis isomer is an anticancer drug.

17.15 How cisplatin works

1. Inside a cell, $[Cl^-]$ is low and $[H_2O]$ is very high, so a **Cl^- ligand is replaced by water** (ligand exchange): $[Pt(NH_3)_2Cl_2] + H_2O \rightarrow [Pt(NH_3)_2(H_2O)Cl]^+ + Cl^-$.
2. The water ligand is then displaced by a **nitrogen atom on a guanine base** in DNA, which donates its lone pair to Pt. The second Cl^- is replaced the same way by a second, **neighbouring** guanine.
3. Pt now cross-links two adjacent bases on the same strand, so the DNA **kinks** and can't be **copied**. The cancer cell can't divide and dies.

- **Why only cis**: the two leaving groups must be on the **same side** to bind two neighbouring bases. In transplatin the second Cl^- is on the opposite side, too far from the DNA strand.
- It is given as **the pure cis isomer**, because the trans form doesn't work and would add side effects without any benefit.
- **Side effects**: cisplatin also attacks healthy, rapidly dividing cells (hair follicles, gut lining), and it damages the kidneys. Doses are balanced against these effects.

MARK-SCHEME LANGUAGE

- "square planar"; bonds drawn from **N** (not H) to Pt (WCH15 Jun 2025).
- "there is a higher concentration of water (than chloride ions) in cells, so the equilibrium moves to the right" (Cl^- replaced by H_2O) (WCH15 Jun 2025).
- Equation "$[Pt(NH_3)_2Cl_2] + H_2O \rightarrow [Pt(NH_3)_2(H_2O)Cl]^+ + Cl^-$" ("do not award products without charges"); "a lone pair on nitrogen is donated to platinum"; trans: "the second chloride is too far from / on the opposite side to the DNA strand" (WCH15 Jan 2024).

Often asked as: shape of cisplatin; 4–7 mark question on how it binds DNA and why the trans isomer is not used.

COMMON MISTAKE

Missing the **+ charge** on $[Pt(NH_3)_2(H_2O)Cl]^+$, or losing an NH_3 ligand instead of Cl^- (WCH15 Jan 2024 examiner report).
Being vague about **which side** the DNA is on for transplatin.

SPEC 17.16

Bidentate and hexadentate ligands

WHAT THIS SPEC POINT ASKS

Understand "bidentate" and "hexadentate" and identify examples such as $NH_2CH_2CH_2NH_2$ and $EDTA^{4-}$.

Ligand	Donor atoms	Dative bonds per ligand	Example complex
Ethane-1,2-diamine, "en", $H_2NCH_2CH_2NH_2$	2 N (one lone pair on each)	**2** (bidentate)	$[Cu(en)_3]^{2+}$, octahedral, CN 6
Ethanedioate, $C_2O_4^{2-}$	2 O (one on each carboxylate)	**2** (bidentate)	$[Fe(C_2O_4)_3]^{3-}$
$EDTA^{4-}$	2 N + 4 O (carboxylate groups)	**6** (hexadentate)	$[Cu(EDTA)]^{2-}$, octahedral
Haem (porphyrin)	4 N	**4** (polydentate)	haemoglobin (17.17)

- **Bidentate**: two donor atoms, two dative bonds. **Hexadentate**: six donor atoms, six dative bonds. **Polydentate / multidentate**: more than one.
- These ligands wrap round the metal ion forming rings, so they are also called **chelating** ligands ("chela" = claw). en forms a stable **five-membered ring** M–N–C–C–N.
- The donor atoms must be **far enough apart** along the chain to reach two sites next to each other on the metal ion.

MARK-SCHEME LANGUAGE

- $EDTA^{4-}$: "a **hexadentate** ligand / can form 6 dative (covalent) bonds"; "complex is more stable / is a chelating agent / traps the Cr^{3+}" (WCH15 Jan 2025).
- en diagram: "6 bonds between N in the diamines and Fe". DO NOT ACCEPT the two N of one en bonded at 180° (WCH15 Oct 2020).
- Use the spec terms: DO NOT ACCEPT "didentate", or "polydentate" when "bidentate" is the answer (WCH15 Jun 2024; Jun 2025).

Often asked as: identify a ligand as bidentate/hexadentate from a structure; draw $[M(en)_3]^{n+}$; why $EDTA^{4-}$ treats metal poisoning.

COMMON MISTAKE

Assuming an unfamiliar ligand is bidentate without counting its donor atoms (salen has 2 O and 2 N, so it is tetradentate) (WCH15 Oct 2022 examiner report).
Giving coordination number 3 for three en ligands.

BEYOND THE SPEC, BUT EXAMINED

Unfamiliar polydentate ligands in context: salen (2 O, 2 N), thioglycolate (S and O), carminic acid, diphenylcarbazide (WCH15 Oct 2022; Jun 2021; Oct 2025; Jun 2022).
Chelation therapy: $EDTA^{4-}$ binds toxic metal ions (e.g. Cr^{3+}, Pb^{2+}) so they can be excreted (WCH15 Jan 2025).

SPEC 17.17 Haemoglobin and carbon monoxide

WHAT THIS SPEC POINT ASKS

Know that haemoglobin is an iron(II) complex with a polydentate ligand, and that ligand exchange happens when CO replaces O_2 bound to haemoglobin.

- **Haem** contains an **Fe^{2+}** ion held by a **polydentate** ring ligand that bonds through **four nitrogen atoms**, all in one plane.
- A fifth dative bond comes from an N in the **globin** protein; the **sixth** site binds **O_2** (or H_2O). Six bonds: **octahedral**.
- O_2 binds **reversibly** and fairly weakly, so it is picked up in the lungs and released in respiring tissues.
- **Carbon monoxide** is a better ligand: it binds to Fe^{2+} about **200 times more strongly**. It **replaces** O_2 by **ligand exchange**:

$HbO_2 + CO \rightleftharpoons HbCO + O_2$

- The equilibrium lies far to the right, so CO is hard to remove: haemoglobin that has bound CO **can't carry oxygen**. Symptoms of poisoning follow from oxygen starvation of tissues.
- (The structure of haem is not assessed.)

MARK-SCHEME LANGUAGE

- "carbon monoxide **replaces / takes the place of the oxygen** molecule / ligand"; "because it **binds more strongly** to the Fe^{2+} ion". ACCEPT "forms a more stable complex", "larger equilibrium constant", "binds (almost) irreversibly" (WCH15 Oct 2023).
- "Prevents oxygen being carried" scores only the second mark: the replacement of O_2 must be stated.

Often asked as: why CO is toxic (2 marks); MCQ on how O_2 bonds to Fe^{2+}.

COMMON MISTAKE

Only saying CO "stops oxygen being carried" without saying it **replaces O_2** by **ligand exchange** (WCH15 Oct 2023 examiner report).

SPEC 17.18 Colours of vanadium's oxidation states

WHAT THIS SPEC POINT ASKS

Know the colours of vanadium in oxidation states +5, +4, +3 and +2.

Oxidation state	Species in acid solution	Name	Colour
+5	**VO_2^+** (VO_3^- in the solid / alkali)	dioxovanadium(V)	**yellow**
+4	**VO^{2+}**	oxovanadium(IV)	**blue**
+3	**V^{3+}**	vanadium(III)	**green**
+2	**V^{2+}**	vanadium(II)	**violet** (lilac)

Memory aid: "**Y**ou **B**etter **G**et **V**anadium" (yellow, blue, green, violet as the number goes down).

- When zinc reduces yellow VO_2^+, a **green** colour appears before the blue: that's a **mixture** of yellow V(+5) and blue V(+4), not V^{3+}. So the full sequence you see is **yellow → green → blue → green → violet**.

MARK-SCHEME LANGUAGE

- "+3 green; +4 blue; +5 yellow" (WCH15 Oct 2020). DO NOT ACCEPT combined colours such as "blue/green".
- The +5 species is **VO_2^+** and the +4 species is **VO^{2+}**, not V^{5+} or V^{4+} (WCH15 Jan 2023).

Often asked as: colours of each oxidation state; MCQ on which colour is +3.

COMMON MISTAKE

Mixing up VO_2^+ (V +5) and VO^{2+} (V +4). Check with oxidation numbers: in VO_2^+, V − 4 = +1, so V = +5. Writing V^{5+} as if it existed as a simple ion (WCH15 Jan 2023 examiner report).

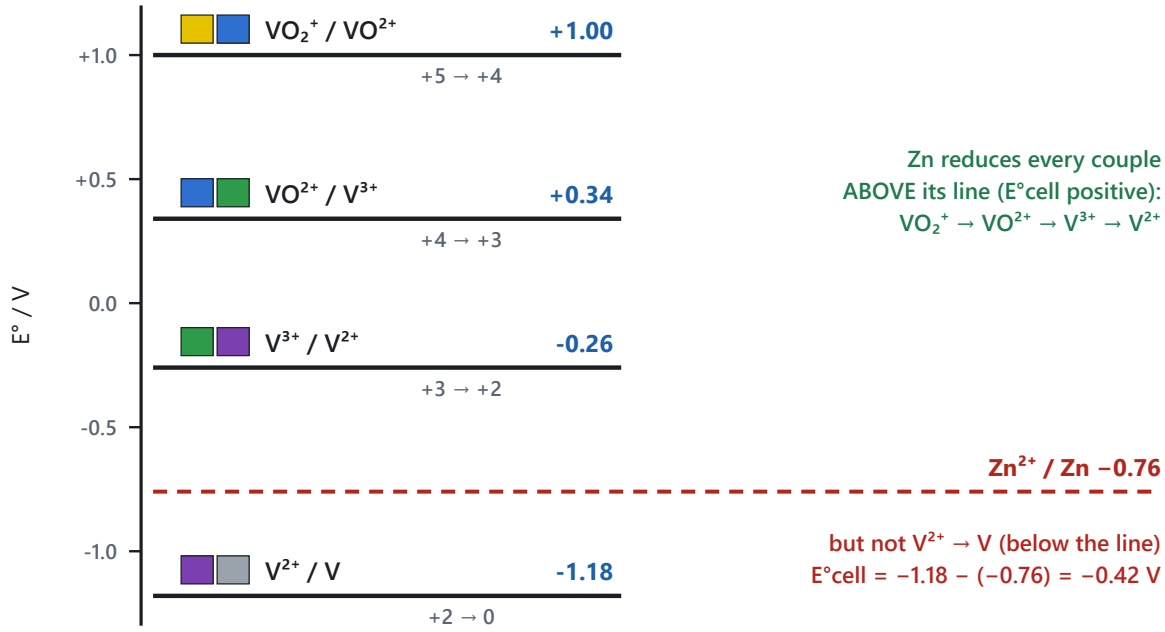
SPEC 17.19

Interconverting vanadium's oxidation states using E°

WHAT THIS SPEC POINT ASKS

Understand the redox reactions that interconvert vanadium's +5, +4, +3 and +2 states, using the relevant E° values.

Half-equation	E° / V
$VO_2^+ + 2H^+ + e^- \rightleftharpoons VO^{2+} + H_2O$	+1.00
$VO^{2+} + 2H^+ + e^- \rightleftharpoons V^{3+} + H_2O$	+0.34
$V^{3+} + e^- \rightleftharpoons V^{2+}$	−0.26
$V^{2+} + 2e^- \rightleftharpoons V$	−1.18



Colours: yellow (+5), blue (+4), green (+3), violet (+2)

Zinc (−0.76 V) is more negative than each of the top three vanadium couples, so it reduces V(+5) all the way to V(+2), but not to the metal.

17.19 Reduction with zinc in acid

Shake ammonium vanadate(V), NH_4VO_3, with **zinc and dilute acid**; put a loose **cotton-wool plug** in the flask so hydrogen can escape but air can't get in (air would oxidise V^{2+} back).

WORKED EXAMPLE: ZINC REDUCES VANADIUM STEP BY STEP

$Zn^{2+} + 2e^- \rightleftharpoons Zn$, E° = −0.76 V

+5 → +4: E°cell = 1.00 − (−0.76) = +1.76 V ✓

$2VO_2^+ + 4H^+ + Zn \rightarrow 2VO^{2+} + Zn^{2+} + 2H_2O$

+4 → +3: E°cell = 0.34 − (−0.76) = +1.10 V ✓

$2VO^{2+} + 4H^+ + Zn \rightarrow 2V^{3+} + Zn^{2+} + 2H_2O$

+3 → +2: E°cell = −0.26 − (−0.76) = +0.50 V ✓

$2V^{3+} + Zn \rightarrow 2V^{2+} + Zn^{2+}$

+2 → 0: E°cell = −1.18 − (−0.76) = −0.42 V ✗ **not feasible**, so the reduction stops at **V^{2+}**.

WORKED EXAMPLE: A WEAKER REDUCING AGENT STOPS SOONER

Fe^{2+} ($Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$, +0.77 V):

$VO_2^+ \rightarrow VO^{2+}$: 1.00 − 0.77 = **+0.23 V** ✓; $VO^{2+} \rightarrow V^{3+}$: 0.34 − 0.77 = **−0.43 V** ✗. Fe^{2+} only takes vanadium to +4.

Iodide ($I_2 + 2e^- \rightleftharpoons 2I^-$, +0.54 V): $VO_2^+ \rightarrow VO^{2+}$ is +0.46 V ✓; $VO^{2+} \rightarrow V^{3+}$ is −0.20 V ✗. Also stops at +4.

Iron metal ($Fe^{2+} + 2e^- \rightleftharpoons Fe$, −0.44 V): +1.44, +0.78 and +0.18 V, so iron takes vanadium down to **V^{2+}**. Tin (Sn^{2+}/Sn, −0.14 V) gives +1.14 and +0.48 V but −0.12 V for $V^{3+} \rightarrow V^{2+}$, so tin stops at **$V^{3+}$**.

17.19 Oxidation back up

- Air ($O_2 + 4H^+ + 4e^- \rightleftharpoons 2H_2O$, +1.23 V) readily oxidises V^{2+} and V^{3+} up to VO^{2+}. Going on to VO_2^+ has E°cell = 1.23 − 1.00 = +0.23 V: feasible on paper but very slow, so blue VO^{2+} solutions are stable in air.
- Acidified **MnO_4^-** (+1.51 V) oxidises every lower state all the way to **VO_2^+**. This is used to titrate vanadium (16.15).

MARK-SCHEME LANGUAGE

- Iron and tin (6-mark, WCH15 Jan 2023): "both reduce V(V) to V(IV)" (Fe +1.44, Sn +1.14 V); "both reduce V(IV) to V(III)" (+0.78, +0.48 V); "only iron reduces V(III) to V(II)" (+0.18 V; tin −0.12 V). Equations for each step.
- "VO^{2+} is not oxidised any further as E°cell is −0.2 V / negative" (WCH15 Oct 2020).

Often asked as: which reagent reduces vanadium to which state (with equations and E°cell values); 6-mark comparing two reducing agents; balanced equation for MnO_4^- with a vanadium ion.

COMMON MISTAKE

Calculating every E°cell but writing no equations, or the reverse (WCH15 Jan 2023 examiner report). Not saying the reduction **stops** at a given state and why.

Pairing the reduction of VO_2^+ with the wrong oxidation half-equation (WCH15 Oct 2022 examiner report).

BEYOND THE SPEC, BUT EXAMINED

Potassium vanadium alum $KV(SO_4)_2{\cdot}12H_2O$ deduced from % composition (WCH15 Jan 2023). Ferrovanadium analysis by back titration (WCH15 Oct 2022). Equation for MnO_4^- with V^{3+} taking it to V(+5): $2MnO_4^- + 5V^{3+} + 2H_2O \rightarrow 2Mn^{2+} + 5VO_2^+ + 4H^+$.

SPEC 17.20 Reducing and making dichromate(VI)

WHAT THIS SPEC POINT ASKS

Understand, using E° values, that (i) zinc in acid reduces $Cr_2O_7^{2-}$ to Cr^{3+} and then Cr^{2+}, and (ii) Cr^{3+} is oxidised to dichromate(VI) by hydrogen peroxide in alkali, followed by acidification.

Half-equation	E° / V
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightleftharpoons 2Cr^{3+} + 7H_2O$	+1.33
$H_2O_2 + 2e^- \rightleftharpoons 2OH^-$ (alkaline)	+0.88
$CrO_4^{2-} + 4H_2O + 3e^- \rightleftharpoons Cr(OH)_3 + 5OH^-$ (alkaline)	−0.13
$Cr^{3+} + e^- \rightleftharpoons Cr^{2+}$	−0.41
$Zn^{2+} + 2e^- \rightleftharpoons Zn$	−0.76

17.20 (i) Zinc reduces dichromate(VI): orange → green → blue

WORKED EXAMPLE: ZINC AND ACIDIFIED DICHROMATE

Cr(+6) → Cr(+3): E°cell = 1.33 − (−0.76) = **+2.09 V** ✓
$Cr_2O_7^{2-} + 14H^+ + 3Zn \rightarrow 2Cr^{3+} + 7H_2O + 3Zn^{2+}$ (orange → green)
Cr(+3) → Cr(+2): E°cell = −0.41 − (−0.76) = **+0.35 V** ✓
$2Cr^{3+} + Zn \rightarrow 2Cr^{2+} + Zn^{2+}$ (green → blue)
Compare Fe^{2+} (+0.77 V): it reduces $Cr_2O_7^{2-}$ to Cr^{3+} (1.33 − 0.77 = +0.56 V) but **not** to Cr^{2+} (−0.41 − 0.77 = −1.18 V).

- Blue Cr^{2+} is **rapidly oxidised by air** back to green Cr^{3+} (O_2/H_2O at +1.23 V gives +1.64 V). So keep air out: a loose cotton-wool plug, or a bung with a valve, lets H_2 escape but stops air entering.

17.20 (ii) Making dichromate(VI) from chromium(III)

1. Add excess **NaOH** to Cr^{3+}(aq): the green precipitate dissolves to dark green **$[Cr(OH)_6]^{3-}$**.
2. Add **hydrogen peroxide** and **warm**: the solution turns **yellow** as chromate(VI) forms.
   - E°cell = 0.88 − (−0.13) = **+1.01 V** ✓ (feasible in alkali)
   - $2[Cr(OH)_6]^{3-} + 3H_2O_2 \rightarrow 2CrO_4^{2-} + 2OH^- + 8H_2O$
   - or, from the hydroxide: $2Cr(OH)_3 + 3H_2O_2 + 4OH^- \rightarrow 2CrO_4^{2-} + 8H_2O$
3. Boil off excess H_2O_2 (otherwise it would reduce the dichromate in acid), then **acidify**: yellow CrO_4^{2-} turns **orange** $Cr_2O_7^{2-}$ (17.21).

EXAM TIP

Why alkaline conditions? In acid the margin is small (1.77 − 1.33 = +0.44 V), and H_2O_2 can also act as a reducing agent towards dichromate, so the product doesn't build up cleanly. In alkali the Cr(VI)/Cr(III) potential is far less positive (−0.13 V), so H_2O_2 oxidises Cr(III) easily.

MARK-SCHEME LANGUAGE

- "zinc converts dichromate(VI) to chromium(III) with E°cell +2.09 V and then to chromium(II) with E°cell +0.35 V"; "iron(II) converts dichromate(VI) to chromium(III) (+0.56 V) but not to chromium(II) as … −1.18 V"; "acidified solution" for the conditions (WCH15 Jun 2024).
- Colours: "$Cr_2O_7^{2-}$ orange and Cr^{3+} green"; DO NOT ACCEPT blue for Cr^{3+} (WCH15 Jun 2022).
- Cr^{2+} cell drift: "Cr^{2+} is (rapidly) oxidised (by oxygen in the air)" (WCH15 Jun 2024).

Often asked as: which reagents reduce dichromate to which state (equations + E°cell); oxidising Cr(III) with H_2O_2; colours at each stage.

COMMON MISTAKE

Leaving H_2O_2 out of the equation; not cancelling H^+, H_2O or e^- (WCH15 Jan 2025 examiner report).
Comparing E° values in words instead of calculating E°cell; using the wrong half-equation from a numbered list (WCH15 Jun 2024 examiner report).
Calling Cr^{3+} "blue" or "blue-green" (WCH15 Jun 2022 examiner report).

BEYOND THE SPEC, BUT EXAMINED

The mark schemes are not consistent on the H_2O_2 equation. One accepted the acid form $2Cr^{3+} + 3H_2O_2 + H_2O \rightarrow Cr_2O_7^{2-} + 8H^+$ (WCH15 Jan 2025). An MCQ used the alkaline form $2Cr(OH)_3 + 3H_2O_2 + 4KOH \rightarrow 2K_2CrO_4 + 8H_2O$ (WCH15 Jun 2023). Learn the alkaline route the spec describes.
Acidified dichromate oxidising ethanol: $3C_2H_5OH + Cr_2O_7^{2-} + 8H^+ \rightarrow 2Cr^{3+} + 3CH_3CHO + 7H_2O$ (WCH15 Jan 2021).

SPEC 17.21

The dichromate–chromate equilibrium

WHAT THIS SPEC POINT ASKS

Know that dichromate(VI) converts to chromate(VI) through the equilibrium $Cr_2O_7^{2-} + H_2O \rightleftharpoons 2CrO_4^{2-} + 2H^+$.

$Cr_2O_7^{2-}(aq) + H_2O(l) \rightleftharpoons 2CrO_4^{2-}(aq) + 2H^+(aq)$
orange ⇌ yellow

- **Add alkali** (OH^-): it removes H^+ ($H^+ + OH^- \rightarrow H_2O$), so the equilibrium shifts **right** → **yellow** chromate(VI).
- **Add acid** (H^+): the equilibrium shifts **left** → **orange** dichromate(VI).
- Chromium is **+6 on both sides**: this is an **acid–base** equilibrium, **not redox**.

MARK-SCHEME LANGUAGE

- "the equilibrium shifts to the chromate(VI) ions because hydrogen ions are removed (as they react with hydroxide)" (WCH15 Oct 2025).
- "not redox because the oxidation number of chromium has not changed: +6 in both" (WCH15 Jan 2025).

Often asked as: write the equation (only 25% got it right, WCH15 Jan 2021); explain the colour change on adding alkali; is it redox?

COMMON MISTAKE

Saying adding alkali "reduces" dichromate. No oxidation number changes.

SPEC 17.22 Reactions of aqueous ions with NaOH and NH_3

WHAT THIS SPEC POINT ASKS

Record observations and write equations for the reactions of Cr^{3+}, Mn^{2+}, Fe^{2+}, Fe^{3+}, Co^{2+}, Ni^{2+}, Cu^{2+} and $Zn^{2+}(aq)$ with aqueous sodium hydroxide and aqueous ammonia, including in excess.

17.22 With sodium hydroxide

Ion (colour)	A few drops of NaOH(aq)	Excess NaOH(aq)
Cr^{3+} (green)	green (grey-green) ppt, $[Cr(OH)_3(H_2O)_3]$	ppt **dissolves**: dark green solution, $[Cr(OH)_6]^{3-}$
Mn^{2+} (very pale pink)	off-white / pale brown ppt, $[Mn(OH)_2(H_2O)_4]$; darkens to brown in air	no change
Fe^{2+} (pale green)	green ppt, $[Fe(OH)_2(H_2O)_4]$; turns orange-brown on standing in air	no change
Fe^{3+} (yellow-brown)	red-brown / orange-brown ppt, $[Fe(OH)_3(H_2O)_3]$	no change
Co^{2+} (pink)	blue ppt, $[Co(OH)_2(H_2O)_4]$; turns pink/beige on standing	no change
Ni^{2+} (green)	green ppt, $[Ni(OH)_2(H_2O)_4]$	no change
Cu^{2+} (blue)	(pale) blue ppt, $[Cu(OH)_2(H_2O)_4]$	no change
Zn^{2+} (colourless)	white ppt, $[Zn(OH)_2(H_2O)_4]$	ppt **dissolves**: colourless solution, $[Zn(OH)_4]^{2-}$

Only the **chromium(III)** and **zinc** hydroxides dissolve in excess NaOH: they are **amphoteric**.

17.22 With aqueous ammonia

Ion	A few drops of $NH_3(aq)$	Excess $NH_3(aq)$
Cr^{3+}	green ppt	ppt dissolves (slowly, best with conc NH_3): **purple / violet** solution, $[Cr(NH_3)_6]^{3+}$
Mn^{2+}	off-white / pale brown ppt (darkens in air)	no change
Fe^{2+}	green ppt (turns brown in air)	no change
Fe^{3+}	red-brown ppt	no change
Co^{2+}	blue ppt	ppt dissolves: **straw / yellow-brown** solution, $[Co(NH_3)_6]^{2+}$; darkens to brown in air
Ni^{2+}	green ppt	ppt dissolves: **blue** (lavender-blue) solution, $[Ni(NH_3)_6]^{2+}$
Cu^{2+}	pale blue ppt	ppt dissolves: **deep blue** solution, $[Cu(NH_3)_4(H_2O)_2]^{2+}$
Zn^{2+}	white ppt	ppt dissolves: **colourless** solution, $[Zn(NH_3)_4]^{2+}$

With a few drops, ammonia acts as a **base** and gives the **same precipitates** as NaOH. In excess, it acts as a **ligand** for Cr, Co, Ni, Cu and Zn.

17.22 Air oxidation on standing

- $Fe(OH)_2$ (green) → $Fe(OH)_3$ (brown): $4Fe(OH)_2 + O_2 + 2H_2O \rightarrow 4Fe(OH)_3$. The brown appears first at the **surface**, where the air is.
- $Mn(OH)_2$ similarly darkens to brown as Mn(II) is oxidised to a higher state.

MARK-SCHEME LANGUAGE

- Cobalt (6-mark, WCH15 Jan 2022): pink solution; with NaOH "blue precipitate / pink on standing"; with NH_3 "straw / yellow / yellow-brown / brown solution (ligand exchange)"; with conc HCl "(dark) blue solution". "The initial colour must be mentioned once."
- Nickel (6-mark, WCH15 Jun 2022): "both alkalis initially give a green precipitate"; "deprotonation"; "with excess ammonia a blue solution, no change with NaOH"; "ligand exchange". DO NOT ACCEPT blue-green or blue-purple.
- Zinc (WCH15 Jun 2022): "white precipitate"; "the precipitate dissolves" in excess NaOH "due to formation of $Zn(OH)_4^{2-}$".
- Iron: "$Fe(OH)_2$ is green (initially)" and "turns brown on standing" (WCH15 Jun 2021).

Often asked as: 6-mark "describe what you would see and write equations" for one metal with both reagents; identify an ion from observations; MCQ on colour sequences.

COMMON MISTAKE

Leaving out the **starting colour** of the solution, or the word "precipitate" (WCH15 Jan 2022 examiner report). The most common score on the cobalt 6-mark was 0.
Copying the copper ammine formula for cobalt or nickel: Cu gives $[Cu(NH_3)_4(H_2O)_2]^{2+}$, but Co and Ni give $[M(NH_3)_6]^{2+}$ (WCH15 Jan 2022 examiner report).
Not recognising a red-brown precipitate as iron(III) hydroxide (WCH15 Oct 2020 examiner report).

SPEC 17.23 Equations: deprotonation, amphoteric behaviour and ligand exchange

WHAT THIS SPEC POINT ASKS

Write ionic equations showing amphoteric behaviour, deprotonation and ligand exchange for the reactions in 17.22.

17.23 1. Deprotonation (a few drops of OH^- or NH_3)

Hexaaqua ions are **weak acids**: the metal ion pulls electron density from the O–H bonds of its water ligands, so a ligand can lose H^+. A base removes **one H^+ per positive charge**, until the complex is **neutral** and precipitates.

- 2+ ions: $\mathbf{[M(H_2O)_6]^{2+}(aq) + 2OH^-(aq) \rightarrow \underline{M(OH)_2(H_2O)_4} + 2H_2O(l)}$
- 3+ ions: $\mathbf{[M(H_2O)_6]^{3+}(aq) + 3OH^-(aq) \rightarrow \underline{M(OH)_3(H_2O)_3} + 3H_2O(l)}$
- With ammonia as the base: $\mathbf{[M(H_2O)_6]^{2+}(aq) + 2NH_3(aq) \rightarrow \underline{M(OH)_2(H_2O)_4} + 2NH_4^+(aq)}$

17.23 2. Amphoteric behaviour (excess OH^- with Cr and Zn)

An **amphoteric** hydroxide reacts with **both acids and bases**.

	As a base (with acid)	As an acid (with excess OH^-)
Cr	$[Cr(OH)_3(H_2O)_3] + 3H^+ \rightarrow [Cr(H_2O)_6]^{3+}$	$[Cr(OH)_3(H_2O)_3] + 3OH^- \rightarrow [Cr(OH)_6]^{3-} + 3H_2O$
Zn	$[Zn(OH)_2(H_2O)_4] + 2H^+ \rightarrow [Zn(H_2O)_6]^{2+}$	$[Zn(OH)_2(H_2O)_4] + 2OH^- \rightarrow [Zn(OH)_4]^{2-} + 4H_2O$

$[Zn(OH)_4]^{2-}$ is tetrahedral, which is why four water ligands are released.

17.23 3. Ligand exchange (excess NH_3)

NH_3 replaces water (and hydroxide) ligands. Write it from the precipitate or directly from the aqua ion:

- **Cr**: $[Cr(OH)_3(H_2O)_3] + 6NH_3 \rightarrow [Cr(NH_3)_6]^{3+} + 3OH^- + 3H_2O$
- **Co**: $[Co(OH)_2(H_2O)_4] + 6NH_3 \rightarrow [Co(NH_3)_6]^{2+} + 2OH^- + 4H_2O$; overall $[Co(H_2O)_6]^{2+} + 6NH_3 \rightarrow [Co(NH_3)_6]^{2+} + 6H_2O$
- **Ni**: $[Ni(OH)_2(H_2O)_4] + 6NH_3 \rightarrow [Ni(NH_3)_6]^{2+} + 2OH^- + 4H_2O$; overall $[Ni(H_2O)_6]^{2+} + 6NH_3 \rightarrow [Ni(NH_3)_6]^{2+} + 6H_2O$
- **Cu**: $[Cu(OH)_2(H_2O)_4] + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 2OH^- + 2H_2O$; overall $[Cu(H_2O)_6]^{2+} + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 4H_2O$
- **Zn**: $[Zn(OH)_2(H_2O)_4] + 4NH_3 \rightarrow [Zn(NH_3)_4]^{2+} + 2OH^- + 4H_2O$

EXAM TIP

Check every equation three ways: **atoms** (count O and H carefully), **charge**, and **brackets**. The number of OH^- in the precipitate always equals the charge on the metal ion.

MARK-SCHEME LANGUAGE

- "$[Ni(H_2O)_6]^{2+} + 2OH^- \rightarrow [Ni(H_2O)_4(OH)_2] + 2H_2O$"; "these are deprotonation reactions"; "$[Ni(H_2O)_6]^{2+} + 6NH_3 \rightarrow [Ni(NH_3)_6]^{2+} + 6H_2O$"; "with excess ammonia it is ligand exchange" (WCH15 Jun 2022).
- Reaction type for a little NH_3: "deprotonation"; ammonia "is a base" (WCH15 Jan 2026).
- Square brackets may be omitted; equations written with simple ions ($Ni^{2+} + 2OH^- \rightarrow Ni(OH)_2$) are often accepted.

Often asked as: write the equation for each observation and name the reaction type; MCQ on the equation for an amphoteric hydroxide dissolving in acid.

COMMON MISTAKE

Mixing up **deprotonation** (precipitate forms) and **ligand exchange** (precipitate dissolves in excess NH_3) (WCH15 Jun 2022 examiner report).
Bracket errors like $[Ni(H_2O)_4(OH_2)]$; $[Zn(OH)_6]^{4-}$ instead of **$[Zn(OH)_4]^{2-}$**; NH_4^+ in an equation with OH^- as the base (WCH15 Jun 2022 examiner report).
Writing $H_2O(aq)$: water is (l) (WCH15 Jun 2024 examiner report).

BEYOND THE SPEC, BUT EXAMINED

Predicting an unfamiliar metal's reactions from the pattern (e.g. working out cobalt by analogy with copper, but with six NH_3) (WCH15 Jan 2022).
Sodium carbonate also gives the hydroxide precipitates with 3+ ions (WCH15 Jun 2024).

SPEC 17.24 Ligand exchange and colour changes with copper and cobalt

WHAT THIS SPEC POINT ASKS

Understand the ligand exchange and colour change in forming (i) $[Cu(NH_3)_4(H_2O)_2]^{2+}$ from $[Cu(H_2O)_6]^{2+}$ via $Cu(OH)_2(H_2O)_4$, (ii) $[CuCl_4]^{2-}$ from $[Cu(H_2O)_6]^{2+}$, (iii) $[CoCl_4]^{2-}$ from $[Co(H_2O)_6]^{2+}$.

17.24 (i) Copper with ammonia

Stage	Species	Observation	Reaction type
start	$[Cu(H_2O)_6]^{2+}$	blue solution	
a few drops of NH_3	$[Cu(OH)_2(H_2O)_4]$	pale blue precipitate	deprotonation (acid–base)
excess NH_3	$[Cu(NH_3)_4(H_2O)_2]^{2+}$	**deep blue** solution	ligand exchange

- Only **four** of the six water ligands are replaced. The coordination number stays **6** (NH_3 and H_2O are similar in size and both neutral) and the shape stays octahedral.

17.24 (ii) Copper with concentrated hydrochloric acid

$[Cu(H_2O)_6]^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons [CuCl_4]^{2-}(aq) + 6H_2O(l)$
blue → (green) → **yellow**

- Coordination number **6** → **4**; octahedral → **tetrahedral** (Cl^- is large). Charge 2+ → 2−. Copper stays **+2**.
- The green seen part-way is a **mixture** of blue and yellow complexes.
- It is **reversible**: adding water shifts the equilibrium back to the blue aqua ion.

17.24 (iii) Cobalt with concentrated hydrochloric acid

$[Co(H_2O)_6]^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons [CoCl_4]^{2-}(aq) + 6H_2O(l)$
pink → blue

- Again 6 → 4, octahedral → tetrahedral, cobalt stays +2. Adding water turns it pink again.
- The forward reaction is endothermic, so warming the mixture makes it bluer. (This is the chemistry of cobalt chloride paper: blue when dry, pink with water.)

MARK-SCHEME LANGUAGE

- Copper 6-mark (WCH15 Jun 2025): "tetrahedral ($[CuCl_4]^{2-}$) to octahedral ($[Cu(H_2O)_6]^{2+}$ / $[Cu(NH_3)_4(H_2O)_2]^{2+}$)"; "the Cl^- ion is larger, so more water ligands can fit"; "copper(II) hydroxide precipitates as the solution becomes alkaline"; "yellow to blue solution, or precipitate to deep blue, is ligand exchange". DO NOT ACCEPT "blue solution to precipitate" as ligand exchange.
- "$[Co(H_2O)_6]^{2+} + 4Cl^- \rightarrow [CoCl_4]^{2-} + 6H_2O$". IGNORE state symbols (WCH15 Oct 2022).
- DO NOT ACCEPT $[Cu(NH_3)_6]^{2+}$ for the deep blue complex (WCH15 Jan 2021).

Often asked as: 6-mark tracing colour and shape changes around copper(II); equations for the chloro complexes; MCQ on the colour of $[CuCl_4]^{2-}$.

COMMON MISTAKE

Calling the precipitate step "ligand exchange" (it is deprotonation). Writing $[Cu(NH_3)_6]^{2+}$.
Saying copper is reduced when the solution goes yellow: the oxidation state is unchanged.

BEYOND THE SPEC, BUT EXAMINED

The order of exchange follows how readily each ligand donates its lone pair: **Cl^- < H_2O < NH_3**, so water replaces chloride and ammonia replaces water (WCH15 Jun 2025).
Cu(I) ammine complexes, colourless $[Cu(NH_3)_2]^+$, oxidised by air to the deep blue Cu(II) complex (WCH15 Jan 2021).

SPEC **17.25**

The chelate effect: entropy and stability

WHAT THIS SPEC POINT ASKS

Understand, in terms of a positive ΔS(system), why replacing a monodentate ligand by a bidentate or hexadentate ligand gives a more stable complex.

17.25 Count the particles

Reaction	Particles before → after
$[Cu(H_2O)_6]^{2+} + 3en \rightarrow [Cu(en)_3]^{2+} + 6H_2O$	4 → **7**
$[Cu(H_2O)_6]^{2+} + EDTA^{4-} \rightarrow [Cu(EDTA)]^{2-} + 6H_2O$	2 → **7**
$[Cu(NH_3)_4(H_2O)_2]^{2+} + 3en \rightarrow [Cu(en)_3]^{2+} + 4NH_3 + 2H_2O$	4 → **7**

- More particles in solution means more ways of arranging them, so **ΔS(system) is positive**.
- **ΔH is about zero**: the same number and the same kind of dative bonds are broken and made (six M–O or M–N bonds each side).
- So ΔS(surroundings) = −ΔH/T ≈ 0, and **ΔS(total) is positive**. Equivalently **ΔG = ΔH − TΔS is negative**. The reaction is feasible with a **large K**, so the chelate complex is **more stable** and the reverse reaction hardly happens.

WORKED EXAMPLE: EDTA AND A HEXAAQUA ION

$[Ni(H_2O)_6]^{2+}(aq) + EDTA^{4-}(aq) \rightarrow [Ni(EDTA)]^{2-}(aq) + 6H_2O(l)$
2 particles on the left, 7 on the right → ΔS(system) is large and positive.
ΔH ≈ 0, because six dative bonds to O are replaced by six dative bonds to N and O.
So ΔS(total) > 0 and ΔG < 0: the EDTA complex forms and is very stable.

MARK-SCHEME LANGUAGE

- "there are **more particles / moles** on the right (of the equation)" and "so **ΔS(system) increases / is positive** (and the reaction is thermodynamically feasible)". Marks are independent (WCH15 Jun 2024; Oct 2023).
- "Allow species or ions for particles". DO NOT ACCEPT "molecules", "atoms" or "compounds" (WCH15 Oct 2023; Jun 2022).
- IGNORE "entropy is positive" (entropy is always positive; it is the **change** that is positive).

Often asked as: write the equation for $EDTA^{4-}$ or en replacing water and explain the stability (2–4 marks); count particles in an unfamiliar ligand exchange.

COMMON MISTAKE

"More molecules on the right" (the products include ions). "Entropy is positive" instead of "the entropy **change** of the system is positive" (WCH15 Jun 2022; Oct 2023 examiner reports).
Talking about bond strength instead of entropy (WCH15 Oct 2022 examiner report). Putting Na^+ inside the complex (WCH15 Jun 2024 examiner report).

BEYOND THE SPEC, BUT EXAMINED

Counting particles when some species are polyatomic anions, e.g. 5 on the left and 7 on the right (WCH15 Jun 2022).

SPEC 17.26 Transition metals as catalysts

WHAT THIS SPEC POINT ASKS

Know that transition metals and their compounds act as heterogeneous and homogeneous catalysts.

- A catalyst provides an **alternative route with a lower activation energy**. It doesn't change ΔH or the position of equilibrium (K), and is unchanged at the end.
- Transition metals are good catalysts because:
  - they have **variable oxidation states**, so they can accept and donate electrons, then return to the original state (homogeneous catalysis and V_2O_5);
  - their **d orbitals** (and partly filled d subshells) let reactant molecules bond to the **surface** (heterogeneous catalysis).

Catalyst	Type	Process
Fe	heterogeneous	Haber process: $N_2 + 3H_2 \rightleftharpoons 2NH_3$
Ni	heterogeneous	hydrogenation of alkenes (e.g. making margarine)
V_2O_5	heterogeneous	Contact process: $2SO_2 + O_2 \rightleftharpoons 2SO_3$
Pt / Pd / Rh	heterogeneous	catalytic converters
Fe^{2+} / Fe^{3+}	homogeneous	$S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2$
Mn^{2+}	homogeneous (autocatalyst)	$MnO_4^- + C_2O_4^{2-}$

MARK-SCHEME LANGUAGE

Compare and contrast (6-mark, WCH15 Oct 2020): both "provide an alternative route / mechanism with a lower activation energy"; "heterogeneous is in a different phase from the reactants, homogeneous in the same phase"; examples (Fe in the Haber process, Ni in hydrogenation, Pt in a catalytic converter; Fe^{2+}/Fe^{3+} with I^- and $S_2O_8^{2-}$); heterogeneous "adsorbed … bonds weakened … reaction … desorbed"; homogeneous "the transition metal ion is oxidised / reduced to a different oxidation state, then changes back".

Often asked as: 6-mark compare heterogeneous and homogeneous catalysis; MCQ on catalyst types.

COMMON MISTAKE

Writing about the two types separately in a "compare" question. Pair them ("**both** lower Ea, **but** …") (WCH15 Oct 2020 examiner report).
Mixing up the Haber (Fe) and Contact (V_2O_5) catalysts.

SPEC 17.27

Heterogeneous catalysts

WHAT THIS SPEC POINT ASKS

Know that a heterogeneous catalyst is in a different phase from the reactants and that the reaction happens on its surface.

- **Heterogeneous**: the catalyst is in a **different phase** (usually a solid with gas or liquid reactants).
- The reaction happens at **active sites** on the **surface**:
  1. **Adsorption**: reactant molecules form weak bonds to the surface.
  2. **Reaction**: bonds within the reactants are **weakened**, and the reactants are held close together in a good orientation, so they react.
  3. **Desorption**: the products leave the surface, freeing the sites.
- **Adsorption strength** must be just right. Too strong (e.g. W) and the products don't leave; too weak (e.g. Ag) and the reactants don't stick. Ni and Pt are in between.
- To get a large **surface area** cheaply, the metal is spread thinly on a **support** (e.g. a ceramic honeycomb).
- **Poisoning**: impurities adsorb strongly and block active sites (lead on catalytic converters; sulfur on the iron in the Haber process). Solid catalysts are easy to separate from products.

MARK-SCHEME LANGUAGE

- "adsorption … to the surface"; "results in the weakening of bonds"; "reaction occurs"; "products desorb". DO NOT ACCEPT "absorption" (penalised once) (WCH15 Oct 2025).
- ACCEPT "easy to separate" as an advantage of heterogeneous catalysts (WCH15 Oct 2020).

COMMON MISTAKE

"Absorb" for **adsorb**. Absorb means soak into; adsorb means stick onto the surface.

SPEC 17.28

Vanadium(V) oxide in the Contact process

WHAT THIS SPEC POINT ASKS

Understand, in terms of oxidation number, how V_2O_5 catalyses the Contact process.

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ (about 450 °C, 1–2 atm, V_2O_5 catalyst)

V_2O_5 is a solid catalyst for gases (heterogeneous), but it works by **changing oxidation state**:

1. **$SO_2 + V_2O_5 \rightarrow SO_3 + V_2O_4$**: V is reduced, **+5 → +4**; S is oxidised, +4 → +6.
2. **$V_2O_4 + ½O_2 \rightarrow V_2O_5$**: V is oxidised back, **+4 → +5**.

Adding the two steps gives $SO_2 + ½O_2 \rightarrow SO_3$: the V_2O_5 is regenerated.

MARK-SCHEME LANGUAGE

- "$V_2O_5 + SO_2 \rightarrow V_2O_4 + SO_3$; $V_2O_4 + ½O_2 \rightarrow V_2O_5$". ACCEPT any balanced pair showing a lower oxide formed by SO_2 then reoxidised by O_2. Two equations in the wrong order score 1 (WCH15 Oct 2022).
- DO NOT ACCEPT V_2O_5 described as a **homogeneous** catalyst (WCH15 Jan 2021).

Often asked as: two equations with oxidation states; MCQ on vanadium's oxidation state during the cycle.

COMMON MISTAKE

Calling V_2O_5 homogeneous because it changes oxidation state. It is a solid catalysing gases: **heterogeneous** (WCH15 Jan 2021 examiner report).

SPEC 17.29 Catalytic converters

WHAT THIS SPEC POINT ASKS

Understand how a catalytic converter cuts CO and NO emissions: CO and NO adsorb on the catalyst surface, bonds weaken and they react; then CO_2 and N_2 desorb.

- **Where the pollutants come from**: CO from incomplete combustion of fuel; NO from N_2 and O_2 in air reacting at the high temperature of the engine.
- **The catalyst**: platinum, palladium and rhodium, coated thinly on a **ceramic honeycomb** (large surface area, little expensive metal).

$2CO(g) + 2NO(g) \rightarrow 2CO_2(g) + N_2(g)$

1. **Adsorption**: CO and NO molecules form bonds to the metal surface.
2. **Bonds weaken** (e.g. the N–O bond), and the adsorbed species react.
3. **Desorption**: CO_2 and N_2 leave the surface, freeing the active sites.

- Unburnt hydrocarbons are also oxidised to CO_2 and H_2O.
- Lead (from old leaded petrol) **poisons** the catalyst, which is why converters need unleaded fuel.

MARK-SCHEME LANGUAGE

- Jan 2021 6-mark (V_2O_5 v Pt in a converter): both heterogeneous; adsorption of reactants (both); weakening of bonds (converter); desorption of products (both). DO NOT ACCEPT "chemisorption" or "bond strongly" (WCH15 Jan 2021).
- CO on Pt (WCH15 Oct 2025): "adsorption of CO to the surface"; "weakening of bonds"; "reaction / oxidation occurs"; "to form CO_2"; "which then desorbs". A reference to the same state or phase loses a mark.

Often asked as: describe how the converter removes CO and NO (3–6 marks); MCQ: "CO and NO are **ad**sorbed".

COMMON MISTAKE

Missing out **desorption** of products (WCH15 Jan 2021 examiner report). Writing "absorbed".

SPEC 17.30 Homogeneous catalysts

WHAT THIS SPEC POINT ASKS

Know that a homogeneous catalyst is in the same phase as the reactants, and that the catalysed reaction goes via an intermediate species.

- **Homogeneous**: catalyst and reactants are in the **same phase**, usually all in aqueous solution.
- The catalyst **reacts** with one reactant to form an **intermediate**, which then reacts to give the products and **regenerate** the catalyst.

- Transition-metal ions are good at this because they switch easily between **oxidation states** (Fe^{2+}/Fe^{3+}, Mn^{2+}/Mn^{3+}).
- The energy profile shows **two humps** (one per step), each lower than the single uncatalysed hump. ΔH is the same.

MARK-SCHEME LANGUAGE

- "same phase"; ACCEPT "same state"; "all species are in aqueous solution" (WCH15 Jun 2021).
- "the transition metal ion is oxidised / reduced to a different oxidation state then changes back" (WCH15 Oct 2020).

SPEC 17.31 Fe^{2+} catalysing the reaction of I^- with $S_2O_8^{2-}$

WHAT THIS SPEC POINT ASKS

Understand how Fe^{2+} ions catalyse the reaction between iodide and peroxodisulfate ions.

$S_2O_8^{2-}(aq) + 2I^-(aq) \rightarrow 2SO_4^{2-}(aq) + I_2(aq)$

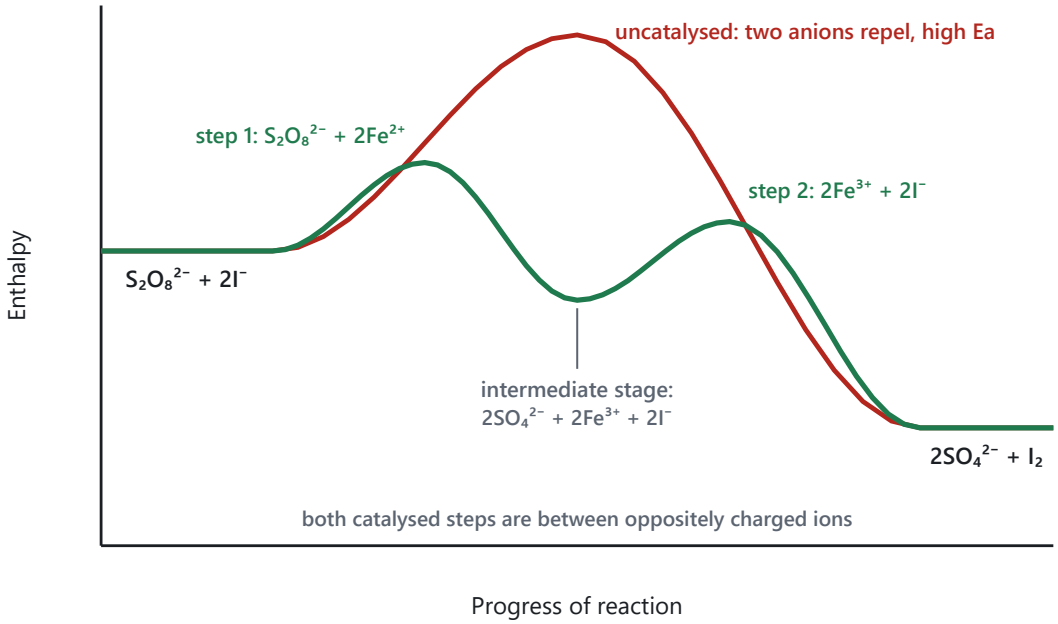
17.31 Why it is slow without a catalyst

E°cell = 2.01 − 0.54 = **+1.47 V**, so it is very feasible. But both reactants are **negative ions**: they **repel**, so the activation energy is high and the reaction is slow (16.12).

17.31 The catalysed route

Step 1: $S_2O_8^{2-} + 2Fe^{2+} \rightarrow 2SO_4^{2-} + 2Fe^{3+}$ E°cell = 2.01 − 0.77 = **+1.24 V**
Step 2: $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$ E°cell = 0.77 − 0.54 = **+0.23 V**

- Each step is between **oppositely charged ions**, which attract, so each has a lower activation energy.
- Fe^{2+} is regenerated in step 2. **Fe^{3+} works equally well**: it just starts with step 2.
- The method works because **E°(Fe^{3+}/Fe^{2+}) = +0.77 V lies between** E°(I_2/I^-) = +0.54 V and E°($S_2O_8^{2-}/SO_4^{2-}$) = +2.01 V, so both steps have positive E°cell.



The catalysed route has two lower humps. The same overall ΔH.

MARK-SCHEME LANGUAGE

- Equations: "$S_2O_8^{2-} + 2Fe^{2+} \rightarrow 2SO_4^{2-} + 2Fe^{3+}$" and "$2I^- + 2Fe^{3+} \rightarrow I_2 + 2Fe^{2+}$" (either order) (WCH15 Jun 2021).
- Why catalysed: "the uncatalysed reaction involves two negatively charged ions reacting, and the catalysed steps involve oppositely charged ions" (WCH15 Jun 2021); "reaction between two negative ions is slow due to repulsion" (WCH15 Oct 2023). DO NOT ACCEPT "negative molecules".
- "addition of either ion results in a reaction that forms the other ion, so either can act as the catalyst" (WCH15 Jan 2026).

Often asked as: two equations for the catalysed route (2 marks); why the uncatalysed reaction is slow; use E° values to show both steps are feasible; why Fe^{3+} also catalyses.

COMMON MISTAKE

Saying Fe^{2+} catalyses iodide with **thiosulfate** (it is **peroxodisulfate**, $S_2O_8^{2-}$) (WCH15 Oct 2020 examiner report).

BEYOND THE SPEC, BUT EXAMINED

Calculating $E°(S_2O_8^{2-}/SO_4^{2-})$ = +2.01 V from a cell measurement, then showing the uncatalysed reaction is feasible but slow (WCH15 Jan 2026). Choosing another possible catalyst: any couple with E° between +0.54 and +2.01 V could in principle work.

SPEC 17.32

Autocatalysis by Mn^{2+}

WHAT THIS SPEC POINT ASKS

Know that Mn^{2+} ions autocatalyse the reaction between MnO_4^- and $C_2O_4^{2-}$ ions.

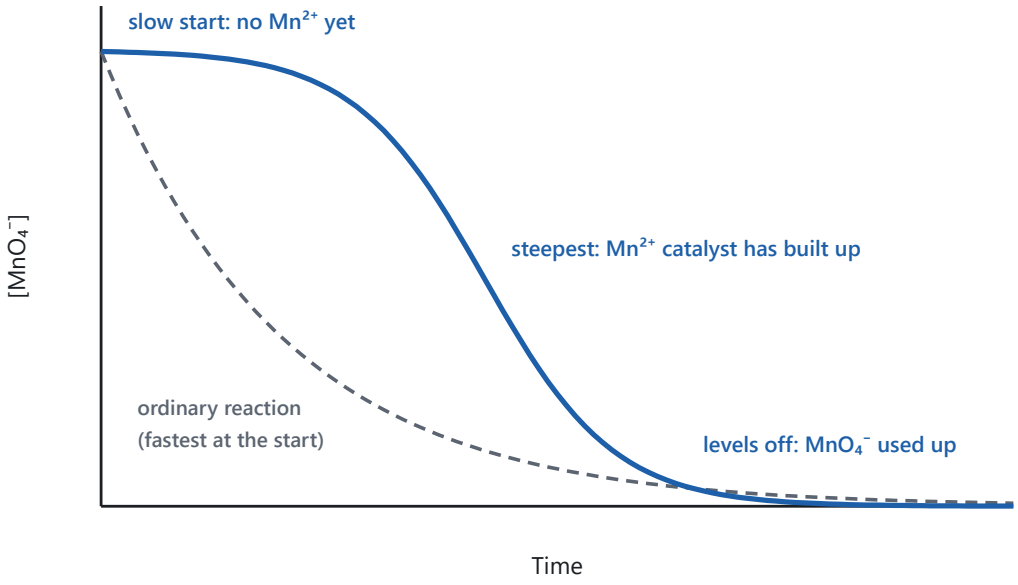
$2MnO_4^-(aq) + 5C_2O_4^{2-}(aq) + 16H^+(aq) \rightarrow 2Mn^{2+}(aq) + 10CO_2(g) + 8H_2O(l)$

- **Autocatalysis**: a **product** of the reaction **catalyses** the reaction. Here the product **Mn^{2+}** is the catalyst.
- **Start**: no Mn^{2+} yet, and both reactants are **anions** that repel, so the reaction is very slow. In a titration the first drops of purple MnO_4^- take a while to fade, even in the warm (≈ 60 °C) flask.
- **Middle**: as Mn^{2+} builds up, the rate **increases**. The purple colour now vanishes instantly.
- **End**: the rate falls as the reactants run out.

17.32 The catalytic cycle

1. $MnO_4^- + 4Mn^{2+} + 8H^+ \rightarrow 5Mn^{3+} + 4H_2O$ (Mn^{2+} is oxidised to Mn^{3+})
2. $2Mn^{3+} + C_2O_4^{2-} \rightarrow 2Mn^{2+} + 2CO_2$ (Mn^{3+} oxidises ethanedioate and is reduced back to Mn^{2+})

Each step is between oppositely charged ions.



An autocatalysed reaction gives an S-shaped (sigmoid) concentration–time curve. Follow it with a colorimeter, since MnO_4^- is the only coloured species.

MARK-SCHEME LANGUAGE

- "this reaction is (auto)catalysed by the Mn^{2+} ions formed"; "experiment 1 starts slowly because there is no Mn^{2+} / catalyst present initially"; "experiment 2 is fastest at the start as Mn^{2+} is already present" (WCH15 Jan 2023).
- Rate–time sketch: "(very) slow initial rate which increases as the catalyst / Mn^{2+} forms", "then decreases (and levels off) as the reactant concentration decreases". DO NOT ACCEPT "when all the reactants are used up the reaction starts to slow down" (WCH15 Oct 2025).
- Step equation: "$10Mn^{3+} + 5C_2O_4^{2-} \rightarrow 10Mn^{2+} + 10CO_2$" (any multiple) (WCH15 Oct 2025).

Often asked as: interpret concentration–time graphs with and without added Mn^{2+} (3 marks); sketch the rate–time curve; write the step equations.

COMMON MISTAKE

A stock definition of a catalyst with no reference to the graphs. Saying Mn^{2+} catalyses only one experiment. Treating dissolved $MnSO_4$ as a heterogeneous catalyst with a "surface area" (WCH15 Jan 2023 examiner report).

BEYOND THE SPEC, BUT EXAMINED

Rate-determining steps and orders in an unfamiliar catalytic cycle (rhodium catalysis in the Cativa process, WCH15 Jan 2026).

SPEC 17.33 Core Practical 14: preparing a transition-metal complex

WHAT THIS SPEC POINT ASKS

Prepare a transition-metal complex.

17.33 Aim and principle

The usual preparation is **tetraamminecopper(II) sulfate**, made by **ligand exchange** of water for ammonia:

$[Cu(H_2O)_6]^{2+}(aq) + 4NH_3(aq) \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+}(aq) + 4H_2O(l)$
$CuSO_4 \cdot 5H_2O + 4NH_3 \rightarrow [Cu(NH_3)_4]SO_4 \cdot H_2O + 4H_2O$

- The first drops of ammonia give a **pale blue precipitate** (deprotonation); excess gives a **deep blue solution** (ligand exchange).
- The complex is much **less soluble in ethanol** than in water, so adding ethanol makes it crystallise out.

CORE PRACTICAL 14: METHOD IN BRIEF

1. Weigh about 2.5 g of $CuSO_4 \cdot 5H_2O$ by difference and dissolve it in the **minimum** volume of distilled water (a few cm^3) in a small beaker.
2. **In a fume cupboard**, add **concentrated ammonia** solution (about 5 cm^3) with stirring: a pale blue precipitate, then a deep blue solution.
3. Slowly add about 8 cm^3 of **ethanol** and cool in an ice bath until deep blue crystals form.
4. Filter by **suction (Buchner) filtration**. **Wash** the crystals with a little **ice-cold ethanol**.
5. Dry between filter papers or in a desiccator. Weigh, and calculate the percentage yield.

17.33 Why each step

Step	Reason
minimum water	more of the product crystallises; less is lost in the filtrate
fume cupboard	concentrated NH_3 releases toxic, corrosive ammonia gas
measuring cylinders are fine	volumes are approximate and NH_3 is in excess
add ethanol	lowers the solubility of the complex
cold ethanol wash	removes soluble impurities without dissolving the product; ethanol evaporates quickly so the crystals dry fast

17.33 Safety

Concentrated ammonia: corrosive, irritating vapour (gloves, goggles, fume cupboard). Ethanol: highly flammable, no flames. Copper(II) sulfate: harmful and toxic to aquatic life, so don't pour it down the sink.

WORKED EXAMPLE: PERCENTAGE YIELD

2.40 g of $CuSO_4 \cdot 5H_2O$ ($M = 249.6\ g\ mol^{-1}$) gives 1.87 g of $[Cu(NH_3)_4]SO_4 \cdot H_2O$ ($M = 245.6\ g\ mol^{-1}$).
$n(CuSO_4 \cdot 5H_2O) = 2.40 \div 249.6 = 9.615 \times 10^{-3}$ mol = n(complex) (1 : 1)
theoretical mass = $9.615 \times 10^{-3} \times 245.6 = 2.361$ g
% yield = $1.87 \div 2.361 \times 100$ = **79.2%**
A "yield" above 100% means the crystals were still **damp** (or impure).

17.33 Errors and improvements

- Low yield: product left in solution (too much water, or the wash solvent not cold); losses on transfer and filtering.
- Impure product: wash properly with cold ethanol; recrystallise.
- Dry to **constant mass** before the final weighing.

MARK-SCHEME LANGUAGE

- "deep / dark blue solution" (not just "blue"); "less soluble in ethanol, so it crystallises"; "wash with cold ethanol to remove soluble impurities, cold so the product doesn't dissolve".
- The method itself has not been examined in WCH15 so far (it is assessed in Unit 6). WCH15 asks for **formula and hydrate calculations** on prepared salts instead: $FeCl_2 \cdot 4H_2O$, $K_2CuCl_4 \cdot 2H_2O$, $KV(SO_4)_2 \cdot 12H_2O$, $KCr(SO_4)_2 \cdot 12H_2O$, Reinecke's salt (WCH15 Oct 2021; Jan 2021; Jan 2023; Jan 2025; Jan 2024).

Often asked as: deduce the formula of a hydrated complex salt from % composition or titration data; observations on adding NH_3.

COMMON MISTAKE

Using atomic numbers instead of relative atomic masses in a formula calculation (WCH15 Oct 2021 examiner report).
Writing the product as $[Cu(NH_3)_6]^{2+}$.

BEYOND THE SPEC, BUT EXAMINED

Other complexes that can be prepared and analysed: potassium tris(ethanedioato)ferrate(III), $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$, and the chromium and vanadium alums. The water of crystallisation is found by heating to constant mass or from titration data (WCH15 Jan 2023; Jan 2025).

Topic 18 – Organic Chemistry: Arenes

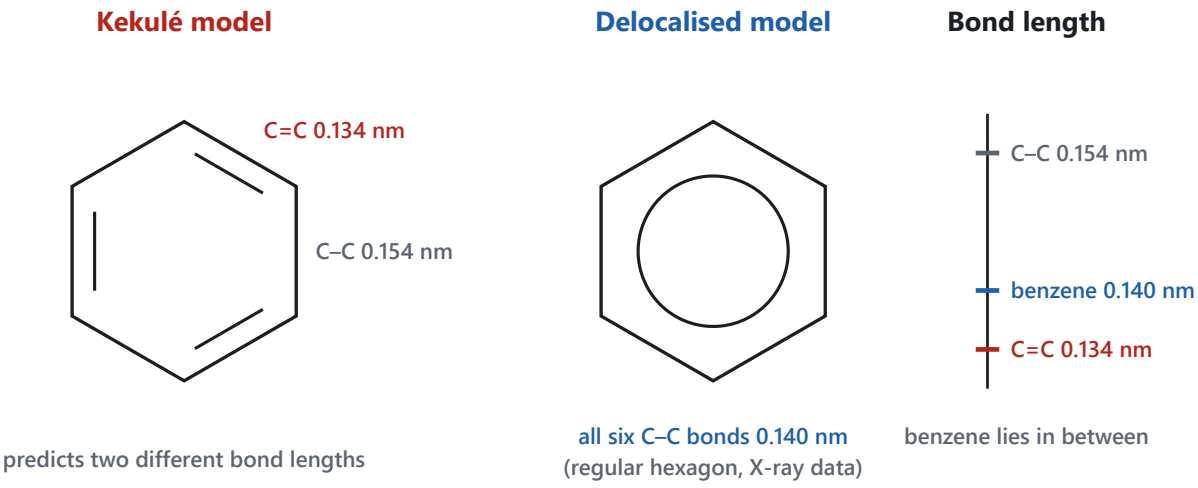
SPEC 18.1 Evidence for the structure and stability of benzene

WHAT THIS SPEC POINT ASKS

Use enthalpy (thermochemical) data, X-ray diffraction data and infrared data as evidence for the structure of the benzene ring and for its extra stability.

18.1 Two models of benzene

- Benzene is C_6H_6: six carbons in a flat ring, each carbon bonded to one hydrogen.
- **Kekulé model**: a ring of alternating C=C double and C–C single bonds (cyclohexa-1,3,5-triene). If this were right, benzene would behave like an alkene with three separate double bonds.
- **Delocalised model**: all six C–C bonds are identical. Six π electrons are spread over the whole ring (18.2). You can draw benzene either way in equations and mechanisms, but the evidence below supports the delocalised model.

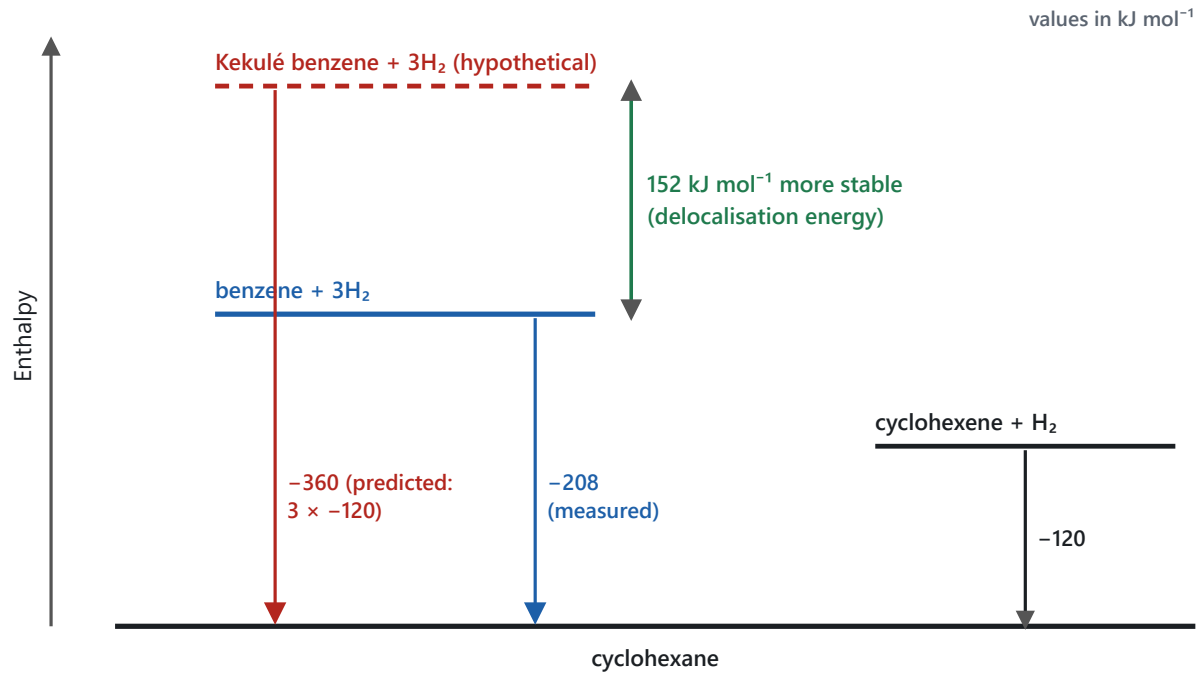


The Kekulé structure predicts two bond lengths. X-ray diffraction finds one: 0.140 nm, between C–C and C=C.

Every answer on this spec point has the same shape: **"Kekulé predicts X, the measurement shows Y, so the bonds are delocalised."**

18.1 Evidence 1: enthalpy change of hydrogenation

- Cyclohexene (one C=C) + H_2 → cyclohexane: ΔH = **−120 kJ mol^{-1}**.
- A Kekulé benzene has three C=C bonds, so it should give 3 × (−120) = **−360 kJ mol^{-1}**.
- The measured value for benzene is only **−208 kJ mol^{-1}**: **152 kJ mol^{-1} less exothermic** than predicted.
- So less energy is released, which means benzene started lower in energy: it is **more stable** than the Kekulé structure. This extra stability is the **delocalisation energy** (also called resonance energy or stabilisation energy).



Real benzene sits 152 kJ mol^{-1} below the hypothetical Kekulé structure, so it releases less energy when hydrogenated.

WORKED EXAMPLE: STABILISATION FROM HYDROGENATION DATA

Data: cyclohexene → cyclohexane, $\Delta H = -118$ kJ mol^{-1}; benzene → cyclohexane, $\Delta H = -205$ kJ mol^{-1}.
Kekulé prediction = 3 × (−118) = −354 kJ mol^{-1}.
Difference = −205 − (−354) = **+149 kJ mol^{-1}**.
Benzene's hydrogenation is 149 kJ mol^{-1} **less exothermic** than expected, so benzene is 149 kJ mol^{-1} more stable than the Kekulé structure, because its π electrons are delocalised.
Papers use slightly different data (−118/−205 or −120/−208). Always use the numbers in the question.

WORKED EXAMPLE: THE SAME IDEA FROM ENTHALPIES OF COMBUSTION

ΔcH: cyclohexene −3752, cyclohexa-1,4-diene −3584, benzene −3267 kJ mol^{-1}.

1. Each extra C=C makes ΔcH less negative, because there are two fewer H atoms to burn. Cyclohexene − diene: −3752 − (−3584) = **−168 kJ mol^{-1}**, so each extra C=C is worth 168 kJ mol^{-1}.
2. Predicted ΔcH for a Kekulé triene = −3584 − (−168) = **−3416 kJ mol^{-1}**.
3. Benzene's real value is −3267, so it releases 3416 − 3267 = **149 kJ mol^{-1} less** energy on combustion.
4. Benzene is 149 kJ mol^{-1} more stable than the Kekulé structure, because of delocalisation (WCH15 Jun 2021).

18.1 Evidence 2: X-ray diffraction (bond lengths)

- X-ray diffraction maps the electron density, so it gives bond lengths.
- Kekulé predicts **two** lengths: three short C=C (0.134 nm) and three long C–C (0.154 nm), giving an irregular hexagon.
- Benzene has **six equal** C–C bonds of **0.140 nm**: a regular hexagon. The length is **in between** a single and a double bond.

18.1 Evidence 3: infrared spectrum

- An alkene C=C stretch absorbs at about **1669–1645 cm^{-1}** (Data Booklet). Benzene shows **no** absorption there.
- Instead it has the arene peaks at about **1600, 1580, 1500 and 1450 cm^{-1}** and an arene C–H stretch near **3030 cm^{-1}**. These show a different kind of carbon–carbon bond from an alkene's.

18.1 Evidence 4: chemical behaviour

- Benzene does **not decolourise bromine water**, and it needs a catalyst to react with bromine at all. It reacts by **substitution**, not addition. An alkene with localised C=C bonds would add bromine at once (see 18.3).

MARK-SCHEME LANGUAGE

- Thermochemistry: Kekulé expected "3 × −118 = −354" (or 3 × −120 = −360); benzene is "149 / 152 kJ mol^{-1} less exothermic than expected" or "more stable than expected"; reason "the π electrons are delocalised". "Delocalisation energy" or "resonance stability" is accepted.
- Use comparative words: "less exothermic", "less negative", "more stable". "Higher" or "lower" on their own are not accepted.
- X-ray: "bond lengths in benzene are **equal**", "in between C=C and C–C" or "longer than C=C in cyclohexene"; and say what Kekulé predicts: "Kekulé structure would have shorter double bonds and longer single bonds". "Regular hexagon" is accepted; "bond lengths are similar" is not.
- IR: specific arene wavenumbers, and no C=C alkene peak.

Often asked as: the 6-mark "evidence for benzene's structure" (WCH15 Jun 2023), a hydrogenation or combustion calculation, explain X-ray data, MCQs on bond lengths.

COMMON MISTAKE

Comparing benzene's −208 with cyclohexene's −120. The comparison is with the **hypothetical Kekulé triene** (−360).
Drawing the enthalpy-level diagram as a reaction profile with a hump. It is a set of horizontal levels with vertical arrows.
Saying X-ray shows "the bonds are the same" without saying what Kekulé would show. The "Kekulé double bonds would be shorter" half-mark is the one most often missed (Jan 2024 report).

BEYOND THE SPEC, BUT EXAMINED

- **Isomer count**: a Kekulé structure would give **two** isomers of 1,2-dichlorobenzene (Cl atoms across a C=C or across a C–C). Only one exists (WCH15 Jan 2024).
- **Conjugated dienes**: cyclohexa-1,3-diene is only about 8 kJ mol^{-1} more stable than 2 × (−120) predicts, because its two C=C bonds are conjugated but not a closed ring. Benzene's continuous ring of delocalisation is much more effective (WCH15 Jan 2026).
- **Bond enthalpy cycle**: the mean aromatic C–C bond enthalpy (about 507 kJ mol^{-1}) comes out higher than the average of C–C and C=C (about 480), again because of delocalisation (WCH15 Jun 2025).
- **Dewar and Ladenburg benzene**: other proposed C_6H_6 structures. Bromine water or NMR shows they are not benzene: benzene's H atoms all appear at about 7.3 ppm (WCH15 Oct 2021).

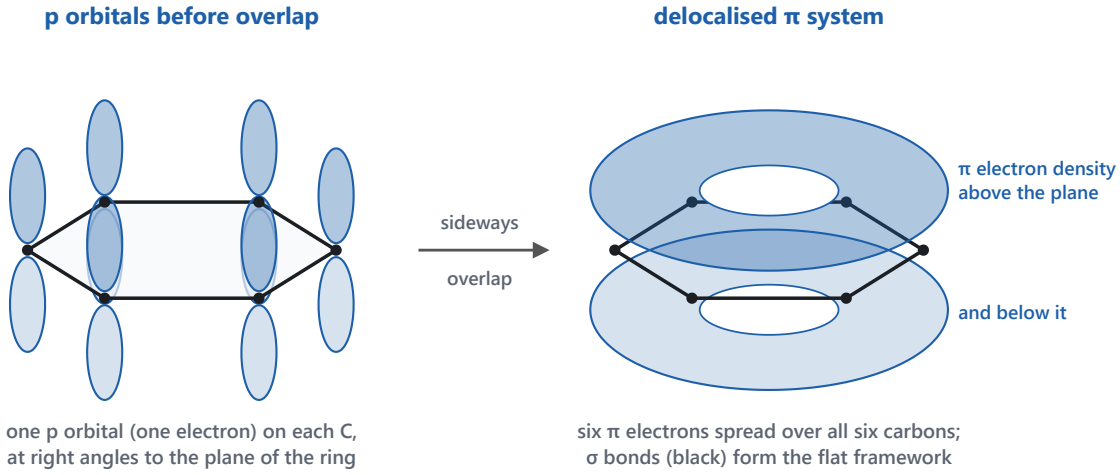
SPEC 18.2 The delocalised model: overlap of p orbitals

WHAT THIS SPEC POINT ASKS

Understand that the delocalised model of benzene involves sideways overlap of p orbitals to form π bonds.

18.2 Building the model

1. Each carbon uses three of its four outer electrons to form **σ bonds**: two to neighbouring carbons and one to a hydrogen. This gives a **flat** (planar) ring with bond angles of **120°**.
2. Each carbon has **one electron left in a p orbital**. The p orbital sticks out **above and below** the plane of the ring, at right angles to it.
3. Each p orbital **overlaps sideways** with the p orbitals on **both** neighbouring carbons, not just one.
4. The result is a continuous **ring of π electron density above and below the plane**, holding **six delocalised π electrons** shared by all six carbons.



Left: one p orbital on each carbon. Right: sideways overlap all round the ring gives a delocalised π system above and below the plane.

- Because every carbon shares in the same π system, all C–C bonds are the same length (0.140 nm).
- Delocalisation spreads the electrons over a bigger region, which lowers the energy. That is the source of the extra stability in 18.1.
- The circle in the hexagon stands for the six delocalised electrons.

EXAM TIP

In an alkene, the p orbitals on **two** carbons overlap to make one **localised** π bond. In benzene, the p orbitals on **six** carbons overlap all round the ring. Use that contrast whenever 18.2 and 18.3 come up together.

MARK-SCHEME LANGUAGE

- "six σ bonds between carbon atoms" (the flat framework), "six p (p_z) orbitals not involved in σ bonding", "overlap (continuously) **above and below** the ring" to form "a (large) π system". All marks can come from a labelled diagram.
- Hybridisation language ("sp^2 orbitals overlap head-on") is accepted although not on the spec. "π orbitals" is not accepted.

Often asked as: describe and explain the bonding (2–3 marks, WCH15 Jan 2024, Jun 2025); MCQs on sideways overlap and on counting bonds.

COMMON MISTAKE

Unlabelled diagrams, and leaving out the σ framework. Say what the σ bonds are as well as the π system (Jan 2024 report).
Saying "p orbitals overlap end-on". End-on overlap makes σ bonds; π bonds come from **sideways** overlap.

BEYOND THE SPEC, BUT EXAMINED

Counting bonds: benzene has **12 σ bonds** (6 C–C + 6 C–H). Drawn as Kekulé it has **3 π bonds** (WCH15 Jun 2022 MCQ).

SPEC 18.3

Why benzene resists bromination

WHAT THIS SPEC POINT ASKS

Explain why benzene reacts with bromine much less readily than an alkene does: delocalised π electrons in benzene, localised π electron density in an alkene.

18.3 Alkene: localised π bond

- In cyclohexene, the two π electrons sit **between just two carbons**, so the electron density there is **high**.
- When a Br_2 molecule approaches, this concentrated negative charge **repels the electrons in Br–Br** and **induces a dipole** (Brδ+–Brδ−).
- The π electrons attack the δ+ bromine. The alkene adds bromine at room temperature: **electrophilic addition**. Bromine water is decolourised at once.

18.3 Benzene: delocalised π system

- Benzene's six π electrons are spread over **six carbons**, so the electron density between any two carbons is **lower**.
- The ring **cannot polarise** Br_2 enough to start a reaction. Benzene doesn't react with bromine water.
- Addition would also **destroy the delocalised ring** and lose about 150 kJ mol^{-1} of stability. So the **activation energy is high** and benzene is **kinetically stable**.
- With a **halogen carrier** ($FeBr_3$) to make a much stronger electrophile, Br^+, benzene does react. It undergoes **electrophilic substitution**: an H atom is replaced and the stable ring is kept (18.4, 18.5).

KEY IDEA

Model answer (benzene v cyclohexene with bromine):

- **Both** react with electrophiles because both have π electrons.
- Cyclohexene has **localised** π electron density in one C=C bond, high enough to **polarise** Br_2. It reacts at room temperature with no catalyst by **electrophilic addition** (product 1,2-dibromocyclohexane).
- Benzene's π electrons are **delocalised** over the ring, so the electron density is **lower** and it cannot polarise Br_2. It needs a **catalyst** ($FeBr_3$) and reacts by **electrophilic substitution** (bromobenzene + HBr), because substitution **keeps the stable delocalised ring**.

MARK-SCHEME LANGUAGE

Indicative points from the 6-mark "cyclohexene, benzene and phenol with bromine" (WCH15 Oct 2022, Oct 2023, Jan 2025):

- "all are attacked by / react with electrophiles";
- "cyclohexene undergoes addition", "benzene and phenol undergo substitution";
- "cyclohexene and phenol react with (aqueous) bromine without a catalyst; benzene needs a catalyst";
- "benzene has delocalised electrons and is (kinetically) stable, so the reaction has a high activation energy";
- "cyclohexene has localised electron density in one π bond";
- phenol: "lone pair on the oxygen overlaps with the ring, increasing electron density" (18.6).

A similarity or a difference given for one compound only, with no comparison, loses the reasoning mark. "Nucleophilic" is never accepted.
Often asked as: a 6-mark compare-and-contrast, or a 4-mark "compare the reactions of benzene and cyclohexene with bromine" (WCH15 Jun 2021).

COMMON MISTAKE

"Benzene doesn't decolourise bromine, but cyclohexene does" is an observation, not an explanation. Give the electron-density reason too.
Leaving out the word **electrophilic** ("benzene undergoes substitution").
Writing cyclohexane or cyclohexa-1,3,5-triene when you mean cyclohexene. Giving dibromobenzene as the product of benzene.
Writing out full mechanisms in a 6-mark answer that didn't ask for them: it wastes time and any error costs a mark (Oct 2022 report).

BEYOND THE SPEC, BUT EXAMINED

Kinetic v thermodynamic stability: the delocalised system makes the activation energy for forming the intermediate high, so benzene is kinetically stable (WCH15 Jun 2021 MCQ). And **HBr** is a by-product of substitution, but not of addition (WCH15 Jan 2025).

SPEC 18.4 Reactions of benzene

WHAT THIS SPEC POINT ASKS

Know these reactions of benzene: (i) combustion in air (smoky flame); (ii) bromine with a catalyst; (iii) a mixture of concentrated nitric and sulfuric acids; (iv) fuming sulfuric acid; (v) halogenoalkanes and acyl chlorides with aluminium chloride (Friedel–Crafts). Know the common uses of the products.

18.4 (i) Combustion

$2C_6H_6(l) + 15O_2(g) \rightarrow 12CO_2(g) + 6H_2O(l)$

- Benzene burns with a **very smoky (sooty), yellow flame**.
- Its **C : H ratio is high** (1 : 1), so there isn't enough oxygen in air for complete combustion. Unburnt **carbon** (soot) forms.
- A sooty flame is a quick test for an **arene** in an unknown (WCH15 Jun 2022).

18.4 (ii) Halogenation (bromination)

$C_6H_6 + Br_2 \rightarrow C_6H_5Br + HBr$

- **Reagent**: bromine (liquid). **Catalyst** (halogen carrier): **$FeBr_3$** (or iron filings, which react with Br_2 to make $FeBr_3$), or $AlBr_3$. Room temperature, dry (anhydrous) conditions.
- **Observations**: the orange-brown colour fades; **steamy fumes of HBr** are given off.
- Chlorination works the same way with Cl_2 and $AlCl_3$ or $FeCl_3$, giving **chlorobenzene**.
- **Product: bromobenzene.** It is a route into Grignard chemistry (C_6H_5MgBr, 20.2).

18.4 (iii) Nitration

$C_6H_6 + HNO_3 \rightarrow C_6H_5NO_2 + H_2O$

- **Reagents**: **concentrated nitric acid** and **concentrated sulfuric acid** (the "nitrating mixture"). The sulfuric acid is a catalyst that generates the electrophile NO_2^+.
- **Conditions**: warm at about **50–55 °C**. Keep it **below about 55 °C** to stop further substitution (dinitrobenzene). Add the benzene slowly and cool the flask, because the reaction is exothermic.
- **Product: nitrobenzene**, a pale yellow oily liquid.
- **Uses**: reduced to **phenylamine** (19.5), which is used to make **dyes** (19.6) and drugs. Multiple nitration of methylbenzene gives the explosive **TNT**.

18.4 (iv) Sulfonation

$C_6H_6 + SO_3 \rightarrow C_6H_5SO_3H$ (or $C_6H_6 + H_2SO_4 \rightarrow C_6H_5SO_3H + H_2O$)

- **Reagent**: **fuming sulfuric acid** (oleum: concentrated H_2SO_4 with dissolved SO_3). Warm, about 40 °C. The electrophile is SO_3 (or HSO_3^+).
- **Product: benzenesulfonic acid**. The $–SO_3H$ group is joined to the ring through **sulfur**.
- **Uses**: alkylbenzenesulfonic acids (as their sodium salts) are **detergents**. Sulfonate groups also make **dyes soluble in water** (WCH15 Jun 2025).

18.4 (v) Friedel–Crafts reactions

Both need **anhydrous aluminium chloride**, $AlCl_3$, as catalyst (a halogen carrier). Heat under reflux in dry conditions, because water destroys $AlCl_3$.

	Alkylation	Acylation
Reagent	**halogenoalkane**, e.g. chloromethane CH_3Cl	**acyl chloride**, e.g. ethanoyl chloride CH_3COCl
Equation	$C_6H_6 + CH_3Cl \rightarrow C_6H_5CH_3 + HCl$	$C_6H_6 + CH_3COCl \rightarrow C_6H_5COCH_3 + HCl$
Product	**methylbenzene** (an alkylbenzene)	**phenylethanone** (an aromatic ketone)
Electrophile	CH_3^+ (a carbocation)	CH_3CO^+ (an acylium ion)

- Friedel–Crafts is the way to **make a new C–C bond to a benzene ring**.
- **Uses**: ethylbenzene (from chloroethane) is turned into phenylethene, the monomer for poly(phenylethene) (polystyrene). The ketone from acylation can be reduced to an alcohol with $NaBH_4$.

18.4 Summary table

Reaction	Reagents and conditions	Electrophile	Organic product	By-product
Bromination	Br_2, $FeBr_3$ catalyst, room temp.	Br^+	bromobenzene	HBr
Nitration	conc. HNO_3 + conc. H_2SO_4, 50–55 °C	NO_2^+	nitrobenzene	H_2O
Sulfonation	fuming H_2SO_4, warm	SO_3	benzenesulfonic acid	(none, with SO_3)
Alkylation	RCl + anhydrous $AlCl_3$, reflux	R^+	alkylbenzene	HCl
Acylation	RCOCl + anhydrous $AlCl_3$, reflux	RCO^+	phenyl ketone	HCl

All except combustion are **electrophilic substitutions**: one H on the ring is swapped for another group, and the delocalised ring survives.

MARK-SCHEME LANGUAGE

- Nitration: "**concentrated** sulfuric acid" (dilute is not accepted) and the electrophile "nitronium ion / NO_2^+" (WCH15 Jan 2024).
- Temperature for nitrating methylbenzene: "below 30 °C ... to prevent further substitutions" (methylbenzene's ring is more reactive than benzene's). "Reagents decompose" is ignored (WCH15 Jun 2022).
- Sulfonation: "(concentrated) sulfuric acid" or "fuming sulfuric acid / oleum / $H_2S_2O_7$"; the product has SO_3H bonded through **S** (WCH15 Jun 2024).
- Friedel–Crafts methylation: "CH_3Cl / chloromethane" + "$AlCl_3$" (CH_3Br, CH_3I, $AlBr_3$, $FeBr_3$, $FeCl_3$ accepted). Chloroethane for a methyl group, or "a halogenoalkane" with no name, is not accepted (WCH15 Jan 2025).
- Sooty flame "indicates an arene / high C : H ratio" (WCH15 Jun 2022).

Often asked as: reagents and conditions for a step in a synthesis, the product of a given reagent, name the electrophile, why the temperature is controlled.

COMMON MISTAKE

Leaving out "concentrated" for the sulfuric acid in nitration (Jan 2024 report).
Choosing chloroethane when the product needs a methyl group: count the carbons (Jan 2025 report).
Calling bromination "Friedel–Crafts". Friedel–Crafts means alkylation or acylation (Jun 2023 report).
Drawing the sulfonic acid group joined through O, or on the wrong carbon (Jun 2024 report).

BEYOND THE SPEC, BUT EXAMINED

- **Substituted rings (methylbenzene, methoxybenzene)**: they react faster than benzene and substitute mainly at the **2- and 4-positions**. The 4-product is often the major one because a group next to the substituent is **sterically hindered** (WCH15 Jan 2026, acylation of methoxybenzene).
- **Side-chain oxidation**: methylbenzene heated under reflux with potassium manganate(VII), then acidified, gives **benzoic acid** (WCH15 Jan 2023).
- **Benzoic acid from benzene**: $Br_2/FeBr_3$ → bromobenzene; Mg in dry ether → C_6H_5MgBr; CO_2 then dilute acid → C_6H_5COOH (WCH15 Jan 2022, Oct 2023).
- **Limiting reagent** in a Friedel–Crafts prep: methoxybenzene 5.25 ÷ 108 = 0.0486 mol against ethanoyl chloride 4.50 ÷ 78.5 = 0.0573 mol, so methoxybenzene is limiting (WCH15 Jan 2026).

SPEC 18.5

Electrophilic substitution mechanisms

WHAT THIS SPEC POINT ASKS

Understand the electrophilic substitution mechanism for halogenation, nitration and Friedel–Crafts reactions of benzene, including how each electrophile is generated.

18.5 The pattern: three stages

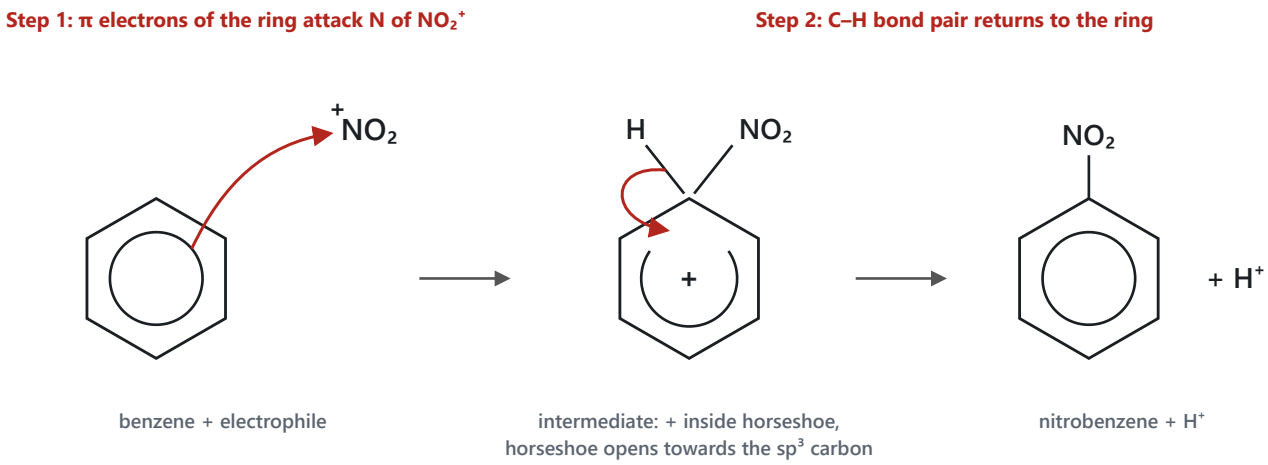
1. **Make the electrophile** (a positive ion, E^+) using the catalyst.
2. **Attack**: the π electrons of the ring attack E^+. A positive **intermediate** forms and the delocalised ring is partly broken.
3. **Restore the ring**: the C–H bond breaks, H^+ is lost, and the delocalised ring re-forms. The H^+ regenerates the catalyst.

18.5 Stage 1: generating each electrophile

Reaction	Equation for the electrophile	Catalyst regenerated by
Nitration	$HNO_3 + 2H_2SO_4 \rightarrow \mathbf{NO_2^+} + 2HSO_4^- + H_3O^+$	$H^+ + HSO_4^- \rightarrow H_2SO_4$
Bromination	$Br_2 + FeBr_3 \rightarrow \mathbf{Br^+} + FeBr_4^-$	$H^+ + FeBr_4^- \rightarrow FeBr_3 + HBr$
Alkylation	$CH_3Cl + AlCl_3 \rightarrow \mathbf{CH_3^+} + AlCl_4^-$	$H^+ + AlCl_4^- \rightarrow AlCl_3 + HCl$
Acylation	$CH_3COCl + AlCl_3 \rightarrow \mathbf{CH_3CO^+} + AlCl_4^-$	$H^+ + AlCl_4^- \rightarrow AlCl_3 + HCl$

- **Nitration**: sulfuric acid is the stronger acid, so it protonates nitric acid. The protonated nitric acid ($H_2NO_3^+$) loses water to give the **nitronium ion, NO_2^+**. A simpler version is also accepted: $HNO_3 + H_2SO_4 \rightarrow NO_2^+ + HSO_4^- + H_2O$.
- **Halogenation**: $FeBr_3$ (or $AlCl_3$) has an **incomplete outer shell**, so it accepts a lone pair from one Br atom. This polarises and breaks Br–Br, leaving Br^+. Writing it as Brδ+–Brδ−···$FeBr_3$ is also accepted.
- **Friedel–Crafts**: $AlCl_3$ takes the chlorine (with its bonding pair) from the C–Cl bond. That leaves a positive carbon: a **carbocation** R^+ or an **acylium ion** RCO^+ (the + is on the carbonyl carbon).

18.5 Stages 2 and 3: the mechanism with curly arrows



Nitration. For the other reactions, replace NO_2^+ with Br^+, CH_3^+ or CH_3CO^+. Everything else is the same.

Step 1: attack on the electrophile

1. Draw a curly arrow **starting inside the hexagon** (on or within the circle) and **ending at the positive atom** of the electrophile: the **N** of NO_2^+, the **Br** of Br^+, the **C** of CH_3^+ or CH_3CO^+.
2. Two of the π electrons form a new bond from one ring carbon to E.

The intermediate

3. The carbon now bonded to both **H and E** is **sp^3 (tetrahedral)**, so it is no longer part of the delocalised system. Draw both atoms on it.
4. The other five carbons share **four π electrons** and a **positive charge**. Draw a **horseshoe** (a partial circle) over these five carbons, with its **gap facing the tetrahedral carbon**, and put the **+ inside the horseshoe**.

Step 2: loss of H^+

5. Draw a curly arrow **from the C–H bond** (the middle of the bond, not the H atom) **back into the ring**. The bonding pair restores the full delocalised ring.
6. Products: the substituted arene (e.g. nitrobenzene) and **H^+**. The H^+ then regenerates the catalyst (table above).

18.5 Why substitution, not addition?

The intermediate could gain a negative ion (addition), but then the product would have lost its delocalised ring. **Losing H^+ restores the delocalised ring**, with its extra stability of about 150 kJ mol^{-1}, so substitution wins.

EXAM TIP

Mechanism checklist (each item is a marking point):

- Arrow 1 starts **inside the hexagon**, not on the edge or outside it.
- Arrow 1 ends on the **positive atom** (N of NO_2^+, C of CH_3^+ or CH_3CO^+), never on an O or on a lone pair.
- Electrophile shown with its **+ charge**.
- Horseshoe covers **at least three carbons** (ideally five), **open towards the sp^3 carbon**, **+ inside**.
- Both H and E on the sp^3 carbon.
- Arrow 2 starts on the **C–H bond** and goes into the ring. Use a **full (double-headed) arrow**: both electrons move.
- **H^+** as a product, then the catalyst-regeneration equation.

MARK-SCHEME LANGUAGE

- "curly arrow from on or within the circle / hexagon to the positively charged atom (N of NO_2^+; C of the acylium ion)";
- "intermediate ... horseshoe covering at least 3 carbon atoms and facing the tetrahedral carbon, with some part of the positive charge within the horseshoe";
- "curly arrow from C–H bond to anywhere in the ring reforming the delocalised structure";
- electrophile equation, e.g. "$CH_3COCl + AlCl_3 \rightarrow CH_3C^+O + AlCl_4^-$", and regeneration "$H^+ + AlCl_4^- \rightarrow AlCl_3 + HCl$".
- Not awarded: an arrow starting on or outside the hexagon; a missing + on the electrophile; an arrow to the O of NO_2^+; dotted C–H bonds; substituents on the wrong carbon.
- Error-spotting (WCH15 Jun 2022): the arrow drawn from CH_3^+ to the ring was "the wrong direction" (the ring is electron-rich, CH_3^+ is the electrophile with no lone pair). A single-headed arrow from the C–H bond is wrong because both electrons move (a half arrow means a radical). "Justify" needs the reason, not just the circled error.

Often asked as: the full 4–5 mark mechanism for nitration, bromination or acylation, nearly every series; find and fix the errors in a drawn mechanism; the equation for making NO_2^+.

COMMON MISTAKE

Starting arrow 1 from the electrophile, or aiming it at the O of NO_2^+ (Jan 2023, Jan 2025 reports).
The horseshoe's gap facing the wrong way, or the + drawn outside it, or left on the NO_2 group (Oct 2020, Jun 2024 reports).
Starting arrow 2 from the H atom instead of the C–H bond (Jan 2025 report).
Adding an extra carbon to the electrophile in Friedel–Crafts, e.g. writing $CH_3CH_2CO^+$ for ethanoyl chloride (Oct 2024 report).
Forgetting to show the catalyst being regenerated.

BEYOND THE SPEC, BUT EXAMINED

- **Unfamiliar electrophiles**: CO + HCl with $AlCl_3$ gives the **formyl cation HCO^+**, which adds a –CHO group to the ring (WCH15 Oct 2022). Treat any new electrophile exactly like NO_2^+: arrow to its positive atom.
- A **carbocation from an alkene** can also be the electrophile, e.g. when $AlCl_3$ and a chloro-acid are used (WCH15 Jan 2021).
- **Substituted rings**: put the substituent and E on the correct carbons in the intermediate. For phenol, the new group goes at position 2 or 4 (WCH15 Jan 2023).

SPEC 18.6

Phenol and bromine water

WHAT THIS SPEC POINT ASKS

Understand the reaction of phenol with bromine water, and why phenol reacts much more easily than benzene.

18.6 The reaction

$C_6H_5OH + 3Br_2 \rightarrow C_6H_2Br_3OH + 3HBr$

- **Conditions**: bromine **water**, **room temperature**, **no catalyst**.
- **Observations**: the bromine water **decolourises** (orange → colourless), and a **white precipitate** of **2,4,6-tribromophenol** forms. It smells of antiseptic.
- Three H atoms are substituted, at **positions 2, 4 and 6** (the carbons next to and opposite the –OH). Carbon 1 carries the OH.

18.6 Why phenol reacts more easily than benzene

1. One **lone pair on the oxygen atom** of the –OH group sits in a p orbital that **overlaps with the π system** of the ring.
2. The lone pair becomes **partly delocalised into the ring**, which **increases the electron density** of the ring (most at positions 2, 4 and 6).
3. The electron-rich ring can **polarise** Br_2 molecules by itself, so no halogen carrier is needed, and it **attracts electrophiles more strongly**.
4. So phenol undergoes **electrophilic substitution** much more readily than benzene, and substitutes three times.

	Benzene	Phenol
Reagent	Br_2 (liquid)	bromine water
Catalyst	$FeBr_3$ needed	none
Product	bromobenzene (one Br)	2,4,6-tribromophenol (three Br)
Observation	slow; HBr fumes	instant decolourisation, white precipitate

MARK-SCHEME LANGUAGE

- "lone pair of electrons on the **oxygen**"; "overlaps / interacts with the (π) electrons of the ring"; "**increasing** the electron density of the ring"; "so phenol is more susceptible to electrophilic attack" (WCH15 Jun 2024).
- Product: "2,4,6-tribromophenol" (Kekulé drawing accepted); equation "$C_6H_5OH + 3Br_2 \rightarrow C_6H_2Br_3OH + 3HBr$".
- Ignored: "increases the reactivity of the ring", "phenol is a nucleophile", "milder conditions" alone, "quicker / easier". Not accepted: "charge density increases", "the ring becomes more electronegative", "electrophilic addition".

Often asked as: name or draw the product (1–2 marks), explain the greater reactivity (2–3 marks), or the 6-mark comparison of cyclohexene, benzene and phenol (WCH15 Oct 2022, Oct 2023, Jan 2025).

COMMON MISTAKE

Write "a lone pair on the **oxygen atom**", not "a lone pair on the OH": one mark scheme refused that wording (WCH15 Jan 2025).
Stating "the lone pair" without saying **where** it is, or not linking the lone pair to **higher electron density in the ring** (Jan 2024, Jan 2023 reports).
Writing "long pair". Drawing the Br atoms at positions 2, 3 and 4.

BEYOND THE SPEC, BUT EXAMINED

- **Phenylamine** does the same thing with bromine water: the lone pair on N activates the ring, giving a white precipitate of 2,4,6-tribromophenylamine.
- **Nitration of phenol** needs only **dilute** nitric acid at room temperature and gives a mixture of **2-nitrophenol and 4-nitrophenol** (WCH15 Jan 2023). Draw the NO_2 on carbon 2 or 4, never on the carbon carrying the OH.
- **Methoxybenzene** ($C_6H_5OCH_3$) is activated in the same way by the O lone pair. The spec suggests it, or methylbenzene, for practical work instead of the more hazardous benzene.

Topic 19 – Organic Nitrogen Compounds: Amines, Amides, Amino Acids and Proteins

SPEC 19.1 Naming and drawing amines, amides and amino acids

WHAT THIS SPEC POINT ASKS

Name amides, amines and amino acids, and draw their structural, displayed and skeletal formulae.

19.1 Amines

- An amine is ammonia with one or more H atoms replaced by carbon groups.
- Classify by the **number of carbon groups on the nitrogen** (not by the carbon the N is on, as you do for alcohols):

Type	General formula	Example	Name
Primary (1°)	RNH_2	$CH_3CH_2CH_2CH_2NH_2$	butylamine (butan-1-amine)
Secondary (2°)	R_2NH	$(CH_3)_2NH$	dimethylamine
Tertiary (3°)	R_3N	$(CH_3)_3N$	trimethylamine
Quaternary ammonium ion	R_4N^+	$(CH_3)_4N^+$	tetramethylammonium ion

- **Naming**: either name the alkyl group + "amine" (methylamine, ethylamine, propylamine), or use the IUPAC suffix "-amine" with a locant: **propan-1-amine**, **propan-2-amine**, **butan-1-amine**.
- Groups on the N in secondary and tertiary amines get the locant **N**: $CH_3NHCH_2CH_3$ is **N-methylethylamine** (or N-methylethanamine).
- When a higher-priority group is present (e.g. –COOH), the NH_2 becomes the prefix **amino-**: 2-aminopropanoic acid.
- **Aliphatic** amines have N joined to an alkyl chain. **Aromatic** amines have N joined directly to a benzene ring: $C_6H_5NH_2$ is **phenylamine** (also called aniline or aminobenzene).

19.1 Amides

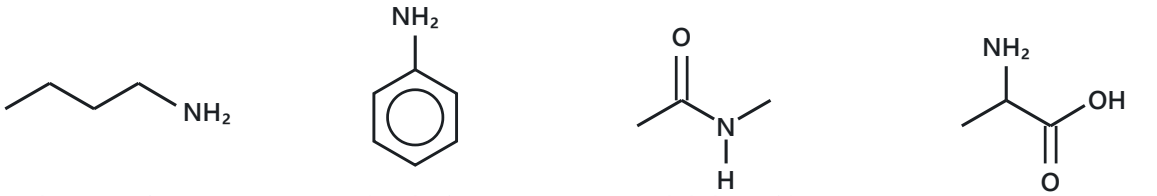
- Amides contain **–$CONH_2$** (or –CONHR, –$CONR_2$). The C=O carbon counts as carbon 1 of the chain.
- Name: stem + **"-amide"**. CH_3CONH_2 is **ethanamide**, $CH_3CH_2CONH_2$ is **propanamide**, $C_6H_5CONH_2$ is **benzamide**, CH_2=$CHCONH_2$ is **prop-2-enamide** (propenamide).
- Groups on the nitrogen use the **N-** locant: $CH_3CONHCH_3$ is **N-methylethanamide**; $CH_3CONHCH_2CH_2CH_2CH_3$ is **N-butylethanamide**; $CH_3CH_2CON(CH_3)_2$ is **N,N-dimethylpropanamide**. Two different groups go in alphabetical order (N-ethyl-N-methyl…).

19.1 Amino acids

- An amino acid has an **amine group (–NH_2)** and a **carboxylic acid group (–COOH)**.
- The natural ones are **2-amino acids**: both groups are on carbon 2 (the carbon next to COOH). General formula **RCH(NH_2)COOH**. About 20 of them make up proteins, differing only in the **R group**.
- Name them as substituted carboxylic acids (the acid takes priority):

Common name	R group	Systematic name
glycine	H	2-aminoethanoic acid (aminoethanoic acid)
alanine	CH_3	2-aminopropanoic acid
serine	CH_2OH	2-amino-3-hydroxypropanoic acid
phenylalanine	$CH_2C_6H_5$	2-amino-3-phenylpropanoic acid
aspartic acid	CH_2COOH	2-aminobutanedioic acid

19.1 Drawing them



butan-1-amine (butylamine) · phenylamine · N-methylethanamide · 2-aminopropanoic acid (alanine)

Skeletal formulae. Hydrogens on N and O are always drawn; hydrogens on C are not.

- **Structural**: $CH_3CH(NH_2)COOH$. **Displayed**: every atom and every bond drawn, including N–H and O–H. **Skeletal**: carbon chain as zigzag lines, with N, O and the **H atoms on N and O** written in.
- To get a **molecular formula from a skeletal formula**, count the carbons at each line end and corner. Then give each carbon enough H atoms to make four bonds.

MARK-SCHEME LANGUAGE

- Classify: "tertiary (amine) / 3°". "3rd / third" and "tertiary **amide**" are not accepted (WCH15 Jan 2022, Jun 2025).
- Names credited: "2-methylbutanamide" (WCH15 Jan 2021), "2-amino-3-methylpentanoic acid" (WCH15 Jun 2022), "2-amino-3-hydroxypropanoic acid" (WCH15 Oct 2024), "N-butylethanamide" (WCH15 Jun 2025), "prop-2-enamide / propenamide" (not "propanamide") (WCH15 Jun 2023).
- Functional groups in a drug molecule: "ester", "(primary) amine", "arene". Not "ketone" or "carboxylic acid" for an ester. "Carbonyl" is kept for aldehydes and ketones only.

Often asked as: name a compound from its skeletal formula; classify an amine; give the molecular formula from a skeletal one; identify functional groups.

COMMON MISTAKE

Classifying amines like alcohols (by the carbon). Count the groups on the **N**.
In skeletal formulae, leaving the H off N or O (–N– instead of –NH–), or an O–H joined to the wrong atom.
Naming an amino acid "aminopropanoic-2 acid" or putting the amino group last. The acid gives the suffix; amino is a prefix with a number.

BEYOND THE SPEC, BUT EXAMINED

IUPAC names of other multifunctional compounds turn up in the same questions: **2-hydroxybenzoic acid** (salicylic acid, WCH15 Oct 2023) and **propane-1,2,3-triol** (glycerol, WCH15 Jun 2024). The suffix names the main group, and everything else is a numbered prefix.

SPEC 19.2 Reactions of primary amines

WHAT THIS SPEC POINT ASKS

Understand the reactions of butylamine (aliphatic) and phenylamine (aromatic) with: (i) water, giving an alkaline solution; (ii) acids, giving salts; (iii) halogenoalkanes; (iv) ethanoyl chloride; (v) copper(II) ions, giving a complex ion.

The key is the **lone pair on the nitrogen**. It lets an amine act as a **base** (accepts H^+), a **nucleophile** (attacks δ+ carbon) and a **ligand** (bonds to a metal ion).

19.2 (i) With water: an alkaline solution

$CH_3CH_2CH_2CH_2NH_2(aq) + H_2O(l) \rightleftharpoons CH_3CH_2CH_2CH_2NH_3^+(aq) + OH^-(aq)$

- The N lone pair accepts H^+ from a water molecule, leaving **OH^- ions**, so the solution is **alkaline** (about pH 11–12). Butylamine is a **weak base**: the equilibrium lies to the left.
- Phenylamine is only slightly soluble and a much weaker base. Its solution is only **very weakly alkaline** (19.3).

19.2 (ii) With acids: salts

$CH_3CH_2CH_2CH_2NH_2 + HCl \rightarrow CH_3CH_2CH_2CH_2NH_3^+Cl^-$ (butylammonium chloride)
$C_6H_5NH_2 + HCl \rightarrow C_6H_5NH_3^+Cl^-$ (phenylammonium chloride)

- The salts are **ionic, white crystalline solids** that are **soluble in water**. Insoluble phenylamine **dissolves** in hydrochloric acid because it turns into the soluble salt.
- The H^+ joins the N by a **dative covalent bond** using the lone pair.
- Adding **NaOH(aq)** gets the amine back: $C_6H_5NH_3^+ + OH^- \rightarrow C_6H_5NH_2 + H_2O$. This step is used in 19.5.

19.2 (iii) With halogenoalkanes: further alkylation

$CH_3CH_2CH_2CH_2NH_2 + CH_3CH_2Br \rightarrow CH_3CH_2CH_2CH_2NHCH_2CH_3 + HBr$

- **Nucleophilic substitution**: the N lone pair attacks the δ+ carbon of C–Br (the same mechanism as NH_3 in Unit 2). The HBr formed reacts with more amine to give a salt.
- The product is itself an amine with a lone pair, so it can react again: **primary → secondary → tertiary amine → quaternary ammonium salt**. You get a **mixture**.
- With **excess halogenoalkane** the main product is the **quaternary ammonium salt**, e.g. $(C_4H_9)(C_2H_5)_3N^+Br^-$. Phenylamine reacts the same way, but more slowly.

19.2 (iv) With ethanoyl chloride: N-substituted amides

$CH_3COCl + CH_3CH_2CH_2CH_2NH_2 \rightarrow CH_3CONHCH_2CH_2CH_2CH_3 + HCl$ (**N-butylethanamide**)
$CH_3COCl + C_6H_5NH_2 \rightarrow CH_3CONHC_6H_5 + HCl$ (**N-phenylethanamide**)

- Vigorous at room temperature; **steamy fumes of HCl**. With excess amine, the HCl is taken up as a salt: $CH_3COCl + 2RNH_2 \rightarrow CH_3CONHR + RNH_3^+Cl^-$.
- **How it happens (addition–elimination)**:
  1. The carbonyl carbon is **δ+**: both O and Cl pull electrons away from it.
  2. The **lone pair on N** attacks this carbon, and the C=O π electrons move onto the O.
  3. The C=O re-forms and the **C–Cl bond breaks**, releasing Cl^- (a good leaving group).
  4. H^+ is lost from the N, giving the amide and HCl.

19.2 (v) With copper(II) ions: a complex

- Add butylamine solution drop by drop to $CuSO_4(aq)$:
  - A little amine: **pale blue precipitate** of copper(II) hydroxide. The amine acts as a base, making OH^-: $[Cu(H_2O)_6]^{2+} + 2RNH_2 \rightarrow [Cu(OH)_2(H_2O)_4] + 2RNH_3^+$
  - Excess amine: the precipitate **dissolves** to a **deep blue solution**. The amine acts as a ligand: $[Cu(H_2O)_6]^{2+} + 4RNH_2 \rightarrow [Cu(RNH_2)_4(H_2O)_2]^{2+} + 4H_2O$
- This is just like ammonia (Topic 17): N lone pairs form dative bonds to Cu^{2+}.
- With **phenylamine** the expected observation is the **pale blue precipitate** (WCH15 Oct 2020).

MARK-SCHEME LANGUAGE

- Amine in water: "the nitrogen has a lone pair", "which can accept a H^+ (from water)", "leaving (an excess of) hydroxide ions" (WCH15 Jan 2022, pyridine).
- Butylamine + ethanoyl chloride: the **displayed** formula of N-butylethanamide **and HCl** (WCH15 Oct 2024).
- Amide-formation explanation, any three: "carbonyl carbon is electron-deficient / δ+", "nitrogen has a lone pair", "which forms a bond to the carbonyl carbon", "C–Cl bond breaks / Cl is a good leaving group" (WCH15 Jun 2022). The name of the mechanism earns nothing on its own.
- Halogenoalkane mechanism (WCH15 Jan 2022): "arrow from lone pair on N to C", "dipole and arrow from C–Cl to Cl", "intermediate with + on N", "arrow from N–H bond to N^+". N^- is never awarded.
- Cu^{2+} + phenylamine: "(pale) blue precipitate"; blue-green is not accepted.
- Salt with HCl: "procaine hydrochloride is ionic", "the ions are hydrated by water", so it is more soluble (WCH15 Jan 2025).

Often asked as: write the equation (displayed or molecular formulae), explain why a solution is alkaline, give observations with Cu^{2+}, explain why a drug is sold as its hydrochloride.

COMMON MISTAKE

Leaving out the HCl in the ethanoyl chloride equation (Oct 2024 report).
Writing that the amine attacks a carbocation. In the acyl chloride reaction there is no carbocation: the N attacks the δ+ carbon of C=O (Jun 2022 report).
Mixing up the words base (takes H^+) and nucleophile (attacks δ+ carbon). Both use the lone pair, but they are different roles.
Putting the H^+ on the wrong N in a molecule with two nitrogens, or drawing the N–H^+ bond without a dative arrow when the question asks for one.

BEYOND THE SPEC, BUT EXAMINED

- **Tertiary and ring amines** (pyridine, trimethylamine) behave the same way: base in water, nucleophile towards chloromethane (WCH15 Jan 2022, Jun 2025).
- **Drugs as salts**: amines are converted into hydrochlorides because the ionic salts are **more soluble in water** (ion–dipole attractions with water) (WCH15 Jan 2025).

SPEC 19.3 Solubility of amines and their strength as bases

WHAT THIS SPEC POINT ASKS

Understand that amines mix with water because of hydrogen bonding, and explain why ammonia, primary aliphatic amines and primary aromatic amines differ in base strength.

19.3 Mixing with water

- Small amines (methylamine to butylamine) are **miscible** with water.
- Two kinds of **hydrogen bond** form:
  - the **lone pair on the amine N** with a δ+ H of water: N···H–O;
  - a δ+ **H on the amine N** with a lone pair on water's O: N–H···O.
- As the hydrocarbon chain gets longer, the non-polar part disrupts the hydrogen bonding, so **solubility falls**. Phenylamine, with its large benzene ring, is only **slightly soluble**.
- Primary amines boil higher than alkanes of similar Mr (they have hydrogen bonds between molecules), but lower than alcohols. N is less electronegative than O, so N–H···N hydrogen bonds are weaker than O–H···O.

EXAM TIP

Hydrogen-bond diagrams: draw the **lone pair** on the N (or O) that accepts the bond, label δ+ on H and δ− on N and O, and keep N···H–O in a **straight line** (about 180°). Use a dashed line for the hydrogen bond, never a solid line or a dative arrow.

19.3 Base strength

A base **accepts a proton (H^+)**, using the **lone pair on nitrogen**. The more available that lone pair is, the stronger the base.

Base	pKa of its conjugate acid	Strength	Reason
butylamine	about 10.6	strongest	alkyl group **releases electrons** (positive inductive effect), raising the electron density on N, so the lone pair is **more available**
ammonia	9.3	in between	reference: only H atoms on N
phenylamine	about 4.6	weakest	lone pair on N **overlaps with the delocalised π system** of the ring, so electron density on N is lower and the lone pair is **less available**

Order: butylamine > ammonia > phenylamine. (A higher pKa of the conjugate acid means a stronger base. You don't need to learn these numbers.)

KEY IDEA

Model answer (why phenylamine is a weaker base than ammonia):

1. Bases accept H^+ using the lone pair on nitrogen.
2. In phenylamine, the **lone pair on N overlaps with (is partly delocalised into) the π electrons of the benzene ring**.
3. So the lone pair is **less available** and nitrogen is **less able to accept H^+** than in ammonia.

MARK-SCHEME LANGUAGE

- Phenylamine: "lone pair on N overlaps with the π cloud / delocalised electrons / interacts with the ring", "so the N is less able to accept a H^+ / proton" (WCH15 Oct 2020). "Lone pair is less available to accept H^+ / form a dative bond" is accepted.
- Butylamine: "acts as a base by accepting a proton", "greater electron density on the nitrogen", "as a result of the alkyl group releasing electron density" (WCH15 Oct 2024). "Positive inductive effect" and "electron pushing" are accepted. "Nitrogen is more negative" and electronegativity arguments are not.
- Hydrogen bond: "lone pair (of electrons)" included; "dipoles on N, H and O"; "linear about the H" (170–190°) (WCH15 Oct 2024, Jun 2025).

Often asked as: rank three bases (MCQ), explain the order (2–3 marks), draw a hydrogen bond between an amine and water.

COMMON MISTAKE

Finishing at "the lone pair is delocalised" without the conclusion: **less able to accept H^+** (Oct 2020 report).
Saying the benzene ring makes the aromatic amine **more** likely to be protonated: the argument goes the other way (Jun 2023 report).
Not defining basicity. The first mark is often just "a base accepts a proton / H^+" (Oct 2024 report).
Hydrogen-bond diagrams with no lone pair, or with the bond bent.

BEYOND THE SPEC, BUT EXAMINED

- **Two nitrogens in one molecule** (e.g. the anaesthetic procaine): the tertiary **alkyl** amine N is the more basic one, and the **aromatic** amine N, whose lone pair is drawn into the ring, is less basic. Discuss **both** nitrogens (WCH15 Jun 2023, Jan 2025).
- **Amides are not basic**: the N lone pair is delocalised onto the C=O group.

SPEC 19.4

Making primary aliphatic amines

WHAT THIS SPEC POINT ASKS

Know the reagents and general conditions for making primary aliphatic amines (i) from halogenoalkanes and (ii) by reducing nitriles.

19.4 (i) From a halogenoalkane and ammonia

$CH_3CH_2CH_2CH_2Br + 2NH_3 \rightarrow CH_3CH_2CH_2CH_2NH_2 + NH_4Br$

- **Reagents and conditions**: **excess concentrated ammonia in ethanol**, heated in a **sealed tube** (under pressure).
- Why each one:
  - **Sealed tube**: ammonia is a gas and would escape from an open flask when heated.
  - **Ethanol** as solvent: in water, OH^- ions would compete and give some alcohol.
  - **Excess ammonia**: the primary amine product is also a nucleophile and can react with more halogenoalkane (19.2 iii). Lots of NH_3 makes it more likely that each halogenoalkane meets NH_3 instead. Even so, you get some secondary and tertiary amine.

19.4 (ii) By reducing a nitrile

$CH_3CH_2CH_2CN + 4[H] \rightarrow CH_3CH_2CH_2CH_2NH_2$

- **Reagent**: **lithium tetrahydridoaluminate(III), $LiAlH_4$, in dry ether**, then dilute acid. (Catalytic hydrogenation with H_2 and a nickel catalyst also works, but one mark scheme refused it, so $LiAlH_4$ is the safe answer.)
- The nitrile comes from a halogenoalkane: **KCN in ethanol, heat under reflux**. $CH_3CH_2CH_2Br + CN^- \rightarrow CH_3CH_2CH_2CN + Br^-$
- So the nitrile route **adds one carbon** to the chain: 1-bromopropane (3 C) → butanenitrile (4 C) → butylamine (4 C).
- It also gives **only the primary amine**, with no further alkylation.

Route	Reagents	Carbon chain	Product purity
Halogenoalkane + NH_3	excess NH_3, ethanol, heat, sealed tube	same length	mixture of amines
Nitrile reduction	$LiAlH_4$ in dry ether	one longer than the halogenoalkane used	primary amine only

MARK-SCHEME LANGUAGE

- "potassium cyanide and (aqueous) ethanol / ethanolic KCN" (NaCN accepted). KCN "in acid" or HCN for a halogenoalkane substitution is not accepted.
- "$LiAlH_4$ / lithium tetrahydridoaluminate(III) in (dry) ether". Oct 2020 also allowed H_2 + Ni or sodium and ethanol, but Jan 2023 refused H_2 + Ni/Pt/Pd.
- Nitrile **hydrolysis** (to an acid, not an amine): "aqueous acid / $H^+(aq)$" and "reflux". Concentrated acid and "oxidation" are not accepted.

Often asked as: reagents for a step in a multi-step synthesis (WCH15 Jan 2023, 5 marks: ethanolic KCN, then $LiAlH_4$, then ethanoyl chloride).

COMMON MISTAKE

Forgetting "ethanolic" for KCN (Jan 2023 report). With aqueous KCN you also get OH^- substitution. Using a reagent that removes or adds the wrong number of carbons. Count carbons at every step.

BEYOND THE SPEC, BUT EXAMINED

Nitrile hydrolysis: heating a nitrile under reflux with dilute HCl gives the **carboxylic acid**: $CH_3CN + HCl + 2H_2O \rightarrow CH_3COOH + NH_4Cl$. Alkaline hydrolysis gives the carboxylate ion, which must then be acidified (WCH15 Jun 2024, Jan 2026). Together with KCN, this is a way to add one carbon and make an acid.

SPEC 19.5 Making aromatic amines: reducing nitro compounds

WHAT THIS SPEC POINT ASKS

Know that aromatic amines are made by reducing aromatic nitro compounds with tin and concentrated hydrochloric acid.

$C_6H_5NO_2 + 6[H] \rightarrow C_6H_5NH_2 + 2H_2O$

19.5 Method

1. Heat **nitrobenzene** with **tin** and **concentrated hydrochloric acid** under **reflux**. Tin is the **reducing agent**: it is oxidised to tin(II)/tin(IV), and supplies the electrons. It is **not a catalyst**.
2. In the acid, the product is the **phenylammonium ion**, $C_6H_5NH_3^+$ (as a salt with chloride and tin ions).
3. Add **excess sodium hydroxide solution** to liberate the free amine: $C_6H_5NH_3^+ + OH^- \rightarrow C_6H_5NH_2 + H_2O$ (tin hydroxides also precipitate).
4. Separate the phenylamine by **steam distillation** (20.5), then purify.

- This is a **reduction** (O atoms removed, H atoms added).
- Overall route from benzene: **nitration** (conc. HNO_3/conc. H_2SO_4, 50–55 °C) → **reduction** (Sn/conc. HCl) → phenylamine.
- **Uses of phenylamine**: making **azo dyes** (19.6), and drugs and other chemicals.

MARK-SCHEME LANGUAGE

- "tin and (concentrated) hydrochloric acid" (Fe or Zn in place of Sn accepted; "heat / reflux" ignored).
- Not accepted: dilute acid, sulfuric or nitric acid, $LiAlH_4$, "Sn is a catalyst", or NaOH **in the same step** as the acid.
- Type of reaction: "reduction" (redox accepted; "hydrogenation" ignored).

Often asked as: reagents for the nitro → amine step (most series), name the type of reaction, write the equation using [H].

COMMON MISTAKE

Only about 40% of candidates knew this reagent in one series (Jan 2022 report). Learn it: **Sn + concentrated HCl**.
Suggesting $LiAlH_4$, or calling tin a catalyst, or listing NaOH with the acid as one step (Jun 2023 report).

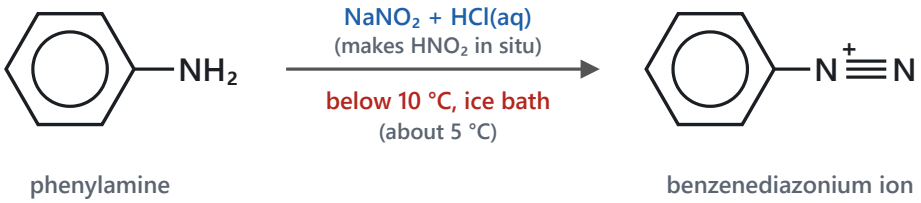
BEYOND THE SPEC, BUT EXAMINED

Tin and HCl reduce the NO_2 group but leave a **COOH** group alone, so 2-nitrobenzoic acid gives **2-aminobenzoic acid** (WCH15 Oct 2021). Making the mixture alkaline afterwards to free the amine is also credited (WCH15 Jun 2022 report).

SPEC 19.6 Diazotisation and azo dyes

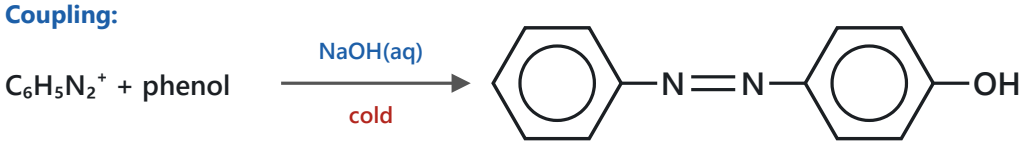
WHAT THIS SPEC POINT ASKS

Describe how an aromatic amine reacts with nitrous acid to give a benzenediazonium ion, and how this then couples with phenol to make a dye.



phenylamine — benzenediazonium ion

Coupling:



4-hydroxyazobenzene: yellow-orange dye (precipitate)
–N=N– is para (position 4) to the OH

Two steps, both kept cold: diazotisation, then coupling.

19.6 Step 1: diazotisation

- **Nitrous acid (HNO_2)** is unstable, so it is **made in the flask**: $NaNO_2 + HCl \rightarrow HNO_2 + NaCl$.
- Dissolve phenylamine in **excess hydrochloric acid**, cool in an **ice bath** to **below 10 °C (about 5 °C)**, and add cold **sodium nitrite (sodium nitrate(III)) solution** slowly.

$C_6H_5NH_2 + HNO_2 + H^+ \rightarrow C_6H_5N_2^+ + 2H_2O$

- The product is the **benzenediazonium ion**, $C_6H_5–N^+\equiv N$. The **+ is on the N joined to the ring**.
- **Why cold?** Above about 10 °C the diazonium ion **decomposes**: it reacts with water to give **phenol** and **nitrogen gas** (you would see bubbles).
  $C_6H_5N_2^+ + H_2O \rightarrow C_6H_5OH + N_2 + H^+$
  Below 0 °C the reaction is too slow and the solution may freeze, so don't go below 0 °C.

19.6 Step 2: coupling

- Add the cold diazonium solution to **phenol dissolved in sodium hydroxide solution** (alkaline), kept cold.
- The diazonium ion is an **electrophile**. It attacks the activated ring of phenol (as the phenoxide ion) at **position 4**, opposite the –OH: electrophilic substitution.

$C_6H_5N_2^+ + C_6H_5OH \rightarrow C_6H_5–N=N–C_6H_4OH + H^+$

- The product, **4-hydroxyazobenzene**, is a **yellow-orange** precipitate: an **azo dye**. The **–N=N–** link is the **azo group**.
- **Why coloured?** The N=N joins the two rings into one large **delocalised system**. This lets the molecule absorb visible light. It also makes the dye **stable**.
- Other coupling partners give other colours: naphthalen-2-ol gives a red dye; N,N-dimethylphenylamine gives methyl red (WCH15 Oct 2021).

MARK-SCHEME LANGUAGE

- Reagents: "sodium nitrite / sodium nitrate(III) / $NaNO_2$ **and** hydrochloric acid" (HNO_2 + HCl accepted). "Sodium nitrate" and "nitric acid" are not accepted.
- Temperature: "at 5 °C / between 0 and 10 °C" (ice bath accepted). "Below 0 °C" is not accepted (WCH15 Jun 2025).
- Why: "diazonium ion decomposes / reacts with water / forms phenol" (above 10 °C); "rate too slow" (below 0 °C). "Side reactions", "yield too low" and "freezes" are ignored (WCH15 Oct 2021).
- Diazonium ion: $ArN^+{\equiv}N$, with the + on the correct N; "N_2^+" is not accepted.
- Dye: ring–N=N–ring with the OH in the right place. "–N_2–" or N≡N between the rings is not accepted.
- Coupling conditions: "phenol dissolved in an alkaline solution" at about 5 °C (WCH15 Jan 2025 MCQ).

Often asked as: reagents and temperature (2 marks, most series); draw the diazonium ion or the dye; explain the temperature; the structure of a dye from an unfamiliar amine and coupling partner.

COMMON MISTAKE

Writing **nitrate** or nitric acid instead of **nitrite** (sodium nitrate(III)).
Drawing N≡N or –N_2– in the dye. The dye has a **double** bond, N=N.
Putting the OH next to the azo link when phenol couples. It ends up **para** (position 4); with 2-hydroxybenzoic acid the positions follow the stem (Jan 2022 report).

BEYOND THE SPEC, BUT EXAMINED

- **E/Z isomerism about N=N**: there is restricted rotation about the N=N double bond, so azo dyes have E and Z isomers (WCH15 Oct 2020).
- **Sulfonate groups** (–SO_3^- Na^+) are added to dyes such as sunset yellow to make them **soluble in water** (WCH15 Jun 2025).
- Dye **purification by recrystallisation** and percentage-yield questions (WCH15 Jun 2025).

SPEC 19.7

Making amides from acyl chlorides

WHAT THIS SPEC POINT ASKS

Understand that amides can be made from acyl chlorides.

- An acyl chloride reacts with **ammonia** to give an **amide**, or with a **primary amine** to give an **N-substituted amide**. Both happen quickly at **room temperature**, giving steamy fumes of HCl.

Reactants	Equation	Product
propanoyl chloride + ammonia	$CH_3CH_2COCl + NH_3 \rightarrow CH_3CH_2CONH_2 + HCl$	propanamide
with excess ammonia	$CH_3CH_2COCl + 2NH_3 \rightarrow CH_3CH_2CONH_2 + NH_4Cl$	propanamide
propanoyl chloride + methylamine	$CH_3CH_2COCl + CH_3NH_2 \rightarrow CH_3CH_2CONHCH_3 + HCl$	N-methylpropanamide
ethanoyl chloride + phenylamine	$CH_3COCl + C_6H_5NH_2 \rightarrow CH_3CONHC_6H_5 + HCl$	N-phenylethanamide

- With excess ammonia, the HCl reacts with ammonia to form **ammonium chloride**, a white solid.
- The mechanism is **addition–elimination** (19.2 iv): N lone pair → δ+ carbonyl carbon; C–Cl breaks; H^+ lost.
- To make an amide from a **carboxylic acid**, first convert the acid to the acyl chloride with **PCl_5** (or $SOCl_2$), then add NH_3. For example, benzoic acid → benzoyl chloride → **benzamide** (WCH15 Oct 2023).

- **Ethanoic anhydride** does the same job more slowly and gives ethanoic acid instead of HCl (see the aspirin practical, 20.4).

MARK-SCHEME LANGUAGE

- "ammonia / NH_3 added to an acyl chloride"; to get the acyl chloride, "phosphorus(V) chloride / PCl_5 (or thionyl chloride)". "Ammonia added to benzoic acid" is not accepted: it only makes the ammonium salt (WCH15 Oct 2023).
- Acylating agent: "ethanoyl chloride or ethanoic anhydride". "Ethanoic acid" is not accepted (WCH15 Jan 2023).

Often asked as: the last step of a synthesis route; the product of an acyl chloride with a given amine; the equation including HCl.

COMMON MISTAKE

Adding ammonia straight to a carboxylic acid to make an amide. That is an acid–base reaction: it gives an **ammonium salt** (Oct 2023 report).
Getting distracted by other functional groups in a big molecule. Find the C–Cl and the N–H that react (Oct 2021 report).

BEYOND THE SPEC, BUT EXAMINED

- **Amide hydrolysis**: amides (and peptide links) are broken by heating under reflux with **acid** (gives the carboxylic acid + ammonium ion) or **alkali** (gives the carboxylate ion + ammonia or amine). This is how proteins are broken into amino acids (19.11).
- **Choosing the acylating agent on safety and atom-economy grounds**: ethanoyl chloride v ethanoic anhydride (WCH15 Jan 2024; see 20.3).

SPEC 19.8

Condensation and addition polymerisation

WHAT THIS SPEC POINT ASKS

Describe (i) condensation polymerisation to make polyamides such as nylon, and proteins; (ii) addition polymerisation, including poly(propenamide) and poly(ethenol).

19.8 (i) Condensation polymers: polyamides

- A **condensation polymer** forms when monomers join and **a small molecule is lost at each link** (H_2O or HCl).
- Each monomer needs **two reactive groups**, one at each end, so the chain can keep growing.
- **Polyamides** contain the amide link **–CONH–**.

Nylon-6,6

- Monomers: **hexane-1,6-diamine** $H_2N(CH_2)_6NH_2$ and **hexanedioic acid** $HOOC(CH_2)_4COOH$. The "6,6" counts the carbon atoms in each monomer.
- Each –COOH reacts with an $–NH_2$, losing **H_2O** and forming an amide link:
  $n\ H_2N(CH_2)_6NH_2 + n\ HOOC(CH_2)_4COOH \rightarrow –[NH(CH_2)_6NHCO(CH_2)_4CO]_n– + 2n\ H_2O$
- In the lab, the **diacyl chloride** (hexanedioyl dichloride) is used instead of the acid. It reacts quickly at room temperature, losing **HCl**: the "nylon rope trick", where nylon is pulled from the boundary between the two solutions.

- **Nylon-6,10** uses decanedioyl dichloride (10 C) with hexane-1,6-diamine.
- **Kevlar**: benzene-1,4-diamine + benzene-1,4-dicarboxylic acid (or its dichloride). It is a polyamide with benzene rings in the chain.

Proteins

- Proteins are natural **polyamides**. Their monomers are **amino acids**, and each amino acid has both an $–NH_2$ and a –COOH, so **one type of monomer** is enough.
- The amide link between amino acids is called a **peptide bond (–CONH–)**. Each one forms with the loss of **H_2O** (19.11).

19.8 (ii) Addition polymers

- An **addition polymer** forms when the **C=C double bonds** of many monomers open and join. **Nothing is lost**: the polymer is the only product (100% atom economy).

Poly(propenamide)

- Monomer: **propenamide** (prop-2-enamide), $CH_2=CHCONH_2$.
- *n* $CH_2=CHCONH_2 \rightarrow –[CH_2–CH(CONH_2)]_n–$
- The many $–CONH_2$ groups **hydrogen-bond with water**, so the polymer **absorbs a lot of water** to form a gel. It is used in water treatment (to clump fine particles together), as a thickener and in gels for separating proteins and DNA. The monomer is toxic; the polymer is not.

Poly(ethenol) (polyvinyl alcohol)

- The "monomer" ethenol, $CH_2=CHOH$, **doesn't exist** in a usable form: it rearranges at once to ethanal.
- So poly(ethenol) is made in two steps:
  1. **Ethenyl ethanoate** ($CH_2=CHOOCCH_3$) is addition-polymerised to **poly(ethenyl ethanoate)**.
  2. The ester groups are swapped for –OH groups by reaction with **methanol** (a hydrolysis-type exchange). How many are swapped controls how soluble the product is (19.10).
- Repeat unit: $–[CH_2–CH(OH)]–$

	Addition	Condensation
Monomers	C=C in each monomer	two reactive groups per monomer (e.g. $–NH_2$ and –COOH)
By-product	none	small molecule (H_2O or HCl) at each link
Main chain	only carbon atoms	contains other atoms (N in polyamides, O in polyesters)
Examples	poly(propenamide), poly(ethenol), poly(ethene)	nylon-6,6, Kevlar, proteins

MARK-SCHEME LANGUAGE

- Condensation polymer: "a very large molecule / formed from a large number of monomers" **and** "a small molecule (H_2O / HCl) splits off" at each link. "Formed by addition" is not accepted (WCH15 Jun 2023).
- Nylon-6,6 monomers: "hexanedioyl dichloride" (or hexanedioic acid) and "hexane-1,6-diamine". Wrong connectivity (e.g. OH–C instead of C–OH) is penalised once.

Often asked as: identify the type of polymerisation or the monomers (MCQ); define a condensation polymer; draw the monomers of a given polymer.

COMMON MISTAKE

Drawing a monomer for an addition polymer **without the C=C**, or with methyl groups where the continuation bonds were (Jan 2022, Jun 2023 reports).
Giving the acid when the question's reaction loses HCl (it needs the acyl chloride), or the other way round.
Saying poly(ethenol) is made from ethenol.

BEYOND THE SPEC, BUT EXAMINED

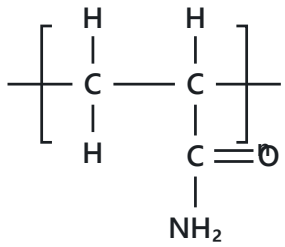
- **Nylon-6** is made by **ring-opening** polymerisation of caprolactam (a cyclic amide). The ring opens at the C–N bond. Repeat unit –[CO(CH$_2$)$_5$NH]– (WCH15 Jun 2023).
- **Hydrogels**: poly(propenamide) (PAM), poly(propenoic acid) (PAA) and PVP absorb water. PAA releases a drug above pH 8, because its –COOH groups become –COO$^-$ and the negative charges **repel** (WCH15 Jun 2023).

SPEC 19.9

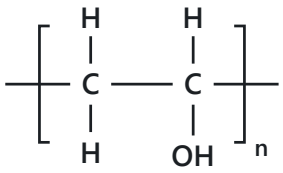
Repeat units of these polymers

WHAT THIS SPEC POINT ASKS

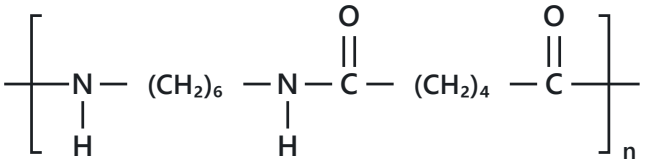
Draw the structural formulae of the repeat units of the polymers in 19.8.



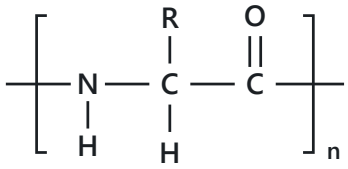
poly(propenamide)
monomer CH$_2$=CHCONH$_2$



poly(ethenol)
made from poly(ethenyl ethanoate)



nylon-6,6 (polyamide)



protein / polypeptide (R varies)

Repeat units. The continuation bonds go out through the brackets, and n follows the closing bracket.

Polymer	Repeat unit
poly(propenamide)	–[CH$_2$–CH(CONH$_2$)]–
poly(ethenol)	–[CH$_2$–CH(OH)]–
nylon-6,6	–[NH(CH$_2$)$_6$NH–CO(CH$_2$)$_4$CO]–
Kevlar	–[NH–C$_6$H$_4$–NH–CO–C$_6$H$_4$–CO]– (rings joined at positions 1 and 4)
polypeptide (general)	–[NH–CHR–CO]–
poly(glycine)	–[NH–CH$_2$–CO]–

19.9 Rules

- **Addition polymer**: the repeat unit is the monomer with its C=C turned into C–C and **two continuation bonds**, one from each of the two former C=C carbons. Substituents hang off the chain.
- **Condensation polymer from two monomers**: the repeat unit contains **one of each monomer** joined by one amide link, with an NH at one end and a CO at the other.
- **Continuation bonds** cross the brackets. Don't put CH_3 or H at the ends.

19.9 Working backwards: monomers from a polymer

- **Addition**: take the two main-chain carbons of the repeat unit, put a **C=C** between them, and remove the continuation bonds.
- **Polyamide**: break each **C–N bond of the amide link**. Add **OH** to the C=O end (acid) or **Cl** (acyl chloride), and **H** to the N end (amine).

WORKED EXAMPLE: DEGREE OF POLYMERISATION

A sample of nylon-6,6 has a mean relative molecular mass of 45 000. How many repeat units are there in an average chain?
Repeat unit: $C_{12}H_{22}N_2O_2$.
Mr = (12 × 12.0) + (22 × 1.0) + (2 × 14.0) + (2 × 16.0) = 144 + 22 + 28 + 32 = **226**.
Number of repeat units = 45 000 ÷ 226 = **about 200**.

WORKED EXAMPLE: PERCENTAGE BY MASS OF OH IN POLY(ETHENOL)

Repeat unit $CH_2CH(OH)$ = C_2H_4O, Mr = 44.0. Mass of OH in it = 17.0.
% OH = 17.0 ÷ 44.0 × 100 = **38.6%**.
The answer is the same for any number of units: for three units, 51 ÷ 132 = 38.6% (WCH15 Jun 2024).

MARK-SCHEME LANGUAGE

- Repeat unit with "continuation bonds" and the amide link correct, e.g. "–$CO(CH_2)_5NHCO(CH_2)_5NH$–" for two units of nylon-6.
- Number of repeat units = polymer Mr ÷ repeat-unit Mr (e.g. 300 000 ÷ 158 = 1900, WCH15 Jan 2022). Number of molecules = moles × 6.02×10^{23} (WCH15 Jun 2023).

Often asked as: draw a repeat unit, give the monomers, calculate the repeat-unit Mr and the number of units.

COMMON MISTAKE

Miscounting the monomer's Mr (extra carbons at the continuation bonds; Jun 2023 report).
Missing the H on the N in a skeletal or displayed polyamide.
Joining a cyclic monomer through the wrong atom (e.g. caprolactam rings joined through the N; Jun 2023 report).

SPEC 19.10 Physical properties of polyamides and poly(ethenol)

WHAT THIS SPEC POINT ASKS

Comment on the physical properties of polyamides, and on the solubility of poly(ethenol) in water, in terms of hydrogen bonding. Include soluble laundry bags and liquid-detergent capsules.

19.10 Polyamides

- The **N–H** of one chain forms **hydrogen bonds** with the **C=O** of a neighbouring chain. There are also London forces along the chains.
- These interchain forces are **much stronger than the London forces alone** between poly(alkene) chains. So polyamides have **higher melting temperatures** and **higher tensile strength**, and make good **fibres** (clothing, ropes, carpets).
- **Kevlar**: flat, rigid chains (benzene rings in the backbone) line up regularly, with many hydrogen bonds between them. This makes it very strong and heat-resistant: used in **bullet-proof vests**, protective gloves and helmets.
- Polyamides can be **hydrolysed** at the amide links by acids or alkalis, which weakens them.

19.10 Poly(ethenol)

- Every second carbon carries an **–OH group**. These form **hydrogen bonds with water** molecules, so poly(ethenol) **dissolves in water**.
- Solubility depends on **how many OH groups** there are:
  - Leaving some ester groups in place (partial hydrolysis) gives a polymer that dissolves in cold or warm water.
  - Almost fully converted poly(ethenol) has so many hydrogen bonds **between its own chains** that it dissolves only in **hot** water.
- **Uses**:
  - **Soluble laundry bags** in hospitals: dirty linen goes into the washing machine still in the bag, which dissolves in the wash. Staff don't handle the linen, so there is **no risk of cross-contamination** (infection).
  - **Liquid-detergent capsules** (liquitabs): the film holds the liquid, which contains very little water, and dissolves when it meets the wash water.
  - Others: dissolving stitches and water-soluble packaging.

MARK-SCHEME LANGUAGE

- Polyamide v polyalkene melting point: "hydrogen bonding (and London forces) between polyamide chains is stronger than the London forces between polyalkene chains, so more energy is needed to separate the chains". Any mention of covalent bonds breaking scores 0. Hydrogen bonding "to water" is not accepted here (WCH15 Oct 2023).
- Poly(ethenol): "OH groups can form hydrogen bonds with water", "high percentage of OH groups, so soluble" (WCH15 Jun 2024).
- PAM absorbs water: "can form many hydrogen bonds with water", "at the NH_2 and the C=O" (both sites), plus a correct hydrogen-bond diagram (WCH15 Jun 2023).
- Laundry bags: "no risk of cross-contamination", "no need to wash (and dry) the bags". "Biodegradable", cost and disposal are ignored.

Often asked as: explain a melting temperature or a solubility; explain the advantage of soluble laundry bags; draw hydrogen bonds between a polymer and water.

COMMON MISTAKE

Saying the amide **bond** is strong. Melting a polymer separates chains: it breaks **intermolecular** forces, not covalent bonds (Oct 2023 report).
Writing "biodegradable" as the advantage of laundry bags (Jun 2024 report).
Showing hydrogen bonds at only one site of the $–CONH_2$ group (Jun 2023 report).

BEYOND THE SPEC, BUT EXAMINED

Plasticisers: glycerol (propane-1,2,3-triol) is added to poly(ethenol) film to make it **more flexible**. "Increases solubility" is not accepted (WCH15 Jun 2024). That paper also included a calculation of polymer density and plasticiser percentage.

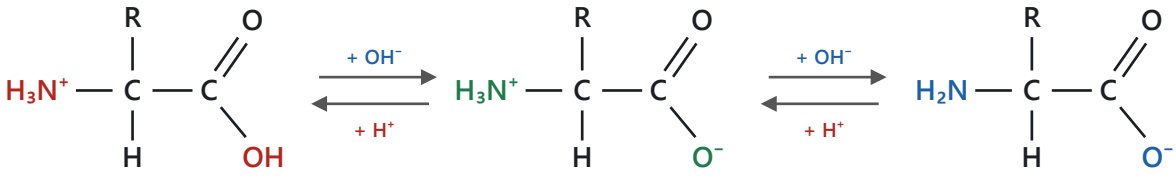
SPEC 19.11 Investigating the behaviour of amino acids

WHAT THIS SPEC POINT ASKS

Describe experiments showing (i) the acid–base behaviour of amino acids and zwitterion formation; (ii) the effect of their aqueous solutions on plane-polarised monochromatic light; (iii) peptide bond formation by condensation.

19.11 (i) Acid, base and zwitterion

- An amino acid has a basic group ($–NH_2$) and an acidic group (–COOH) in the same molecule. The –COOH can pass its H^+ to the $–NH_2$ **inside the molecule**, giving a **zwitterion**: **$^+H_3N–CHR–COO^-$**. It has both charges but **no overall charge**.
- In the **solid**, amino acids exist as zwitterions. The ions attract each other strongly (like an ionic lattice), so amino acids:
  - have **high melting temperatures** (often over 200 °C; many decompose instead of melting);
  - **dissolve in water** but not in non-polar solvents.
- Amino acids are **amphoteric**: they react with both acids and bases.



The form depends on pH. Adding acid protonates the COO^-; adding alkali removes H^+ from the NH_3^+.

Conditions	Form	Formula (glycine)
Acid (low pH)	**cation**	$^+H_3NCH_2COOH$
Isoelectric point (and the solid)	**zwitterion**	$^+H_3NCH_2COO^-$
Alkali (high pH)	**anion**	$H_2NCH_2COO^-$

Equations (glycine):

- with acid: $^+H_3NCH_2COO^- + H^+ \rightarrow {}^+H_3NCH_2COOH$
- with alkali: $^+H_3NCH_2COO^- + OH^- \rightarrow H_2NCH_2COO^- + H_2O$

Experiments

- Measure the pH of a glycine solution with a **pH meter**: close to neutral (about 6), because the zwitterion has both an acidic group and a basic group.

- Add a few drops of dilute HCl, then of dilute NaOH, to glycine solution, and to the same volume of water. The pH of the glycine solution changes **much less**, because the zwitterion **reacts with both H^+ and OH^-** (it acts as a buffer).
- Compare melting temperatures, or solubility in water and in hexane, with a similar-sized molecule that isn't ionic.

19.11 (ii) Plane-polarised light (optical activity)

- Every amino acid except **glycine** has a **chiral carbon**: carbon 2 is bonded to **four different groups** (H, NH_2, COOH and R). So it has two **optical isomers** (enantiomers): non-superimposable mirror images.
- A solution of **one** enantiomer **rotates the plane of plane-polarised light**. Its mirror image rotates it by the **same angle** in the **opposite direction**.
- **Experiment (polarimeter)**:
  1. Shine **monochromatic light** (one wavelength, e.g. a sodium lamp) through a **polariser**, giving plane-polarised light.
  2. Pass it through a tube of the amino acid solution.
  3. Turn an **analyser** (a second polariser) until the light is brightest (or darkest). Compare the angle with a tube of pure water.
  4. The difference is the **angle of rotation**: clockwise (+) or anticlockwise (−).
- Monochromatic light is used because the angle of rotation depends on the wavelength.
- **Results**: natural amino acids (from proteins) are **single enantiomers**, so they rotate the light. **Glycine** has no chiral carbon, so no rotation. An amino acid made in the lab is usually a **racemic mixture** (equal amounts of both enantiomers), so the rotations cancel and there is **no overall rotation**.

19.11 (iii) Peptide bonds

- The –COOH of one amino acid reacts with the $–NH_2$ of another. **Water is lost** and a **peptide bond, –CONH–**, forms: a **condensation** reaction.

$H_2NCH_2COOH + H_2NCH(CH_3)COOH \rightarrow H_2NCH_2CONHCH(CH_3)COOH + H_2O$

- Two **different** amino acids can join in **two ways** (Gly–Ala and Ala–Gly), so there are two possible dipeptides. Each still has a free $–NH_2$ at one end and a free –COOH at the other, so more amino acids can join: **polypeptides** and **proteins**.
- **Experiment (reverse direction)**: **hydrolyse** a protein or dipeptide by heating under reflux with **6 mol dm^{-3} HCl** for several hours. This breaks the peptide bonds. Then identify the amino acids by **paper or thin-layer chromatography**: spray with **ninhydrin**, which shows the spots as purple, and compare the **R_f values** with known amino acids.
- On hydrolysis, H goes back on the N and OH goes back on the C=O at each peptide bond.

WORKED EXAMPLE: HYDROLYSIS AND NUMBER OF MOLECULES (ASPARTAME)

Aspartame ($C_{14}H_{18}N_2O_5$, Mr 294) is the methyl ester of a dipeptide of aspartic acid and phenylalanine. Full hydrolysis breaks the peptide bond **and** the ester link. Products: **2-aminobutanedioic acid** (aspartic acid), **2-amino-3-phenylpropanoic acid** (phenylalanine) and **methanol**.
A drink contains 200 mg of aspartame. How many molecules?
Moles = 0.200 g ÷ 294 g mol^{-1} = 6.80×10^{-4} mol.
Molecules = $6.80 \times 10^{-4} \times 6.02 \times 10^{23}$ = **4.10×10^{20}**.
Convert mg to g first: this conversion is the usual slip (WCH15 Jan 2023).

MARK-SCHEME LANGUAGE

- Zwitterion: "+ anywhere on the NH_3 group"; carboxylate drawn as COO^- (or with the charge delocalised). Leaving the amine as NH_2 loses the mark.
- Explaining zwitterion formation: "amine group can accept a proton / become $–NH_3^+$" and "carboxylic acid group can lose a proton / become $–CO_2^-$" (WCH15 Oct 2025).
- Chiral carbon: "attached to four different **groups**" ("four different molecules" is not accepted); a mirror-image pair of 3D diagrams scores both marks.
- Hydrolysis products: drawn with no trailing bonds; NH_3^+ accepted for NH_2 (WCH15 Jan 2023).

Often asked as: draw the zwitterion, cation or anion; draw a dipeptide and circle the peptide bond; identify chiral carbons; give the hydrolysis products; MCQs on pH and form.

COMMON MISTAKE

Drawing the "zwitterion" with an unchanged NH_2 (Jan 2023 report).
Reading a dipeptide as a polymer repeat unit, or cutting hydrolysis products into fragments with dangling bonds (Jan 2023 report).
Saying a chiral molecule has cis–trans isomers, or that a C=C carbon can be chiral (Oct 2022 report).
Saying the racemic mixture "doesn't rotate light because it isn't chiral". Each molecule is chiral; the **equal amounts** of opposite rotations cancel.

BEYOND THE SPEC, BUT EXAMINED

- **Chromatography**: R_f = distance moved by the spot ÷ distance moved by the solvent, both measured **from the baseline** (WCH15 Jan 2023).
- **Optical isomerism outside amino acids** (coniine, gingerol, metal complexes): the same "four different groups" rule. Wedge and dash bonds show the 3D arrangement (WCH15 Oct 2022, Jan 2024).

SPEC 19.12

Core Practical 15: analysis of inorganic and organic unknowns

WHAT THIS SPEC POINT ASKS

Core Practical 15: identify unknown inorganic and organic substances using test-tube reactions.

19.12 How to approach it

- Work from **observations** to **conclusions**: each test either confirms or rules out a group or ion.
- Record **exactly what you see**, with colours **before and after** ("orange solution turns green"), and whether a gas, precipitate or solution forms.
- Combine evidence: one test rarely proves a structure. Data such as formula and spectra (20.1) finish the job.

CORE PRACTICAL 15: APPROACH

1. **Cations**: add NaOH(aq) drop by drop, then in excess; repeat with NH_3(aq). Note precipitate colours and whether they dissolve in excess (full colour table in 17.22). Flame tests for Group 1 and 2 ions.
2. **Anions**: test for carbonate, sulfate and halides (table below).
3. **Organic**: test for each functional group in turn (table below), using small samples and a fresh sample for each test.

4. Warm only with a **water bath** (most organic liquids are flammable). Work in a fume cupboard for volatile, toxic or corrosive reagents, e.g. PCl_5 and bromine.

19.12 Key inorganic tests

Ion	Test	Positive result
CO_3^{2-}	dilute acid; gas through limewater	fizzing; limewater turns cloudy (CO_2)
SO_4^{2-}	acidify with HCl, add $BaCl_2$(aq)	white precipitate ($BaSO_4$)
Cl^- / Br^- / I^-	acidify with HNO_3, add $AgNO_3$(aq)	white / cream / yellow precipitate. Then dilute NH_3 dissolves AgCl, conc. NH_3 dissolves AgBr, AgI doesn't dissolve
NH_4^+	warm with NaOH(aq)	gas turns damp red litmus blue (NH_3)
Cu^{2+}	NaOH(aq); NH_3(aq)	blue precipitate; with excess NH_3, deep blue solution
Fe^{2+}	NaOH(aq)	green precipitate, turns orange-brown at the surface in air
Fe^{3+}	NaOH(aq)	orange-brown precipitate
Cr^{3+}	NaOH(aq)	grey-green precipitate; dissolves in excess to a dark green solution

19.12 Key organic tests

Group	Test	Positive result
C=C (alkene)	bromine water, shake	orange → **colourless**
arene	burn a small sample	**very smoky (sooty)** flame
–OH (alcohol, acid)	PCl_5 (dry)	**steamy fumes** of HCl
1° or 2° alcohol, aldehyde	acidified $K_2Cr_2O_7$, warm	**orange** → **green** (no change with 3° alcohols or ketones)
C=O (aldehyde or ketone)	2,4-dinitrophenylhydrazine	**orange (or yellow) precipitate**
aldehyde (not ketone)	Tollens' reagent, warm	**silver mirror**
aldehyde (not ketone)	Fehling's or Benedict's solution, warm	blue solution → **red precipitate**
CH_3CO– or $CH_3CH(OH)$–	iodine + NaOH(aq), warm (iodoform)	**pale yellow precipitate** of CHI_3 (antiseptic smell)
–COOH	Na_2CO_3 or $NaHCO_3$ solution	**fizzing**; CO_2 turns limewater cloudy
halogenoalkane	warm with NaOH(aq), acidify with HNO_3, add $AgNO_3$	white / cream / yellow precipitate shows Cl / Br / I
phenol	bromine water	decolourises; **white precipitate**
amine	damp red litmus; smell; Cu^{2+}(aq)	litmus blue; fishy smell; pale blue precipitate, then deep blue solution with excess (butylamine)

MARK-SCHEME LANGUAGE

- Unknown deduction (WCH15 Jun 2022, 6 marks): "sooty flame indicates benzene ring / arene / high C : H ratio"; "orange precipitate with 2,4-DNPH indicates carbonyl"; "silver precipitate with Tollens' indicates aldehyde"; "decolourises bromine water indicates C=C"; "exists as geometric isomers indicates only 1 H on each carbon of the C=C".
- Alcohol class: "does not react with acidified potassium dichromate(VI), so tertiary alcohol" (WCH15 Oct 2020).
- Iodoform: CHI_3 alone scores for $CH_3CO–$ or $CH_3CH(OH)–$ (WCH15 Jan 2023).
- Inorganic: "pale green" solution means **iron(II)**; then use the masses to show it is a hydrate (WCH15 Oct 2021).

Often asked as: deduce a structure from a set of test results plus formula data; give the reagent and the observation for a named group; an inorganic salt identified from tests and percentage composition.

COMMON MISTAKE

Drawing a structure that contradicts one of the results you were given, e.g. a primary alcohol when dichromate stayed orange (Oct 2020 report).
Giving only the final colour. Write "orange to green", not "green".
Not linking a pale green solution to Fe^{2+}, or missing the water of crystallisation (Oct 2021 report).

BEYOND THE SPEC, BUT EXAMINED

- **Cerium(IV) ammonium nitrate** turns from yellow to **red** with an alcohol (WCH15 Oct 2020).
- **Iron(III) chloride** solution gives a **purple** colour with phenols. It is useful for checking aspirin for unreacted 2-hydroxybenzoic acid (20.4).

Topic 20 – Organic Synthesis

SPEC 20.1 Deducing formulae and structures from data

WHAT THIS SPEC POINT ASKS

Deduce empirical, molecular and structural formulae from combustion analysis, percentage composition, functional-group tests, infrared spectra, mass spectra and ^{13}C and ^{1}H NMR spectra.

20.1 A four-stage method

1. **Empirical formula** from combustion data or percentage composition.
2. **Molecular formula** from Mr (usually the M^+ peak in the mass spectrum).
3. **Functional groups** from chemical tests (19.12) and the IR spectrum.
4. **Full structure** from the NMR spectra and mass-spectrum fragments. Check that your structure fits **every** piece of data.

20.1 Stage 1a: combustion analysis

A known mass of the compound (C, H and maybe O) is burned in excess oxygen. The CO_2 and H_2O are collected and weighed.

- mass of C = mass of CO_2 × **12 ÷ 44**
- mass of H = mass of H_2O × **2 ÷ 18** (each H_2O contains **two** H atoms)
- mass of O = mass of sample − mass of C − mass of H (by difference)
- Then convert each mass to moles and find the simplest whole-number ratio.

WORKED EXAMPLE: COMBUSTION ANALYSIS

1.44 g of compound Q (C, H and O only) burns completely to give 3.52 g of CO_2 and 1.44 g of H_2O.
Mass of C = 3.52 × 12 ÷ 44 = 0.960 g. Mass of H = 1.44 × 2 ÷ 18 = 0.160 g.
Mass of O = 1.44 − 0.960 − 0.160 = 0.320 g.
Moles: C = 0.960 ÷ 12 = 0.0800; H = 0.160 ÷ 1 = 0.160; O = 0.320 ÷ 16 = 0.0200.
Divide by the smallest (0.0200): C 4, H 8, O 1. Empirical formula **C_4H_8O** (Mr 72).
The molecular ion is at m/z 72, so the molecular formula is also **C_4H_8O**.

20.1 Stage 1b: percentage composition

WORKED EXAMPLE: FROM PERCENTAGES

A compound contains 40.0% C, 6.7% H and 53.3% O by mass, and its Mr is 60.
Moles in 100 g: C 40.0 ÷ 12 = 3.33; H 6.7 ÷ 1 = 6.7; O 53.3 ÷ 16 = 3.33.
Ratio 1 : 2 : 1, so the empirical formula is **CH_2O** (Mr 30).
60 ÷ 30 = 2, so the molecular formula is **$C_2H_4O_2$** (e.g. ethanoic acid or methyl methanoate).

20.1 Stage 3: infrared spectrum

Look up wavenumbers in the **Data Booklet** and quote a value **within its range**. The ones used most:

Bond	Wavenumber / cm^{-1}	Notes
O–H (alcohol, phenol)	3750–3200	broad if hydrogen-bonded
O–H (carboxylic acid)	3300–2500	very broad
N–H (amine)	3500–3300	
C–H (aldehyde)	2900–2820 and 2775–2700	two weak peaks
C≡N (nitrile)	2260–2215	
C=O (ester)	1750–1735	
C=O (aldehyde)	1740–1720	
C=O (ketone)	1720–1700	aryl ketone 1700–1680
C=O (carboxylic acid)	1725–1700	
C=C (alkene)	1669–1645	
C=C (arene ring)	about 1600, 1580, 1500, 1450	

20.1 Stage 4: mass spectrum

- The **molecular ion peak M^+** (the last major peak) gives the Mr.
- **Fragments** show pieces of the structure. Write them **with a + charge**: m/z 15 CH_3^+, 29 $C_2H_5^+$ (or CHO^+), 43 CH_3CO^+ (or $C_3H_7^+$), 57 $C_2H_5CO^+$, 77 $C_6H_5^+$.
- **Chlorine** has isotopes ^{35}Cl : ^{37}Cl = 3 : 1. One Cl gives peaks at M and M+2 in a **3 : 1** ratio; two Cl give M, M+2 and M+4 in a **9 : 6 : 1** ratio.
- A small **M+1** peak comes from molecules containing one ^{13}C atom.

20.1 Stage 4: NMR spectra

^{13}C NMR: the number of peaks = the number of **different carbon environments**. Look for symmetry: in 1,4-disubstituted benzene the ring has fewer environments than in the 1,2- or 1,3-isomers.

^{1}H NMR gives three kinds of information:

Feature	Tells you
number of peaks	number of different H environments
chemical shift δ / ppm	the type of environment (use the Data Booklet). Roughly: alkyl 0.8–1.5; H–C next to C=O 2.0–2.8; H–C–O 3.3–4.3; arene H 6.5–8.0; aldehyde H 9–10; COOH 10–12
relative peak area (**integration**)	the ratio of H atoms in each environment
splitting	**n + 1 rule**: a peak is split into (n + 1) lines by n H atoms on the **neighbouring carbon(s)**. Singlet = 0 neighbours, doublet = 1, triplet = 2, quartet = 3

- An **O–H** (or N–H) proton usually gives a **singlet**: it neither splits nor is split.
- Equivalent H atoms don't split each other.

WORKED EXAMPLE: PUTTING THE EVIDENCE TOGETHER

Q is C_4H_8O (from the combustion example above).

- Tests: an **orange precipitate** with 2,4-DNPH (C=O); **no** silver mirror with Tollens' reagent (not an aldehyde), so it is a **ketone**. A **yellow precipitate** with iodine and NaOH shows $CH_3CO–$.
- IR: strong peak at 1715 cm^{-1} (ketone C=O); no broad O–H.
- Mass spectrum: M^+ at 72; fragments at 43 (CH_3CO^+) and 57 ($CH_3CH_2CO^+$).
- ^{13}C NMR: 4 peaks, so 4 carbon environments.
- ^1H NMR: three peaks, areas 3 : 2 : 3. A **singlet** at 2.1 ppm (CH_3 next to C=O, no H neighbours); a **quartet** at 2.4 ppm (CH_2, 3 H neighbours); a **triplet** at 1.0 ppm (CH_3, 2 H neighbours).
- Structure: $CH_3COCH_2CH_3$, **butanone**.

MARK-SCHEME LANGUAGE

- Combustion: "mass of C = 4.95 × (12 ÷ 44)", "mass of H = 0.90 × (2 ÷ 18)", "mass of O by difference", then moles and ratio. A correct answer with working scores all marks (WCH15 Oct 2025, Jun 2022).
- Rounding moles: "carbons per molecule = 95.6 ÷ 24 = 3.98, so 4" (WCH15 Jun 2024).
- ^1H NMR: "4 peaks means 4 (different) hydrogen environments"; "peak at 7.5 ppm is due to H on the benzene ring"; quartet "3 H on the neighbouring carbon"; singlet "no H on neighbouring carbon"; "area 5 for the ring shows C_6H_5" (WCH15 Jan 2022).
- "The OH proton neither splits nor causes splitting" (WCH15 Jan 2021). Unusual wording such as "two splits" is accepted if clear.
- Chemical shifts and IR: "allow a single value or a range **within** the Data Booklet range". A value outside the range scores 0.
- Fragments need the **+ charge** ("penalise omission of the positive charge"); a radical instead of an ion is wrong.
- Isotopes: "two atoms of chlorine, so a 9 : 6 : 1 ratio"; M+1 "due to one ^{13}C atom" (WCH15 Jun 2024).

Often asked as: empirical formula from combustion data (most series); a 6–7 mark "deduce the structure" from combined data; count ^{13}C environments; assign ^1H NMR peaks with splitting; identify fragment ions.

COMMON MISTAKE

Taking the moles of H_2O as the moles of H. That **halves** the hydrogen (Jun 2022 report: the commonest error).
Using trial and error instead of the systematic C, H, O route (Jan 2022 report).
Suggesting N-containing groups for a compound with no N in its formula.
Counting the C=O carbon as a proton environment, or counting both halves of a symmetrical molecule as different (Jan 2021, Oct 2024 reports).
Quoting IR or ^{13}C data when the question asked about ^1H NMR (Oct 2021 report).
Writing "^{36}Cl" or "$^{35.5}Cl$" (Jun 2024 report).

BEYOND THE SPEC, BUT EXAMINED

- **Inverse calculation**: percentage composition from a given formula, and values in ppm or ppb (WCH15 Oct 2022).
- **Gas chromatography and GC–MS**: components separate by retention time (how strongly each is attracted to the stationary phase), then each goes into the mass spectrometer (WCH15 Jun 2025).
- **Empirical formula from a skeletal formula** (about a third of candidates managed it in WCH15 Oct 2024). Count every carbon and hydrogen carefully.

SPEC 20.2 Lengthening the carbon chain: Grignard reagents

WHAT THIS SPEC POINT ASKS

Understand how to lengthen a carbon chain by using magnesium to make Grignard reagents, and how these react with carbon dioxide and with carbonyl compounds in dry ether.

20.2 Making a Grignard reagent

$CH_3CH_2Br + Mg \rightarrow CH_3CH_2MgBr$ (ethylmagnesium bromide)

- **Reagents and conditions**: a halogenoalkane (or halogenoarene, e.g. bromobenzene) with **magnesium** turnings, in **dry ether** (ethoxyethane), warmed gently under reflux.
- **Why dry?** Grignard reagents react with water at once, giving an alkane: $CH_3CH_2MgBr + H_2O \rightarrow CH_3CH_3 +$ Mg(OH)Br. All apparatus and reagents must be **anhydrous**.
- The C–Mg bond is polar with the carbon **δ−** (Mg is much less electronegative). So the carbon acts as a **nucleophile**: it attacks δ+ carbon atoms and forms a **new C–C bond**.

20.2 Reactions (two stages each: add, then dilute acid)

1. The Grignard reagent adds to the δ+ carbon of C=O (CO_2, an aldehyde or a ketone) in dry ether.
2. **Dilute acid** (e.g. dilute HCl or H_2SO_4) is added to hydrolyse the magnesium salt and release the product.

Grignard reacts with	Product (after dilute acid)	Example with CH_3CH_2MgBr
carbon dioxide (dry ice)	**carboxylic acid**, one C longer	propanoic acid, CH_3CH_2COOH
methanal	**primary** alcohol	propan-1-ol, $CH_3CH_2CH_2OH$
any **other aldehyde**	**secondary** alcohol	with ethanal: butan-2-ol, $CH_3CH_2CH(OH)CH_3$
a **ketone**	**tertiary** alcohol	with propanone: 2-methylbutan-2-ol, $CH_3CH_2C(OH)(CH_3)_2$

Equations, e.g.:

$CH_3CH_2MgBr + CO_2 \rightarrow CH_3CH_2COOMgBr$, then $CH_3CH_2COOMgBr + H^+ \rightarrow CH_3CH_2COOH + Mg^{2+} + Br^-$

$CH_3CH_2MgBr + CH_3CHO \rightarrow CH_3CH_2CH(CH_3)OMgBr$, then $+ H_2O \rightarrow CH_3CH_2CH(OH)CH_3 +$ Mg(OH)Br

20.2 Planning a Grignard route (working backwards)

1. Find the **carbon carrying the OH** (or the COOH carbon).
2. That carbon was the **C=O carbon** (or the CO_2 carbon) of the starting carbonyl.
3. **One** of the groups attached to it came from the Grignard reagent. Split the molecule there.

WORKED EXAMPLE: MAKING 2-METHYLBUTAN-2-OL, $(CH_3)_2C(OH)CH_2CH_3$

The OH carbon has two CH_3 groups and one CH_2CH_3 group, so it is a tertiary alcohol: make it from a **ketone**.

Route 1: CH_3CH_2MgBr + **propanone** (CH_3COCH_3), then dilute acid.

Route 2: CH_3MgBr + **butanone** ($CH_3COCH_2CH_3$), then dilute acid.

Each Grignard is made from the bromoalkane + Mg in **dry ether**.

20.2 Other ways to add carbon atoms

Method	Reagents	Gives	Carbons added
Grignard + CO_2	RMgX, then CO_2, then H^+	carboxylic acid	1
Grignard + carbonyl	RMgX + aldehyde or ketone, then H^+	alcohol	the R group
Cyanide + halogenoalkane	KCN in ethanol, reflux	nitrile (then acid or amine)	1
Cyanide + carbonyl	HCN + KCN	hydroxynitrile	1
Friedel–Crafts on an arene	RCl or RCOCl + $AlCl_3$	alkyl- or acylbenzene	R or RCO

MARK-SCHEME LANGUAGE

- "magnesium (powder) and (**dry**) ether"; "structure of the Grignard reagent" CH_3CH_2MgBr (CH_3CH_2BrMg is not accepted); "reaction with CO_2" then "hydrolysis with dilute acid / H^+". CO_2 and acid can be two labelled parts of one arrow.
- Missing "dry" ether is penalised (WCH15 Jan 2024).
- Grignard + methanal → primary alcohol; + other aldehyde → secondary; + ketone → tertiary; + CO_2 → carboxylic acid (WCH15 Oct 2020 MCQ). The Grignard acts as a **nucleophile** (WCH15 Jun 2024 MCQ).
- Dehydrating a Grignard alcohol product: "concentrated phosphoric acid or concentrated sulfuric acid" (or alumina) (WCH15 Jan 2021).

Often asked as: a 5–7 mark synthesis route using a Grignard step (benzene → benzoic acid, WCH15 Jan 2022; alcohol → acid one carbon longer, WCH15 Jan 2025); which Grignard and carbonyl give a named alcohol (MCQ).

COMMON MISTAKE

Forgetting **dry** ether, or merging CO_2 and the acid hydrolysis into one step with no acid (Jan 2025 report).
Making the Grignard from a compound that still has an –OH group: the OH would destroy it (Oct 2024 report).
Wrong number of carbons. Count before and after every step.
Putting Br on the ring as well as Mg in the aryl Grignard. It is C_6H_5MgBr: the Mg goes **between** the ring carbon and the Br (Jan 2022 report).

BEYOND THE SPEC, BUT EXAMINED

Grignard reagents and $LiAlH_4$ both react with water, which is why both need dry conditions (WCH15 Jan 2025 report). Choosing between a Grignard route and a cyanide route to the same acid is also examined (WCH15 Jun 2022).

SPEC 20.3 Problem solving in organic chemistry

WHAT THIS SPEC POINT ASKS

Use the organic chemistry in the specification to: (i) predict the properties of unfamiliar compounds with one or more known functional groups; (ii) plan reaction schemes of up to four steps; (iii) choose suitable practical procedures; (iv) choose control measures to reduce risk, using hazard data.

20.3 (i) Predicting properties of an unfamiliar compound

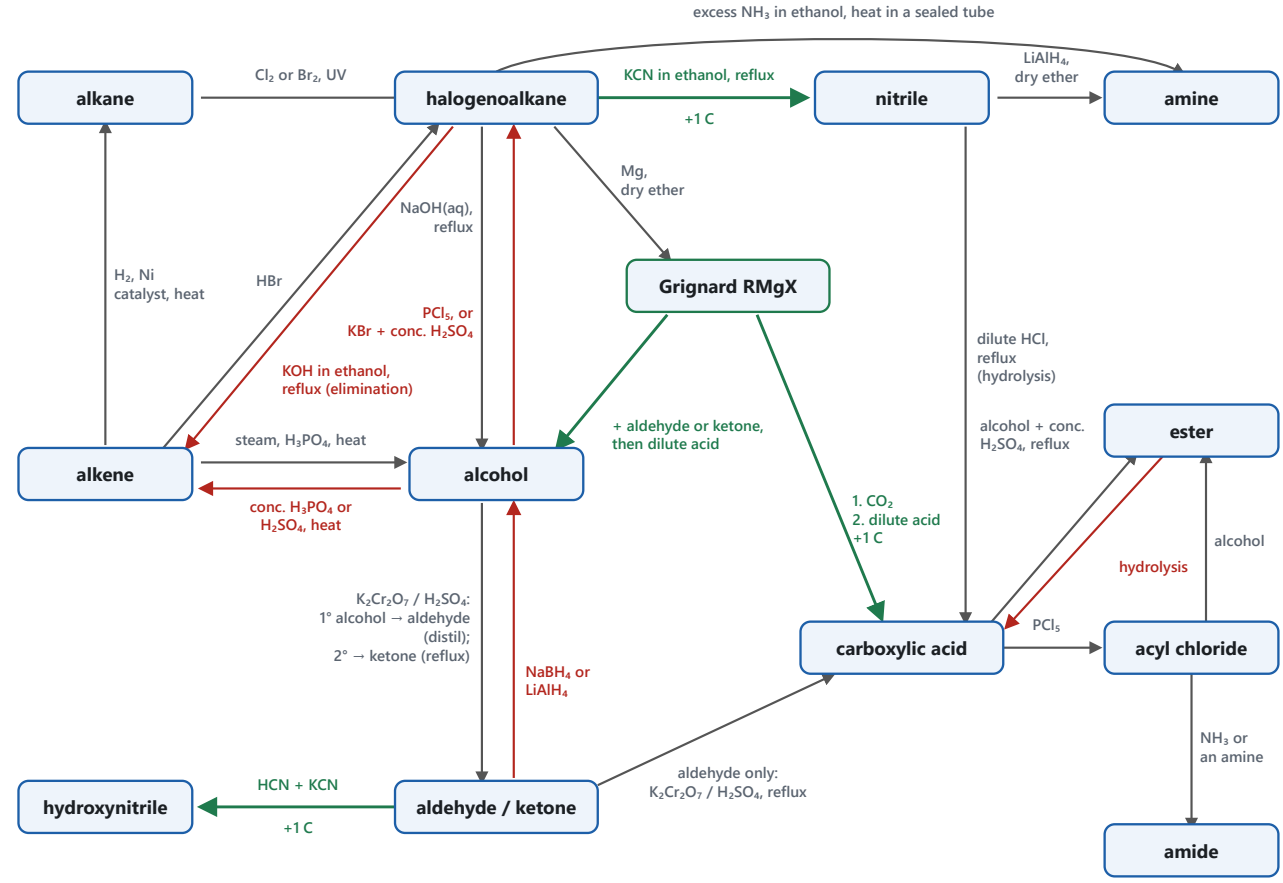
Find **each functional group** and apply its chemistry one at a time. Then check for things the groups do together (isomerism, polymers, hydrogen bonding).

WORKED EXAMPLE: PREDICTING THE REACTIONS OF $CH_2{=}CHCH_2CH(OH)COOH$

The compound is 2-hydroxypent-4-enoic acid. Its groups: an **alkene**, a **secondary alcohol** and a **carboxylic acid**.

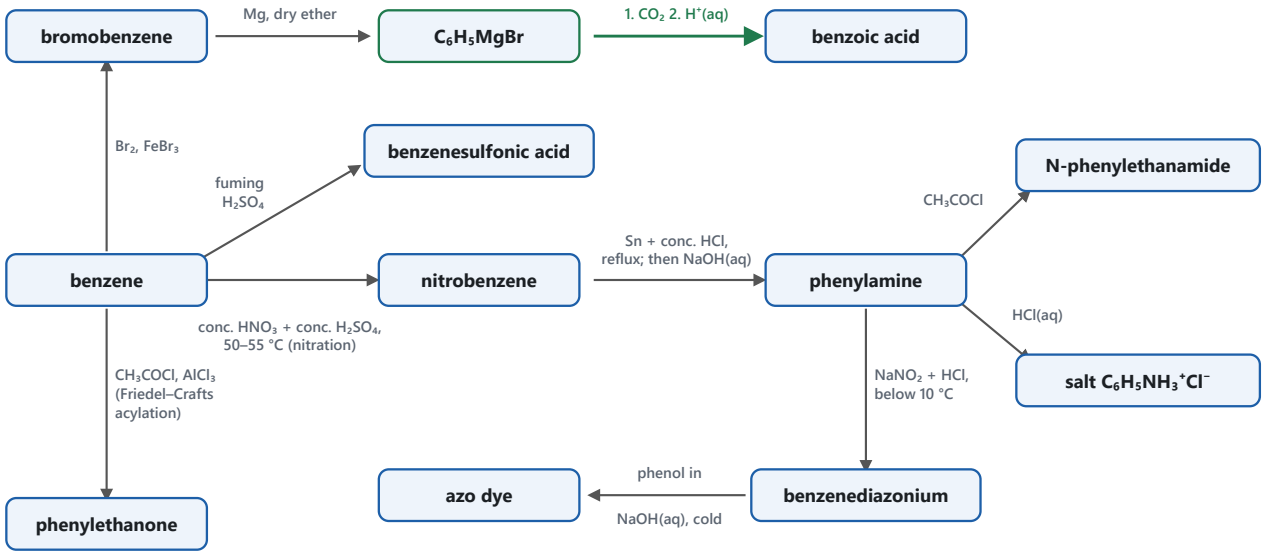
- **Bromine water**: decolourises (C=C addition), giving $CH_2BrCHBrCH_2CH(OH)COOH$.
- **Na_2CO_3(aq)**: fizzing, CO_2 given off (COOH).
- **PCl_5**: steamy fumes of HCl (both OH and COOH react).
- **Acidified $K_2Cr_2O_7$, warm**: orange → green. The secondary alcohol is oxidised to a ketone.
- **Iodine + NaOH**: no yellow precipitate. There is no $CH_3CH(OH)–$ or $CH_3CO–$ group.
- **Isomerism**: C2 is chiral (H, OH, COOH and $CH_2CH{=}CH_2$), so it has optical isomers. No E/Z isomers: the end carbon of the C=C has two H atoms.
- **Polymers**: OH and COOH in one molecule, so it can form a **polyester** by condensation; the C=C allows **addition** polymerisation too.
- **Solubility**: soluble in water, because the OH and COOH groups hydrogen-bond with water.

20.3 (ii) Planning a synthesis (up to four steps)



Green arrows add a carbon atom. Red arrows are the reverse of the black arrow beside them.

Aliphatic reaction map. Green arrows add a carbon atom.



Also: benzene + RCl and $AlCl_3$ → alkylbenzene (alkylation). Phenol + Br_2(aq) → 2,4,6-tribromophenol (no catalyst).

Aromatic reaction map (Topics 18 and 19).

Method

1. Write out the **start** and the **target**. Compare the **number of carbons** and the **functional groups**.
2. If the chain must get longer, you need a **cyanide** or **Grignard** step (or Friedel–Crafts on a ring). If it must get shorter by one, use the **iodoform** reaction (beyond the spec).
3. **Work backwards** from the target: "what could I make this from?" Then work forwards from the start, and find where the two meet.
4. For **every step** give the **reagent(s)**, the **conditions** (heat/reflux, distil, catalyst, solvent, temperature) and the **intermediate's structure**.

20.3 Reagents and conditions: master table

Change	Reagents and conditions
alkane → halogenoalkane	Cl_2 or Br_2, UV light
alkene → alkane	H_2, Ni catalyst, heat
alkene → halogenoalkane	HBr (or HCl), room temperature
alkene → dibromoalkane	Br_2, room temperature
alkene → alcohol	steam, H_3PO_4 catalyst, high temperature and pressure
halogenoalkane → alcohol	NaOH(aq) or KOH(aq), heat under reflux
halogenoalkane → alkene	KOH in ethanol, heat under reflux
halogenoalkane → nitrile	KCN in ethanol, heat under reflux (+1 C)
halogenoalkane → primary amine	excess NH_3 in ethanol, heat in a sealed tube
halogenoalkane → Grignard	Mg, dry ether
alcohol → chloroalkane	PCl_5, room temperature, dry
alcohol → bromoalkane	KBr + concentrated H_2SO_4 (makes HBr), heat
alcohol → alkene	concentrated H_3PO_4 or H_2SO_4, heat (or Al_2O_3)
1° alcohol → aldehyde	$K_2Cr_2O_7$ + dilute H_2SO_4, warm, **distil off** the aldehyde as it forms
1° alcohol → carboxylic acid	excess $K_2Cr_2O_7$ + dilute H_2SO_4, heat under **reflux**
2° alcohol → ketone	$K_2Cr_2O_7$ + dilute H_2SO_4, heat under reflux
aldehyde or ketone → alcohol	$NaBH_4$ (in water or methanol), or $LiAlH_4$ in dry ether
aldehyde or ketone → hydroxynitrile	HCN + KCN (+1 C)
carboxylic acid → primary alcohol	$LiAlH_4$ in dry ether, then dilute acid
carboxylic acid → acyl chloride	PCl_5 (or $SOCl_2$), dry
acid + alcohol → ester	concentrated H_2SO_4 catalyst, heat under reflux
acyl chloride → ester / amide / acid	alcohol / NH_3 or amine / water, room temperature
ester → acid + alcohol	dilute acid, reflux (or NaOH(aq), reflux, gives the carboxylate)
nitrile → carboxylic acid	dilute HCl, heat under reflux
nitrile → primary amine	$LiAlH_4$ in dry ether
benzene → nitrobenzene	conc. HNO_3 + conc. H_2SO_4, 50–55 °C
nitrobenzene → phenylamine	Sn + conc. HCl, reflux; then NaOH(aq)
phenylamine → diazonium ion	$NaNO_2$ + HCl, below 10 °C

WORKED EXAMPLE: ETHANOL TO PROPANOIC ACID (ONE MORE CARBON)

Ethanol has 2 C and propanoic acid has 3 C, so a chain-lengthening step is needed.

Route A (cyanide):

1. Ethanol → bromoethane: KBr + concentrated H_2SO_4, heat.
2. Bromoethane → propanenitrile, CH_3CH_2CN: KCN in ethanol, heat under reflux.
3. Propanenitrile → propanoic acid: dilute HCl, heat under reflux.

Route B (Grignard): steps 1 as above; then Mg in dry ether → CH_3CH_2MgBr; then CO_2, followed by dilute acid → CH_3CH_2COOH.

WORKED EXAMPLE: BENZENE TO N-PHENYLETHANAMIDE, WITH OVERALL YIELD

1. Benzene → nitrobenzene: conc. HNO_3 + conc. H_2SO_4, 50–55 °C. Yield 70%.
2. Nitrobenzene → phenylamine: Sn + conc. HCl, reflux, then NaOH(aq). Yield 60%.
3. Phenylamine → N-phenylethanamide ($C_6H_5NHCOCH_3$, Mr 135): ethanoyl chloride, room temperature. Yield 85%.

Overall yield = 0.70 × 0.60 × 0.85 = 0.357 = **35.7%**. Multiply the decimal yields; never add them.
What volume of benzene (Mr 78, density 0.879 g cm^{-3}) is needed for 5.00 g of product?
Moles of product = 5.00 ÷ 135 = 0.03704 mol.
Moles of benzene needed = 0.03704 ÷ 0.357 = 0.1037 mol. **Divide** by the yield: you need more than the theoretical amount.
Mass = 0.1037 × 78 = 8.09 g. Volume = 8.09 ÷ 0.879 = **9.21 cm^3**.

20.3 (iii) Choosing a procedure

Situation	Procedure and why
slow reaction with volatile liquids	**heat under reflux**: heats for a long time without losing material
product must be removed as it forms (aldehyde from a 1° alcohol)	**distil** it off, so it isn't oxidised further
flammable solvent or reactant (ether, ethanol)	**water bath or electric heating mantle**, no naked flame
unstable intermediate (diazonium ion), exothermic nitration	**ice bath**; add reagents slowly
Grignard, $LiAlH_4$, $AlCl_3$, PCl_5, acyl chlorides	**dry** apparatus and reagents (react with water)
only one group should react	a **selective reagent**: H_2/Ni reduces C=C but not C=O; $NaBH_4$ reduces C=O but not C=C

20.3 (iv) Hazards and control measures

Hazard in the data	Control measure
flammable	no naked flames; heat with a water bath or electric heating mantle
toxic or harmful vapour, or a toxic gas formed (HCl, HCN, Br_2)	**fume cupboard**
corrosive (conc. acids, PCl_5, acyl chlorides, NaOH)	gloves and eye protection; add slowly
irritant	gloves, eye protection; avoid skin contact
carcinogen (benzene)	use a less harmful substitute (methylbenzene or methoxybenzene) if possible, or work in a fume cupboard with gloves
reacts violently with water	keep dry; add carefully, in small portions

Control measures must **match the hazard**: "wear goggles" doesn't help with a flammable solvent.

KEY IDEA

Model 6-mark points: ethanoyl chloride v ethanoic anhydride for acylating phenylamine (WCH15 Jan 2024).

- Ethanoyl chloride gives **HCl**, a toxic, corrosive gas; the anhydride gives **ethanoic acid**, which is less hazardous (though flammable).
- Ethanoyl chloride reacts **violently** (with water too), so it is **harder to control**; the anhydride reacts more **slowly**.
- Use the hazard information the question gives for each reagent.
- **Atom economy** for making N-phenylethanamide (Mr 135):
  - with ethanoyl chloride: 135 ÷ (93 + 78.5) × 100 = **78.7%**
  - with ethanoic anhydride: 135 ÷ (93 + 102) × 100 = **69.2%**
- A higher atom economy means less waste: an advantage of the acyl chloride.

MARK-SCHEME LANGUAGE

- Oxidation: "$K_2Cr_2O_7$ and H_2SO_4 / acidified dichromate"; aldehyde "with distillation" (reflux is not accepted for the aldehyde); acid "reflux". Missing H^+, or HCl used for the acid, is penalised once (WCH15 Jan 2024, Oct 2024).
- Reduce only C=C: "H_2 and Ni / Pt". $LiAlH_4$ is not accepted, because it would reduce the ketone too (WCH15 Jan 2024, Oct 2024).
- Intermediates score even if a reagent is wrong. Give only **one** route: if you give two, the one scoring more marks is credited (WCH15 Jan 2024).
- Comparing two routes, any three: "number of steps", "purification required", "availability / cost of reagents", "toxicity of reagents / intermediates / by-products", "atom economy", "yield of each step", "side reactions". Unjustified "cost", "safety", "yield" or "rate" are ignored (WCH15 Jun 2024).

Often asked as: a 4–7 mark synthesis (almost every series); choose a selective reagent; a 6-mark hazards-and-atom-economy comparison; reverse yield calculations; pick a hazard symbol.

COMMON MISTAKE

Not knowing where to start. Work **backwards** from the target (Oct 2020, Jun 2022 reports).
Leaving out the acid catalyst in esterification (Jun 2022 report: the commonest slip).
Adding cyanide to benzene as a first step: that reaction doesn't happen (Oct 2023 report).
Oxidising a primary alcohol under reflux when you need the aldehyde. Distil it off (Oct 2024 report).
Hazard answers that just say "safety" or "harmful", with no atom-economy numbers (Jan 2024 report).
Show the molar masses.
Multiplying by the percentage yield in a reverse calculation. **Divide** (Jun 2022 report).

BEYOND THE SPEC, BUT EXAMINED

- **Descending the series**: the iodoform reaction (I_2 + NaOH) turns $CH_3CO–R$ into $RCOO^-$ + CHI_3, removing one carbon (WCH15 Jan 2021).
- **Unfamiliar mechanisms** given in the stem (radical steps with half-arrows, additions to C=O, catalytic cycles): follow the pattern shown and keep the arrows from bonds or lone pairs (WCH15 Oct 2021, Jan 2026).
- **Industrial context**: a factory route can't use some lab reagents (cost, hazard, scale), and more steps usually means a lower overall yield (WCH15 Jun 2025).

SPEC 20.4

Core Practical 16: preparation of aspirin

WHAT THIS SPEC POINT ASKS

Core Practical 16: prepare aspirin, and purify it.

20.4 The reaction

2-Hydroxybenzoic acid (salicylic acid) + ethanoic anhydride → 2-ethanoyloxybenzoic acid (aspirin) + ethanoic acid

$HOC_6H_4COOH + (CH_3CO)_2O \rightarrow CH_3COOC_6H_4COOH + CH_3COOH$

- The **phenol –OH** group of salicylic acid is **acylated**, forming an **ester**. Phenols react only slowly with carboxylic acids, so a more reactive **acid anhydride** is used. An **acid catalyst** (a few drops of concentrated H_2SO_4 or 85% H_3PO_4) speeds it up.
- Aspirin contains an **ester** group, a **carboxylic acid** group and a **benzene ring**.

CORE PRACTICAL 16: METHOD IN BRIEF

Preparation

1. Weigh about 2–3 g of 2-hydroxybenzoic acid into a dry conical (or pear-shaped) flask. Record the exact mass.
2. In a **fume cupboard**, add about 4–5 cm^3 of **ethanoic anhydride** (in **excess**) and 5 drops of concentrated sulfuric or phosphoric acid.
3. Warm in a **water bath** at about 60–80 °C for about 15–20 minutes, swirling, until the solid dissolves.
4. Carefully add a little **cold water** to react with (hydrolyse) the excess ethanoic anhydride, then more cold water. Cool in an **ice bath** until crystals form.
5. Collect the crude crystals by **filtration under reduced pressure** (Büchner funnel) and wash with a little **ice-cold water**.

Purification by recrystallisation

6. Dissolve the crude product in the **minimum volume of hot solvent** (hot water, or a little hot ethanol then warm water).
7. If there are insoluble bits, filter while **hot**.
8. Let the solution **cool slowly**, then in ice, so that pure crystals form. Soluble impurities stay in solution.
9. Filter under reduced pressure, wash with a little ice-cold water, and **dry** (between filter papers or in a desiccator). Weigh.

Purity check: measure the **melting temperature**. Pure aspirin melts sharply at about 135 °C; an impure sample melts lower and over a wider range.

20.4 Treating the results

WORKED EXAMPLE: PERCENTAGE YIELD AND LIMITING REAGENT

3.00 g of 2-hydroxybenzoic acid (Mr 138) reacts with 4.00 cm^3 of ethanoic anhydride (Mr 102, density 1.08 g cm^{-3}). 2.85 g of pure aspirin (Mr 180) is obtained.
Moles of acid = 3.00 ÷ 138 = 0.02174 mol.
Mass of anhydride = 4.00 × 1.08 = 4.32 g; moles = 4.32 ÷ 102 = 0.04235 mol. Ratio 1 : 1, so the anhydride is in excess and **2-hydroxybenzoic acid is limiting**.
Theoretical mass of aspirin = 0.02174 × 180 = 3.913 g.
Percentage yield = 2.85 ÷ 3.913 × 100 = **72.8%**.

WORKED EXAMPLE: ATOM ECONOMY

Atom economy = Mr of the desired product ÷ total Mr of all the reactants (from the balanced equation) × 100.
With ethanoic anhydride: 180 ÷ (138 + 102) × 100 = **75.0%** (by-product ethanoic acid).
With ethanoyl chloride (by-product HCl): 180 ÷ (138 + 78.5) × 100 = **83.1%**.
The anhydride is still chosen: it is cheaper, less violent, and doesn't give off corrosive HCl gas.

Why the yield is less than 100%

- Product is lost on transfer, and on the filter paper and glassware.
- Some aspirin stays **dissolved in the solvent** during recrystallisation and washing.
- The reaction may be incomplete, or some aspirin may be hydrolysed back to salicylic acid.

Sources of error and improvements

- Product still **damp** when weighed gives a mass that is too high. Dry it to constant mass.
- Too much recrystallisation solvent lowers the yield. Use the **minimum** volume of hot solvent.
- Washing with warm water dissolves product. Use **ice-cold** water, in small amounts.

MARK-SCHEME LANGUAGE

- Equation: "salicylic acid + ethanoic anhydride → acetylsalicylic acid + ethanoic acid" (skeletal accepted; molecular formulae not) (WCH15 Oct 2023).
- % yield: 2.00 g salicylic acid (Mr 138) gives a theoretical 2.6087 g of aspirin (Mr 180); 1.9513 g obtained, so 74.8% (WCH15 Oct 2023).
- Recrystallisation: "dissolve in the **minimum** volume of **hot** solvent, filter (hot) to remove **insoluble** impurities, cool, and filter to collect crystals". Soluble impurities stay in the solution. The solvent must dissolve the product well **when hot** and poorly **when cold** (WCH15 Jan 2025, Oct 2022 MCQs).
- Melting-point bath: **mineral oil**, because water boils below aspirin's melting temperature. The likely impurity is unreacted **2-hydroxybenzoic acid** (WCH15 Jun 2024 MCQs).

Often asked as: the equation, percentage yield, recrystallisation and melting-point MCQs, and a whole Section C on aspirin (WCH15 Oct 2023, 19 marks).

COMMON MISTAKE

Using the anhydride (in excess) for the theoretical yield. Use the **limiting** reagent.
Using Mr values for the wrong compound. The Mr of aspirin is 180; salicylic acid is 138.
Forgetting ethanoic acid when balancing the equation (Oct 2023 report).
Naming 2-hydroxybenzoic acid wrongly: only a quarter of candidates got it (Oct 2023 report).

BEYOND THE SPEC, BUT EXAMINED

- **Back titration for aspirin content**: each aspirin uses **2 NaOH** (one to neutralise the COOH, one to hydrolyse the ester) (WCH15 Oct 2023).
  A 0.330 g tablet is heated with 25.0 cm^3 of 0.200 mol dm^{-3} NaOH (5.00×10^{-3} mol). The excess NaOH needs 16.70 cm^3 of 0.100 mol dm^{-3} HCl (1.670×10^{-3} mol).
  NaOH used by aspirin = $5.00 \times 10^{-3} - 1.670 \times 10^{-3} = 3.330 \times 10^{-3}$ mol. Aspirin = $3.330 \times 10^{-3} \div 2 = 1.665 \times 10^{-3}$ mol.
  Mass = $1.665 \times 10^{-3} \times 180 = 0.2997$ g. Percentage of the tablet = $0.2997 \div 0.330 \times 100$ = **90.8%**.
- **Aspirin in the body**: it is a weak acid, so in the acidic stomach (extra H^+) its dissociation equilibrium shifts **left**, and in the alkaline small intestine (OH^- removes H^+) it shifts **right**. Don't say Ka changes (WCH15 Oct 2023).
- **Oil of wintergreen** (methyl 2-hydroxybenzoate) is made from 2-hydroxybenzoic acid and **methanol** with an acid catalyst. In ^1H NMR, aspirin's COOH proton is at 10–12 ppm (WCH15 Oct 2023).
- **Iron(III) chloride** test: a purple colour shows unreacted phenol (salicylic acid) left in the product.

SPEC 20.5 Techniques for preparing and purifying organic compounds

WHAT THIS SPEC POINT ASKS

Understand these techniques: (i) refluxing; (ii) washing, including with water and sodium carbonate solution; (iii) solvent extraction; (iv) recrystallisation; (v) drying; (vi) distillation; (vii) steam distillation; (viii) melting temperature determination; (ix) boiling temperature determination.

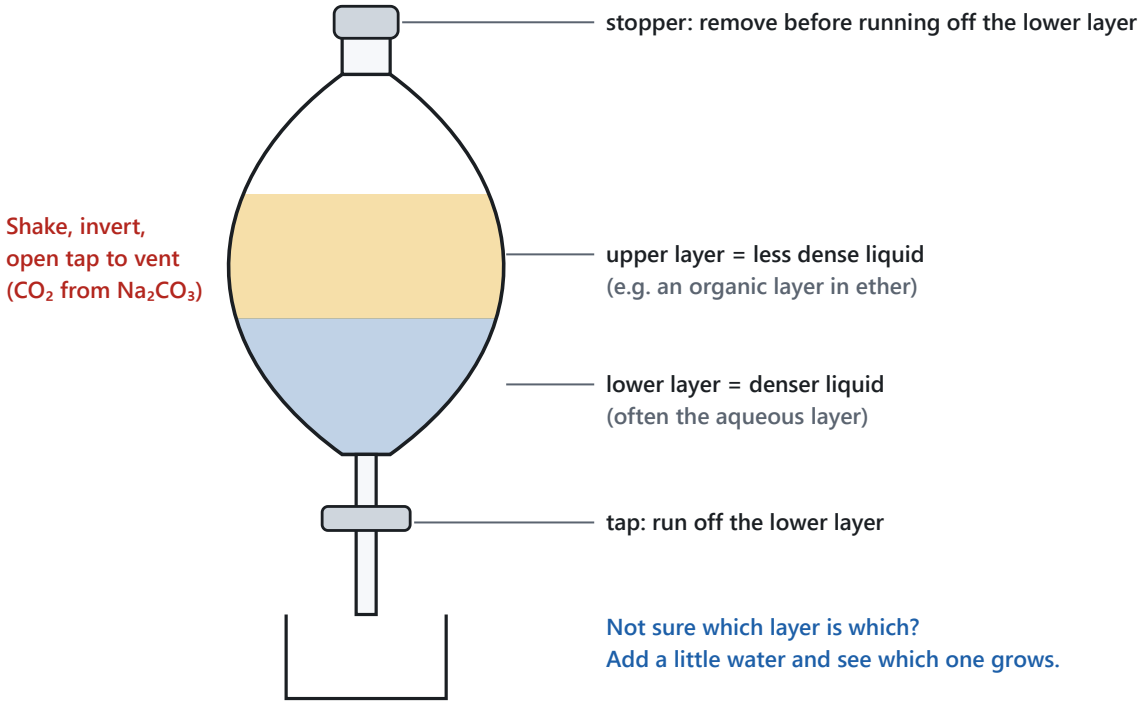
20.5 (i) Heating under reflux

- A round-bottomed or pear-shaped flask with **anti-bumping granules**, and a **vertical condenser** with its top **open**. Water enters the condenser at the **bottom** and leaves at the top.

- **Why**: vapours condense and drip back, so the mixture can be **heated for a long time** without losing volatile reactants or products. This lets slow organic reactions go to completion.
- Heat with an **electric heating mantle** or **water bath** (flammable liquids). Never stopper the top: pressure would build up.
- Anti-bumping granules make the liquid **boil smoothly**, without large bubbles.

20.5 (ii) Washing

- The crude organic liquid is shaken with an aqueous solution in a **separating funnel**, and the layers are separated.
- **Water** removes water-soluble impurities (e.g. acids, salts, small alcohols).
- **Sodium carbonate solution** (or $NaHCO_3$) **neutralises acidic impurities** (e.g. HCl, H_2SO_4, carboxylic acid), turning them into water-soluble salts. CO_2 is given off, so **invert the funnel and open the tap often** to release the pressure.
- Sometimes a NaCl solution (brine) wash follows, to remove most of the water.



Separating funnel. Run off the lower layer through the tap, then pour the upper layer out of the top.

20.5 (iii) Solvent extraction

- Shake the aqueous mixture with an **organic solvent that is immiscible with water** (e.g. ethoxyethane or dichloromethane) and in which the **product is much more soluble** than in water. The product moves into the organic layer.
- Separate the layers. Use **several small portions** of solvent rather than one large portion: this extracts more product.
- Combine the organic layers, dry them (v), and remove the solvent by distillation.
- Adding salt to the water ("**salting out**") lowers the organic compound's solubility in water and helps it move into the organic layer (used for phenylamine, 19.5).

20.5 (iv) Recrystallisation (purifying a solid)

1. Choose a solvent in which the product is **very soluble when hot** but **only slightly soluble when cold**.
2. Dissolve the impure solid in the **minimum volume of hot solvent**, so the solution is saturated when hot.

3. **Filter hot** (warmed funnel) to remove **insoluble** impurities.
4. **Cool slowly**: the product crystallises. **Soluble** impurities are present in small amounts, so they stay in solution.
5. Filter under **reduced pressure** (Büchner funnel), wash with a little **ice-cold** solvent, and dry.

Repeating recrystallisation gives a purer product but a lower yield.

20.5 (v) Drying

- **Liquids**: add an anhydrous drying agent such as **anhydrous magnesium sulfate, calcium chloride or sodium sulfate**. Swirl and add more until some powder moves freely and the liquid is **clear**, not cloudy. Then **decant or filter** off the solid.
- **Solids**: press between filter papers, leave in a **desiccator**, or warm in a low oven, well below the melting temperature.

20.5 (vi) Distillation

- Separates liquids by **boiling temperature**, or a volatile liquid from non-volatile substances.
- Thermometer **bulb level with the side arm** (measures the temperature of the vapour that is distilling). Condenser sloping down, water in at the lower end.
- Collect the fraction that distils **within about 2 °C of the expected boiling temperature**.
- Also used to **remove a product as soon as it forms**, e.g. an aldehyde from a primary alcohol, before it is oxidised further.

20.5 (vii) Steam distillation

- Steam is passed through the mixture (or it is boiled with water). A **volatile compound that is immiscible with water** distils over with the steam **below 100 °C**, well below its own boiling temperature.
- **Why**: the compound can be separated **without heating it to its boiling temperature**, where it might **decompose**. It also leaves behind non-volatile tars and inorganic salts.
- The distillate is cloudy: two layers of water and the compound. Separate them with a separating funnel or by solvent extraction.
- Used for **phenylamine** (from the tin/HCl mixture, 19.5) and to extract **essential oils** (e.g. lavender oil, limonene) from plants.

20.5 (viii) Melting temperature determination

- Pack a little **dry, finely powdered** solid into a capillary tube sealed at one end. Heat it in a melting-point apparatus or an **oil bath** (Thiele tube), **slowly** (about 1–2 °C per minute) near the melting temperature.
- Record the temperature range from first melting to completely liquid.
- **Pure**: melts **sharply** (over 1–2 °C) at the Data Book value. **Impure**: melts at a **lower** temperature, over a **wider range**.

20.5 (ix) Boiling temperature determination

- Distil the liquid (or heat it under reflux with a thermometer in the vapour). A **pure** liquid boils at a **constant temperature** that matches the Data Book value.
- Impurities make it boil **over a range** of temperatures. A dissolved non-volatile impurity **raises** the boiling temperature.

Technique	Removes or does
reflux	heats for a long time without loss of volatile substances
washing with water	water-soluble impurities
washing with Na_2CO_3(aq)	acidic impurities
solvent extraction	moves product from water into an immiscible solvent
recrystallisation	purifies a solid (removes soluble and insoluble impurities)
drying agent	water from an organic liquid
distillation	separates by boiling temperature; collects a volatile product
steam distillation	separates a volatile, water-immiscible compound below 100 °C
melting / boiling temperature	checks identity and purity

MARK-SCHEME LANGUAGE

- Solvent extraction: "the two solvents are immiscible / two layers are formed" and "the solute is more soluble in the extracting solvent" (WCH15 Oct 2025).
- Distillation: "the aldehyde is distilled off to prevent its further oxidation" (not "the temperature is too high to reach") (WCH15 Oct 2022).
- Anti-bumping granules: "so the solvent boils smoothly" (WCH15 Oct 2022 MCQ).
- Washing with NaOH or Na_2CO_3: the acid is "converted to an (ionic) salt that is more soluble in water" because the "ions are solvated by water" (WCH15 Jun 2022).
- Recrystallisation: "insoluble impurities are removed by filtering the hot solution" (WCH15 Jan 2026 MCQ).

Often asked as: MCQs on each technique; "suggest why" a step is done; name the technique for purifying a solid product; choose and justify a heat source.

COMMON MISTAKE

"Wash to remove impurities" without saying **which**: Na_2CO_3 removes **acidic** impurities.
Forgetting to **vent** the separating funnel when using carbonate.
Using lots of solvent in recrystallisation, or washing crystals with warm solvent: both dissolve product.
Stoppering the top of a reflux condenser.
Thinking an impurity **raises** the melting temperature. It lowers it and widens the range (Oct 2021 MCQ).

BEYOND THE SPEC, BUT EXAMINED

- **Purity from data**: a damp sample gives a mass too high and an apparent purity above the real value (WCH15 Oct 2024 MCQ; only a minority answered it correctly).
- **Chromatography**: R_f values from TLC or paper chromatography, and GC retention times, as purity checks (WCH15 Jan 2023, Jun 2025).